L05 — study notes

Contents

Classes 5+6 — Introduction to causal inference · Map of the notes **4**
Reading order . . . 5
Part I — the language: graphs and potential outcomes . . . 5
Part II — what goes wrong in a regression . . . 5
Part III — designs . . . 5
Support . . . 5
The one result to hold . . . 6
Numbers and cases to have memorized . . . 6
Figures . . . 6
Interactive companions . . . 7
Scope and ceiling . . . 7
Course-wide position . . . 8
Errata found in the sources . . . 8
Bibliography actually used in these notes . . . 9

Slides §2–7 · Simpson's paradox, and the graph that decides which table to read **10**
1 · U — three paradoxes from one book (slides pp. 5–7) . . . 10
2 · C — why the reversal happens: a pooled rate is a mix . . . 11
3 · C — the three junctions (slides pp. 8–9) . . . 12
3.1 · C — Gaussian algebra for the collider (why conditioning *creates* dependence) . . . 13
4 · C — paths and junctions on a larger DAG (slides pp. 9–10) . . . 14
5 · C — simulating data from a DAG (slide p. 10) . . . 15
6 · C — d-separation (slide p. 11) . . . 15
6.1 · Worked example 1 (slide pp. 11–12): $Z \to W \leftarrow X \to Y$, $W \to U$ . . . 15
6.2 · Worked example 2 (slide p. 12): add $T \to Z$ and $T \to Y$ . . . 16
7 · C — the marketing campaign (slides pp. 12–13) . . . 16
8 · G — what the slides do not ask . . . 16
Links . . . 16

Slides §8 · Potential outcomes: what a causal effect is, and the four assumptions that let data estimate it **18**
1 · U — the question the slides start from (p. 14) . . . 18
2 · C — potential outcomes and the estimands (slide p. 15) . . . 18
3 · C — the naive comparison: ATT plus selection bias (MHE p. 11) . . . 19
4 · C — the identifying assumptions (slide p. 16) . . . 20
5 · C — identification: the adjustment formula . . . 21
5.1 · Worked example: the slides' drug table, gender as L . . . 21
6 · C — the propensity score and inverse weighting (slides pp. 16–17) . . . 22

6.1 · Worked example: IPW on the drug table . . . 22
7 · C — regression as an adjustment device (MHE §3.2.1, pp. 38–44) . . . 23
8 · G — beyond the slides . . . 23
Links . . . 23

omitted-variable.html · MHE §3.2.2–3.2.3 · Short equals long plus omitted times included, and the controls you must not add **24**
1 · C — the site's model: a trivariate Gaussian . . . 24
2 · C — the omitted-variables bias formula . . . 25
3 · C — the site's simulation, made exact . . . 25
4 · C — reading MHE Table 3.2.1 through the formula (p. 46) . . . 26
5 · C — bad control (MHE §3.2.3, pp. 47–49) . . . 27
5.1 · C — proxy control (MHE pp. 49–50) . . . 27
Links . . . 28

error-in-variables.pdf · W §9.4 · A noisy regressor shrinks the slope towards zero, by exactly the reliability ratio **29**
1 · C — the slide's derivation (error-in-variables.pdf) . . . 29
2 · C — Wooldridge's two polar assumptions (W §9.4b, pp. 289–291) . . . 30
2.1 · Error in the dependent variable is harmless for consistency (W §9.4a, pp. 287–289) 30
3 · C — other regressors make it worse (W eq. 9.36, p. 291) . . . 30
4 · C — the cure: an instrument that is a second measurement (W §15.4, pp. 479–481; MHE p. 84) . . . 31
Links . . . 32

Slides §9 · Instrumental variables: a variable that moves the treatment and nothing else identifies the effect as a ratio **33**
1 · U — the IV graph and the examples (slide p. 17) . . . 33
2 · C — the two-equation system and the IV ratio (slides pp. 17–18) . . . 34
3 · C — why adding z to the outcome regression does not work (slides pp. 18–19; G p. 173) 35
4 · C — the Wald estimator (MHE §4.1.2, pp. 94–96) . . . 35
5 · C — two-stage least squares (MHE §4.1.1, pp. 89–91; W §15.2–15.3, pp. 471–477) . . . 36
6 · C — the price and the fragility of IV (W pp. 466–470, 478) . . . 36
7 · C — the worked example: Card's college proximity (slide p. 19; example1-iv-card.html; W Example 15.4) . . . 37
8 · G — heterogeneous effects: LATE (MHE §4.4.1, Theorem 4.4.1, p. 114) . . . 38
Links . . . 39

Slides §10 · Difference-in-differences: subtract the control group's change, and assume it is the treated group's counterfactual change **40**
1 · U — two back doors, one design (slides pp. 21–22) . . . 40
2 · C — the identifying assumption (MHE §5.2, p. 170) . . . 40
3 · C — the slides' two strategies and the regression (slide p. 22; W Table 13.3) . . . 41
4 · C — the worked example: the site's Card–Krueger file . . . 42
5 · C — checking and extending (MHE pp. 172–178) . . . 43
6 · G — the slides' Bayesian version (slide p. 25) . . . 43
Links . . . 43

Slides §11 · Regression discontinuity: when a rule switches treatment on at a cutoff, the jump in the outcome is the effect at the cutoff **45**

1 · U — birthdays and funerals (slides pp. 26–28) . . . 45
2 · C — the sharp design (slide p. 28; MHE §6.1, p. 189) . . . 45
3 · C — the linear model (slide p. 29; MHE eq. 6.1.2, p. 190) . . . 46
4 · C — identification without functional form (MHE eq. 6.1.7, p. 193) . . . 46
5 · C — polynomials and interactions (slides pp. 30–31; MHE eqs. 6.1.4–6.1.6, p. 192) . . . 46
6 · C — the worked example: the site's MLDA data . . . 47
7 · C — local linear regression and the bandwidth (slide p. 36; MHE p. 193) . . . 48
8 · G — beyond the sharp parametric design . . . 49
Links . . . 49

Classes 5+6 · Formula sheet: every definition, result and number of the unit 50
0 · The exam specification, item by item . . . 50
1 · Definitions and results . . . 51
Simpson and DAGs (L05-01-simpson-dags) . . . 51
Potential outcomes (L05-02-potential-outcomes-identification) . . . 51
Omitted variables, bad control (L05-03-omitted-variables-bad-control) . . . 52
Errors in variables (L05-04-errors-in-variables) . . . 52
Instrumental variables (L05-05-instrumental-variables) . . . 52
DiD (L05-06-difference-in-differences) . . . 53
RDD (L05-07-regression-discontinuity) . . . 53
2 · Every number in the unit . . . 54
Links . . . 55

Classes 5+6 · Pitfalls and a True/False self-test 56
Part I — the confusions . . . 56
1 · "Control for everything" vs control for confounders only . . . 56
2 · A collider blocks; conditioning on it opens . . . 56
3 · d-separation is not "no arrow between them" . . . 56
4 · Simpson: the arithmetic never decides . . . 56
5 · Naive difference = ATE . . . 56
6 · QTE = quantile of individual effects . . . 56
7 · Ignorability can be tested with enough data . . . 57
8 · OVB means the long regression is causal . . . 57
9 · Measurement error biases towards zero, always . . . 57
10 · Add the instrument to the regression . . . 57
11 · A significant first stage is a strong first stage . . . 57
12 · IV is "less biased" than OLS . . . 57
13 · DiD needs equal levels . . . 57
14 · The DiD regression's β_2 is the effect . . . 57
15 · RD identifies the average effect . . . 57
16 · A jump in a polynomial fit is a discontinuity . . . 57
Part II — True/False self-test (original items) . . . 58
Block A — Simpson and DAGs . . . 58
Block B — potential outcomes and identification . . . 58
Block C — omitted variables, bad control, measurement error . . . 59
Block D — instrumental variables . . . 59
Block E — DiD and RDD . . . 60
Links . . . 60

Classes 5+6 · Python — the reference implementation, asserted against the sources 61

0 · Set-up: the comparison helper . . . 61
1 · DAGs and d-separation (L05-01-simpson-dags §3–6) . . . 62
2 · Potential outcomes, adjustment and IPW (L05-02-potential-outcomes-identification §5–6) 65
3 · Omitted-variable bias (L05-03-omitted-variables-bad-control §1–5) . . . 66
4 · Errors in variables (L05-04-errors-in-variables §1–4) . . . 67
5 · Instrumental variables on card.csv (L05-05-instrumental-variables §2–7) . . . 68
6 · Difference-in-differences (L05-06-difference-in-differences §3–6) . . . 70
7 · Regression discontinuity (L05-07-regression-discontinuity §5–7) . . . 71
8 · Running it . . . 72
Links . . . 72

Classes 5+6 — Introduction to causal inference · Map of the notes

One claim. A causal effect is a contrast of potential outcomes, and data identify it only through an assumption the data cannot check. The graph decides what to condition on: forks must be blocked, chains out of the treatment and colliders must not be — which is why the same Simpson table is read stratified for gender and pooled for post-treatment blood pressure. **Regression inherits every failure of that choice:** leaving out a confounder moves the coefficient by $\gamma \cdot \delta$, a noisy regressor shrinks it by $\sigma_x^2/(\sigma_x^2 + \sigma_u^2)$, and a bad control reintroduces selection bias. **When no set of observables will do, design supplies the variation:** an **instrument** that moves the treatment and nothing else (Card: 0.1315 vs OLS 0.0747), a **control group** with the same trend (Card–Krueger: DiD +2.75 FTE), or a **cutoff** with a smooth counterfactual (MLDA: +4.53 deaths per 100,000 at 21). Each design trades one untestable assumption for another.

Three layers per topic:

Layer	What it means here
U — intuition	the three Simpson stories; the IV graph; Huntington-Klein's open-office story; birthdays and funerals
C — course (examinable)	the slides' and MHE/W results with their numbering: pooled rate as a mix, Rules 1–3, d-separation; $Y(a)$, ATE/ATT/QTE/CATE, SUTVA, consistency, ignorability, positivity, adjustment formula, IPW identity; OVB (MHE 3.2.11), bad and proxy control; attenuation (W 9.33, 9.36); IV ratio, reduced form, the failure of y on (x, z), Wald, 2SLS, IV variance and weak instruments; DiD (MHE 5.2.1–5.2.3, W 13.10–13.11); sharp RD (MHE 6.1.1–6.1.7) and local linear — **all with the site's own printouts**
G — graduate	the inverse-Wishart variance of the long regression; the LATE theorem (MHE 4.4.1); fuzzy RD as IV (MHE 6.2.5); the slides' Bayesian DiD/RDD and Bayesian IV (Greenberg §11.2). **Named or proved, not examinable**

Reading order

Part I — the language: graphs and potential outcomes

1. L05-01-simpson-dags — slides pp. 5–14. **The drug wins in both strata (93 vs 87, 73 vs 69) and loses pooled (78 vs 83) because a pooled rate is a mix: composition −10.5 points beats the within effect +5.9.** Chains, forks, colliders; Harrison's 5 paths and 15 junctions; d-separation on the slides' two examples, checked exhaustively; the collider Z_3 that makes $\{Z_3\}$ alone a wrong adjustment set; the slide's R code that does not propagate noise.
2. L05-02-potential-outcomes-identification — slides pp. 14–17; MHE Ch. 2 (pp. 10–18), §3.2.1 (pp. 38–44). **Naive = ATT + selection bias; randomisation removes it; ignorability + positivity give the adjustment formula, and IPW is the same thing proved line by line.** Drug table: naive −4.57, ATE +5.37, ATT +4.80. Links the existing notebook and the three potential-outcome companions.

Part II — what goes wrong in a regression

3. L05-03-omitted-variables-bad-control — omitted-variable.html; MHE §3.2.2–3.2.3 (pp. 44–50). **Short = long + γδ; the site's simulation is** $0.75+1.2\rho$**, with exact sds; the long regression is centred but its variance explodes as** $\rho \to 1$**.** MHE Table 3.2.1 read through OVB; bad control (3.2.12) and proxy control (3.2.15).
4. L05-04-errors-in-variables — error-in-variables.pdf; W §9.4 (pp. 287–292), §15.4. **Attenuation factor** $\sigma_x^2/(\sigma_x^2+\sigma_u^2)$**; error in** y **is harmless; controls make attenuation worse (0.39 vs 0.64); a second measurement as instrument restores** β**.**

Part III — designs

5. L05-05-instrumental-variables — slides pp. 17–20; MHE §4.1 (pp. 84–96); W §15.1–15.5; G §11.2. $\beta_1 = \theta_1/\delta_1 = \mathrm{Cov}(z,y)/\mathrm{Cov}(z,x)$**; adding** z **to the outcome regression leaves the bias** σ_{xy}/σ_x^2**; Wald 0.0891 (MHE Table 4.1.2); Card: OLS 0.0747, IV 0.1315 (SE 0.0550), first stage 0.320,** $F = 13.26$**; a weak first stage magnifies an exclusion violation as** φ/δ_1**.**
6. L05-06-difference-in-differences — slides pp. 21–25; MHE §5.2 (pp. 169–178); W §13.2 (pp. 407–411). **Common trends ⇒ DD =** δ**; four cells = four coefficients; Card–Krueger DiD 2.750 (SE 1.731), the opposite sign of the textbook labour demand story; a differential trend moves it one for one.** MHE's p. 170 bracket order corrected.
7. L05-07-regression-discontinuity — slides pp. 26–38; MHE §6.1–6.2 (pp. 189–198). **Continuity ⇒ jump = effect at the cutoff; MVA +4.534 (3.411 residual sds), quadratic 4.663, local linear 4.76 at** $b = 1$**; internal causes and fake cutoffs show nothing on the full sample — but one window of the placebo outcome does.**

Support

8. L05-08-formulas — the class goals mapped to notes; every numbered result; every number.
9. L05-09-pitfalls — sixteen confusions with minimal counterexamples and an original 25-item True/False self-test.
10. L05-10-python — the reference implementation (`L05_python_checks.py`): d-separation, IPW, OVB, attenuation, IV/2SLS on card.csv against `linearmodels`, DiD, RDD — all asserted against the sources' numbers.

The one result to hold

$$\underbrace{E[Y \mid A=1] - E[Y \mid A=0]}_{\text{what the data compare}} = \underbrace{E[Y(1) - Y(0) \mid A=1]}_{\text{ATT}} + \underbrace{E[Y(0) \mid A=1] - E[Y(0) \mid A=0]}_{\text{selection bias}}$$

Every method in the unit is a way to make the last term vanish. Randomisation sets it to zero by design; adjustment for confounders (note 02) zeroes it within cells of L; OVB (note 03) is its regression form; IV compares groups defined by the instrument, whose untreated outcomes are equal by exclusion; DiD assumes the difference in $Y(0)$ is constant over time and differences it out; RD assumes it vanishes at the cutoff.

Numbers and cases to have memorized

Case	Number	Meaning
Simpson drug table	93/87, 73/69, **78/83**; ATE **+5.37** points	pooled ≠ stratified; adjust for the fork
Collider, unit coefficients	$\text{Corr}(X, Z \mid W) = -0.5$	conditioning creates dependence
OVB simulation	$E\hat{\beta}^s = 0.75 + 1.2\rho$: 1.89 at $\rho = 0.95$	short = long + γρ
Attenuation	$\lambda = 0.5$ when $\sigma_u = \sigma_x$	half the slope
Wald (MHE Table 4.1.2)	$-0.01349 / -0.1514 =$ **0.0891**	reduced form over first stage
Card	OLS **0.0747**, IV **0.1315** (SE 0.0550), first stage 0.320, t 3.64, F 13.26	IV almost twice OLS, SE 15.7 times
Card–Krueger	PA −2.283, NJ +0.467, DiD **2.750** (SE 1.731)	no employment loss
MLDA (site data)	MVA jump **4.534** (SE 0.768, 3.411 sds); internal 0.392	the effect at 21
MLDA (slide, all deaths)	$\hat{\rho} =$ **7.7**; +100 deaths/day on 150	Mastering 'Metrics

---

Figures

Built by `figures/make_figures.py` (run: `python Notas/L05/figures/make_figures.py`; every figure ends in `assert` checks on its plotted numbers).

Figure	Shows	Lives in
L05-simpson-weights	pooled recovery rate vs share of women for drug / no drug; observed vs common mix; exercise–cholesterol reversal	L05-01-simpson-dags §2
L05-collider-bias	exact partial correlation given a collider and given its descendant	L05-01-simpson-dags §3.1

Figure	Shows	Lives in
L05-ovb-sampling	seeded re-run of the site's simulation with exact means and sds, long vs short	L05-03-omitted-variables-bad-control §3
L05-attenuation	attenuation factor vs noise-to-signal, with controls; seeded scatter	L05-04-errors-in-variables §3
L05-iv-plims	OLS, IV and long-regression plims vs ρ at the Card calibration; IV vs OLS bias vs δ_1 with $F = 10$	L05-05-instrumental-variables §6
L05-did-card-krueger	four cell means and the common-trend counterfactual; effect vs a differential trend	L05-06-difference-in-differences §4
L05-rdd-mlda	MLDA data with linear and quadratic fits; local linear jump vs bandwidth	L05-07-regression-discontinuity §7

Interactive companions

New for this unit (verified by `companions/check_companions.py`: each page's JavaScript under `node` against numpy at both ends of every range and at the defaults):

- The Instrument Dial (`companions/companion-iv-plims.html`) — move the true effect β_1, the endogeneity ρ, the first stage δ_1, an exclusion violation φ and n; read the OLS and IV biases (defaults reproduce Card's 0.0747 and 0.1315), the long-regression plim, SEs and the first-stage F against 10. Linked from note 05.
- The Discontinuity Window (`companions/companion-rdd-window.html`) — choose the outcome, bandwidth, polynomial order, separate slopes and the cutoff (placebo tests off 21); read $\hat{\rho} \pm 1.96$SE, the jump in residual sds and $\hat{\rho}$ against every bandwidth. Linked from note 07.

Already built for potential outcomes (in `Aula/class-05-06/`, linked from note 02): Selection Bias Decomposer (`../../Aula/class-05-06/companion-selection-bias.html`) · Bigger Sample, Same Bias (`../../Aula/class-05-06/companion-sample-size.html`) · The Eight-Person Table (`../../Aula/class-05-06/companion-eight-person-table.html`).

PDF of these notes: L05-notes.pdf (`python -I ~/.claude/commands/templates/notes_pdf.py Notas/L05`).

Scope and ceiling

- **Examinable (C):** the results of notes 01–07 as derived there. The final exam has no coding (Aula/COURSE): expect the bias formulas (OVB, attenuation, OLS under endogeneity), the IV ratio and the failure of adding z, the DiD and RD identification arguments, d-separation on a small graph, and readings of the site's Card, Card–Krueger and MLDA printouts.
- **Proved but above the line (G):** the exact sampling sds of the OVB simulation; the LATE theorem's statement.

- **Named only:** matching and doubly robust estimators; causal discovery; fuzzy RD; data-driven bandwidths; Bayesian nonparametric RD (NBA draft); the slides' Bayesian DiD/RDD; Bayesian IV (Greenberg §11.2, the Lopes papers on slide p. 20).
- **Belongs to other classes:** posterior simulation and priors (Classes 3+4); panel fixed and random effects (W Ch. 14, MHE §5.1: syllabus topic 03, not on the site schedule) and hierarchical models (Class 7).

Course-wide position

Class	Relation to Classes 5+6
1 — Review	OLS as Cov / Var and as a projection: every bias formula here is that ratio with a contaminated covariance
2 — Statistical learning	prediction vs inference: a model can predict well with a confounded coefficient; the `default` student coefficient that flips sign is a Simpson reversal; polynomial order and CV mirror the RD bandwidth choice
3+4 — Bayesian learning	the tools behind the slides' Bayesian DiD/RDD and Greenberg's Bayesian IV and treatment models
5+6 — Causal inference	**Simpson, DAGs, potential outcomes, OVB, measurement error, IV, DiD, RDD**
7 — More econometric models	panel and hierarchical models generalise the state and period effects of DiD; probit/Tobit for limited outcomes
8–9 — Time series	VAR identification (structural VAR) is the time-series version of the simultaneous-equations problem IV began with (MHE p. 84)

Rules packs: rules/02 (potential outcomes, selection bias, OVB, bad controls) · rules/01 (endogeneity, measurement error, 2SLS, Wald, LATE, weak instruments) · rules/03 (DiD and parallel trends).

Errata found in the sources

Source, place	Slip	Correct	Note
slides p. 7 (Example 3)	"if we don't segregate by gender"	by **age**	L05-01-simpson-dags §1
slides p. 9	collider picture labels both parents "Y"	$X \to Z \leftarrow Y$ (Rule 3 beneath)	L05-01-simpson-dags §3
slides p. 10, R code	noise added after all nodes are computed, so it never propagates	generate each node from its parents' noisy values, as the equations say	L05-01-simpson-dags §5

Source, place	Slip	Correct	Note
slides p. 15	heading "STUVA"	SUTVA	L05-02-potential-outcomes-identification §4
omitted-variable.html	$\mathrm{Var}(y \mid x, z)$ numerator "$1 - \rho - ...$"	$1 - \rho^2 - ...$ (equal only at $\rho = 0$)	L05-03-omitted-variables-bad-control §1
slides p. 18	"$z_i \to x_i \to y_i$ is a fork"	a **chain** (Rule 1)	L05-05-instrumental-variables §2
W p. 470	"Corr(z,u) must be less than one-fifth of Corr(z,u)"	… of Corr(**x**,u)	L05-05-instrumental-variables §6
MHE p. 170	population DD written as (PA change) − (NJ change) = β	(NJ change) − (PA change) = β	L05-06-difference-in-differences §2
slides p. 24	"NJ (0) and PA (1)"	`treatment = 1` is New Jersey (cell means match MHE Table 5.2.1)	L05-06-difference-in-differences §4
slides p. 24, R plot	y-axis "Employment growth"	employment level (FTE per store)	L05-06-difference-in-differences §4
slides p. 33, R code	`nsig1 = round(delta/sigma1,3)`	`delta1/sigma1` (0.380, not 4.394); the site page has it right	L05-07-regression-discontinuity §6

Bibliography actually used in these notes

Book / source	Where
The class materials (`Aula/class-05-06/materials/`): notes-on-causal-inference.pdf, error-in-variables.pdf, the four site pages and three data files	everywhere; the spine of each note
MHE (Angrist & Pischke 2009) Ch. 2, §3.2, §4.1, §4.4.1, §5.2, §6.1–6.2	notes 02, 03, 05, 06, 07
Wooldridge 6e §9.4, §13.2, §15.1–15.5	notes 04, 05, 06
Greenberg 1e §11.1–11.2	notes 02, 05 (Bayesian treatment and IV, named)

Related project material. Potential outcomes (built from the old draft plan, kept and linked here): notebook; quizzes 1, 2, 3; practice plan pratica-class-01-potential-outcomes; narration class-01-narrated.txt; video `Videos/class-01-potential-outcomes/`; NotebookLM audio prompts part 1 and part 2. Readings, practice and Python per class: Map/class-plan. Practice for this class (statements in the books): W 13.2 (p. 425), 13.6 (p. 426), 13.C3 (p. 427), 15.C3 (p. 492), 15.C5 (p. 493); G 11.1–11.2 (p. 180), 11.3–11.4 (p. 181); MHE Tables 2.2.1 (p. 14), 2.2.2 (p. 15), 3.2.1 (p. 46), 4.1.1 (p. 92), 4.1.2 (p. 96), 4.1.3 (p. 97), 5.2.1 (p. 171), Fig. 6.1.2 (p. 195), Table 6.2.1 (p. 202) as worked examples — each note's `## Links` gives the ones it serves.

Slides §2–7 · Simpson's paradox, and the graph that decides which table to read

> **Claim.** The same four recovery rates favour the drug inside each stratum (93% vs 87%, 73% vs 69%) and oppose it in the pooled table (78% vs 83%), because a pooled rate is a **weighted average of the stratum rates with group-specific weights**. Arithmetic cannot say which table answers the causal question; the **causal graph** can. If the stratifying variable is a common cause (a **fork**, gender) read the stratified table; if it is a consequence of the treatment (a **chain**, blood pressure after treatment) read the pooled one. Three junctions and one blocking rule, **d-separation**, turn that judgement into a mechanical test of which variables to condition on, including the trap of conditioning on a **collider**. Prev: L05-00-map · Next: L05-02-potential-outcomes-identification · Map: L05-00-map

Notation.

- $D \in \{0,1\}$: drug taken; $R \in \{0,1\}$: recovered; G: gender ($G = 1$ woman); π: share of women in a group.
- $r_{dg} = P(R = 1 \mid D = d, G = g)$: stratum recovery rate; $\bar{r}_d$: pooled rate of group $D = d$.
- $\pi_d = P(G = 1 \mid D = d)$: share of women among those with $D = d$; $\pi^* = P(G = 1)$: share in all 700 patients.
- In a DAG, $A \to B$: A is a direct cause of B. $A \perp B \mid S$: A and B independent given the set S.
- X, Z, W, U, T, Y: the generic nodes of the slides' examples; a, b, s, t: coefficients and noise sds of the Gaussian illustrations (§6).

---

1 · U — three paradoxes from one book (slides pp. 5–7)

The slides open with three examples quoted from Pearl, Glymour and Jewell, *Causal Inference in Statistics*, ch. 1. The paradox, in the slides' words: "a statistical association that holds for an entire population is re[v]ersed in every subpopulation" (p. 5).

Example 1–2, the drug table (700 patients, 350 chose the drug; p. 6):

	Drug	No drug
Men (or: low BP)	81 of 87 = **93%**	234 of 270 = **87%**
Women (or: high BP)	192 of 263 = **73%**	55 of 80 = **69%**
Combined	273 of 350 = **78%**	289 of 350 = **83%**

The drug wins in both rows and loses in the total. The slides then put **the same numbers** under two stories:

- **Gender** (p. 6): estrogen lowers recovery whatever the treatment, and women are much more likely to take the drug. Gender causes both: $D \leftarrow G \to R$, plus $D \to R$.
- **Blood pressure measured after treatment** (p. 7): the drug works *by* lowering BP but is also toxic. BP is caused by the drug: $D \to BP \to R$, plus $D \to R$.

  > **"Remarkably, though the numbers are the same in the gender and blood pressure examples, the correct result lies in the segregated data for the former and the aggregated data for the latter"** (slide p. 7). Nothing in the table distinguishes the two cases; only the arrow between the third variable and the drug does.

Example 3 (p. 7): within each age group, more exercise goes with lower cholesterol; pooled over ages the slope turns positive, because older people exercise more and have higher cholesterol regardless. Age is "a common cause of both treatment (exercise) and outcome (cholesterol)": exer ← age → chol. (The slide says "if we don't segregate by gender" here; the variable is **age**.)

---

2 · C — why the reversal happens: a pooled rate is a mix

Pooled rate as a weighted average. By the law of total probability, for group $D = d$:

$$\bar{r}_d = P(R = 1 \mid D = d) = \sum_g P(G = g \mid D = d)\, r_{dg} = (1 - \pi_d)\, r_{d0} + \pi_d\, r_{d1}. \tag{1}$$

Plug in the drug group ($\pi_1 = 263/350 = 0.75143$, $r_{10} = 81/87 = 0.93103$, $r_{11} = 192/263 = 0.73004$):

$$\bar{r}_1 = (1 - 0.75143)(0.93103) + (0.75143)(0.73004) = 0.23143 + 0.54857 = 0.78000.$$

Plug in the no-drug group ($\pi_0 = 80/350 = 0.22857$, $r_{00} = 234/270 = 0.86667$, $r_{01} = 55/80 = 0.68750$):

$$\bar{r}_0 = (1 - 0.22857)(0.86667) + (0.22857)(0.68750) = 0.66857 + 0.15714 = 0.82571.$$

Decompose the pooled gap. Add and subtract $(1 - \pi_0)r_{10} + \pi_0 r_{11}$, the drug's rate at the no-drug group's gender mix:

$$\bar{r}_1 - \bar{r}_0 = \underbrace{\left[(1 - \pi_0)(r_{10} - r_{00}) + \pi_0(r_{11} - r_{01})\right]}_{\text{drug vs no drug, same mix } \pi_0} + \underbrace{(\pi_1 - \pi_0)(r_{11} - r_{10})}_{\text{composition}}. \tag{2}$$

Numbers in (2):

same mix: $(0.77143)(0.93103 - 0.86667) + (0.22857)(0.73004 - 0.68750) = 0.04965 + 0.00972 = 0.05938,$

composition: $(0.75143 - 0.22857)(0.73004 - 0.93103) = (0.52286)(-0.20099) = -0.10509,$

$$\bar{r}_1 - \bar{r}_0 = 0.05938 + (-0.10509) = -0.04571.$$

The pooled table compares a mostly-female drug group with a mostly-male control group. The composition term (−10.5 points) swamps the within-gender benefit (+5.9 points). Equation (2) is the categorical twin of the selection-bias decomposition of L05-02-potential-outcomes-identification (naive gap = effect + bias).

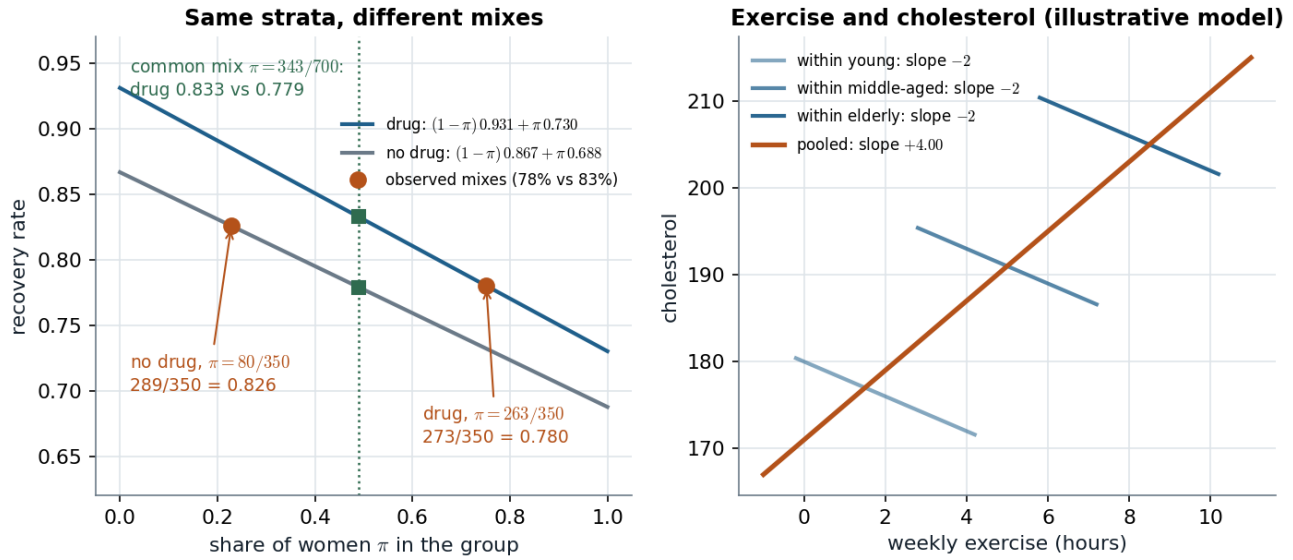
Standardise instead. Evaluate (1) for both groups at the **same** mix, the population share $\pi^* = 343/700 = 0.49$:

$$\bar{r}_1(\pi^*) = (0.51)(0.93103) + (0.49)(0.73004) = 0.47483 + 0.35772 = 0.83255,$$

$$\bar{r}_0(\pi^*) = (0.51)(0.86667) + (0.49)(0.68750) = 0.44200 + 0.33688 = 0.77888,$$

$$\bar{r}_1(\pi^*) - \bar{r}_0(\pi^*) = 0.05367.$$

This is the **adjustment formula** of note 02 (eq. (12) there): when gender is the only confounder, the drug raises recovery by 5.4 points on average.



Left: eq. (1) for each group as a function of the mix π; slide p. 6 counts. Right: an illustrative linear model of slide p. 7's Example 3 (group means of exercise 2, 5, 8 hours; intercepts 180, 201, 222; within slope −2), exact population lines.

On the left the drug line lies above the no-drug line at **every** mix; the reversal comes only from reading the two lines at different points (orange dots). At a common mix (green squares) the drug wins. On the right the pooled slope is $\mathrm{Cov}(x, y)/\mathrm{Var}(x)$ with $\mathrm{Var}(x) = s^2 + \mathrm{Var}(m_g) = 1 + 6 = 7$ and $\mathrm{Cov}(x, y) = -2(7) + \mathrm{Cov}(m_g, c_g) = -14 + 42 = 28$, so the pooled slope is $28/7 = +4$ while every within-age slope is -2.

---

3 · C — the three junctions (slides pp. 8–9)

A **DAG** (directed acyclic graph) draws each variable as a node and each direct cause as an arrow; "acyclic" means no node causes itself through a loop. Every path between two nodes is a sequence of **junctions** of three nodes, and there are only three kinds.

Junction	Shape	Unconditionally	Given the middle node	Slides' rule
Chain	$X \to Y \to Z$	dependent	**independent**	Rule 1 (p. 8)
Fork	$Y \leftarrow X \to Z$	dependent	**independent**	Rule 2 (p. 8)
Collider	$X \to Z \leftarrow Y$	**independent**	dependent (also given any descendant of Z)	Rule 3 (p. 9)

The rules, as stated on the slides:

- **Rule 1** (chains): "X and Z are conditionally independent given Y, if there is only one unidirectional path between X and Z and Y is any set of variables that intercepts that path."
- **Rule 2** (forks): "If X is a common cause of Y and Z, and there is only one path between Y and Z, then Y and Z are independent conditional on X."
- **Rule 3** (colliders): "If Z is a collision between X and Y and there is only one path between X and Y, then X and Y are unconditionally independent but are conditionally dependent given Z and any descendants of Z."

(The collider picture on p. 9 labels both parents "Y"; Rule 3 beneath it names them X and Y.)

Back to the paradox. Gender is the middle of a **fork** $D \leftarrow G \rightarrow R$: conditioning on it blocks the non-causal path, so the stratified table is the causal one. Post-treatment BP is the middle of a **chain** $D \rightarrow BP \rightarrow R$: conditioning on it blocks part of the causal effect itself, so the pooled table is the causal one. Age in Example 3 and firm size in the marketing example (§7) are forks.

3.1 · C — Gaussian algebra for the collider (why conditioning *creates* dependence)

Take X, Z independent $N(0,1)$ and the collider $W = aX + bZ + e$, $e \sim N(0, s^2)$ independent.

Moments. From the definition of W:

$$\mathrm{Var}(W) = a^2 + b^2 + s^2, \qquad \mathrm{Cov}(X, W) = a, \qquad \mathrm{Cov}(Z, W) = b, \qquad \mathrm{Cov}(X, Z) = 0. \tag{3}$$

Partial covariance and variances (normal conditioning, $\mathrm{Cov}(A, B \mid W) = \mathrm{Cov}(A, B) - \mathrm{Cov}(A, W)\,\mathrm{Cov}(B, W)/\,\mathrm{Var}(W)$). Substituting (3):

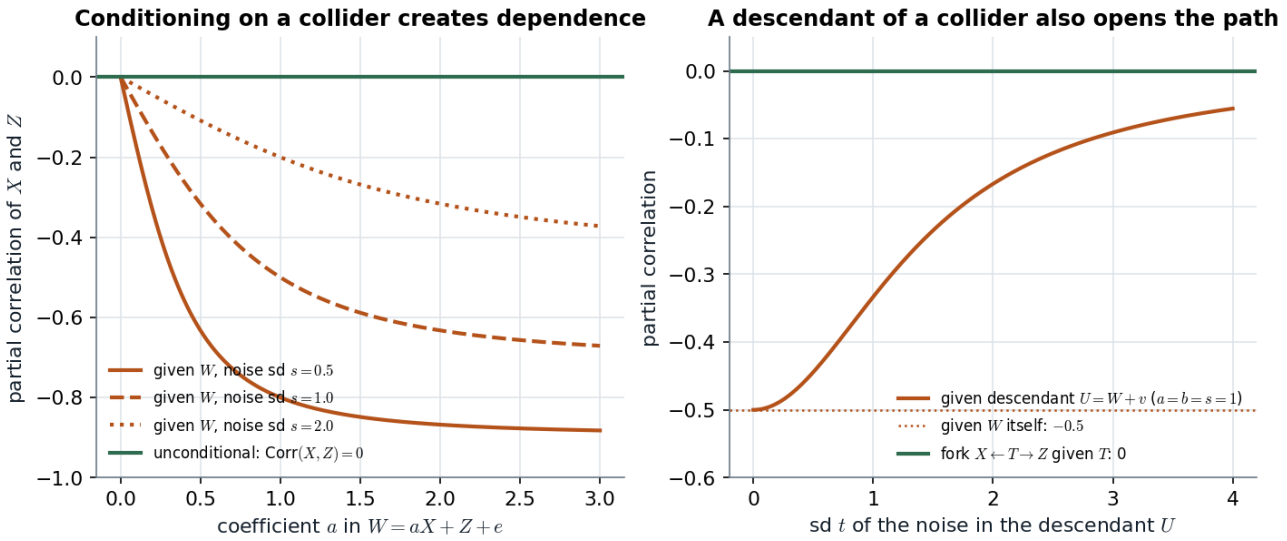
$$\mathrm{Cov}(X, Z \mid W) = 0 - \frac{(a)(b)}{a^2 + b^2 + s^2}, \qquad \mathrm{Var}(X \mid W) = 1 - \frac{a^2}{a^2 + b^2 + s^2} = \frac{b^2 + s^2}{a^2 + b^2 + s^2},$$

$$\mathrm{Var}(Z \mid W) = \frac{a^2 + s^2}{a^2 + b^2 + s^2}.$$

Partial correlation. Divide; the common denominator cancels:

$$\mathrm{Corr}(X, Z \mid W) = \frac{-ab}{\sqrt{(a^2 + s^2)(b^2 + s^2)}}. \tag{4}$$

With $a = b = s = 1$: $-1/\sqrt{(2)(2)} = -0.5$. **A descendant** $U = W + v$, $v \sim N(0, t^2)$, replaces s^2 by $s^2 + t^2$ in (4): with $t = 1$, $-1/\sqrt{(3)(3)} = -1/3$. The dependence is weaker but not zero, which is why Rule 3 says "and any descendants".



Eq. (4) and its descendant version; slide p. 9 (Rule 3) and p. 11 (conditioning on U).

Reading: the green line is the truth before conditioning, zero. Every orange curve is a dependence the analyst **manufactures** by conditioning on W: the more both causes matter relative to the noise, the stronger it is. On the right, a noisy descendant of the collider still opens the path, fading only as the descendant stops carrying information about W. A fork conditioned on its middle node, by contrast, gives exactly zero.

---

4 · C — paths and junctions on a larger DAG (slides pp. 9–10)

Harrison's DAG (slide p. 9): $Z_1 \to X$, $Z_1 \to Z_3$, $Z_2 \to Z_3$, $Z_2 \to Y$, $Z_3 \to X$, $Z_3 \to Y$, $X \to W \to Y$. "Paths always start at the treatment (X), always end at the outcome (Y) and are acyclic" (p. 9). The five paths:

#	Path	Junctions	Open with no conditioning?
1	$X \to W \to Y$	chain at W	yes: **the causal path**
2	$X \leftarrow Z_1 \to Z_3 \to Y$	fork at Z_1, chain at Z_3	yes: back door
3	$X \leftarrow Z_1 \to Z_3 \leftarrow Z_2 \to Y$	fork, **collider at** Z_3, fork	**no**: blocked by the collider
4	$X \leftarrow Z_3 \to Y$	fork at Z_3	yes: back door
5	$X \leftarrow Z_3 \leftarrow Z_2 \to Y$	chain at Z_3, fork at Z_2	yes: back door

The slide lists all 15 junctions of the graph: 7 chains (junctions 1–7), 3 forks (8–10), 5 colliders (11–15) — counts confirmed by brute force in L05-10-python §1.

The adjustment puzzle this graph sets. To estimate $X \to Y$ you must block paths 2, 4 and 5 without blocking 1 or opening 3. Z_3 sits on paths 2, 4, 5 (a chain or fork node: conditioning blocks) **and** is the collider of path 3 (conditioning opens). So $\{Z_3\}$ alone blocks three back doors and opens a fourth; $\{Z_1, Z_3\}$ or $\{Z_2, Z_3\}$ block everything. In the slide's SEM (§5) the population regression of Y on X gives **5.4381** with no controls, **6.0051** with Z_3 alone and exactly **6** (the causal 3×2) with $\{Z_1, Z_3\}$ or $\{Z_2, Z_3\}$ (L05-10-python §1). The Z_3-only bias is small here, but it is not zero, and in other graphs it is large.

5 · C — simulating data from a DAG (slide p. 10)

The slide writes the graph as a linear structural model:

$$Z_1 \sim N(4.75, 1.72^2), \quad Z_2 \sim N(3.29, 1.89^2), \tag{5}$$

$$Z_3 = 3Z_1 - 1.5Z_2 + \varepsilon_{z3}, \qquad X = 2Z_1 + 2.5Z_3 + \varepsilon_x, \qquad W = 3X + \varepsilon_w, \qquad Y = 2W + 2Z_2 - 3Z_3 + \varepsilon_y. \tag{6}$$

Total effect of X on Y. Substitute W from (6) into the Y equation:

$$Y = 2(3X + \varepsilon_w) + 2Z_2 - 3Z_3 + \varepsilon_y = 6X + 2Z_2 - 3Z_3 + 2\varepsilon_w + \varepsilon_y,$$

so the causal effect is 6, and a regression recovers it only if the remaining terms $2Z_2 - 3Z_3$ are uncorrelated with X given the controls — exactly the back-door condition of §4.

(G) **A slip in the slide's R code.** The code first computes every node noise-free (`z3 = 3*z1 - 1.5*z2`, `x = 2*z1 + 2.5*z3`, …) and only afterwards adds noise to each one, so the noise of Z_3 never reaches X, W or Y. The data it prints do not follow (6). L05-10-python §1 uses the equations as written, with each noise variance equal to the variance of its node's systematic part (the code's `sig.*` rule).

Causal discovery vs validation (p. 10, quoted): discovery generates a DAG from data; validation tests a proposed DAG; "it is typically possible for more than one DAG to satisfy the causal validation tests". Named only.

6 · C — d-separation (slide p. 11)

Definition (slide p. 11, verbatim in substance). A path p is **blocked** by a set of nodes Z if and only if

1. p contains a chain $A \to B \to C$ or a fork $A \leftarrow B \to C$ whose middle node B is in Z, or
2. p contains a collider $A \to B \leftarrow C$ such that B is **not** in Z and **no descendant** of B is in Z.

"If Z blocks every path between X and Y, then X and Y are d-separated, conditional on Z, and thus are independent conditional on Z." Two consequences on the slide: "d-connected" = some path is open; and "if we are not conditioning on any variable, then only colliders can block a path".

6.1 · Worked example 1 (slide pp. 11–12): $Z \to W \leftarrow X \to Y$, $W \to U$

The only path from Z to Y is $Z \to W \leftarrow X \to Y$: a collider at W, then a fork at X.

Conditioning set	Collider W	Fork X	Path	Verdict (slide)
$\emptyset$	blocks	open	blocked	$Z \perp Y$
$\{W\}$	**opened**	open	open	dependent
$\{U\}$ (U descends from W)	**opened**	open	open	dependent
$\{W, X\}$	opened	**blocks**	blocked	$Z \perp Y \mid W, X$

6.2 · Worked example 2 (slide p. 12): add $T \to Z$ and $T \to Y$

Now two paths: (a) $Z \leftarrow T \to Y$ (fork at T), (b) $Z \to W \leftarrow X \to Y$ (as above). $Z \perp Y \mid S$ needs **both** blocked: (a) needs $T \in S$; (b) needs $W, U \notin S$ or $X \in S$.

S	path (a)	path (b)	Verdict
$\emptyset$	open	blocked	dependent: "$Z \not\perp Y$, since path between Z and Y with no colliders"
$\{T\}$	blocked	blocked	$Z \perp Y \mid T$
$\{T, W\}$	blocked	open ($X \notin S$)	dependent
$\{T, W, X\}$	blocked	blocked	$Z \perp Y \mid T, W, X$

The slide's two full lists — d-connected given W, U, $\{W, U\}$, $\{W, T\}$, $\{U, T\}$, $\{W, U, T\}$, $\{W, X\}$, $\{U, X\}$, $\{W, U, X\}$; d-separated given T, $\{X, T\}$, $\{W, X, T\}$, $\{U, X, T\}$, $\{W, U, X, T\}$ — follow from the two conditions, and the five separating sets are **all** the separating subsets of $\{T, W, U, X\}$ (checked exhaustively in L05-10-python §1).

> **The test is mechanical.** List every path, mark each junction, apply the two blocking conditions. The only judgement is the graph itself — and that, as the slides say of ignorability, "cannot be assessed from the observed data" (p. 16).

---

7 · C — the marketing campaign (slides pp. 12–13)

A seller of computer equipment offered discounts to some business customers; larger firms were more likely to get them, and larger firms buy more (Kögel's example). Graph: disc ← size → sales, disc → sales. Firm size is the middle of a **fork** on the back-door path, so "to isolate the causal effect, we need to control for or hold constant the confounding variables" (p. 13): compare discounted and non-discounted firms **of the same size**, exactly as in the gender table. The naive comparison mixes the discount effect with the size effect, upwards.

---

8 · G — what the slides do not ask

The back-door criterion in general, do-calculus and causal discovery algorithms are named in the slides' sources (Pearl et al.; Neal 2020; *The Mixtape*, linked from Aula/COURSE) but not developed. Not examinable beyond the junction rules and d-separation above.

Links

- Neighbours: L05-00-map · L05-02-potential-outcomes-identification (next: the same confounding in potential-outcome language; adjustment formula = §2's standardisation).
- Unit: L05-03-omitted-variables-bad-control (a confounder left out = OVB; a mediator put in = bad control, the BP chain) · L05-05-instrumental-variables (the IV DAG) · L05-06-difference-in-differences (back doors through Time and group) · L05-08-formulas · L05-09-pitfalls · L05-10-python (d-separation by brute force).

- Class 2: L02 §4 — the `default` student coefficient that flips sign once balance is controlled is a Simpson reversal through a fork.
- Rules pack: rules/02 (bad controls, confounding).
- Existing material: potential-outcomes notebook.
- **Practice:** no book on disk has DAG exercises; the slides' two d-separation examples (pp. 11–12) and Harrison's path list (pp. 9–10) are the drill. Redo §6.2 for Z and U.

Slides §8 · Potential outcomes: what a causal effect is, and the four assumptions that let data estimate it

> **Claim.** Each unit has two potential outcomes, $Y(1)$ and $Y(0)$, and we see one. Causal estimands are contrasts of their distributions — **ATE, ATT, CATE, QTE, causal risk ratio**. A naive treated-vs-untreated comparison equals the **ATT plus selection bias** (MHE p. 11). **Randomisation** kills the bias; without it, **SUTVA + consistency + ignorability given L + positivity** identify the ATE through the **adjustment formula**, equivalently through **inverse propensity weighting**, whose identity is proved here line by line. On the slides' drug table, adjusting for gender turns a naive -4.57-point gap into an ATE of $+5.37$ points (ATT $+4.80$). Prev: L05-01-simpson-dags · Next: L05-03-omitted-variables-bad-control · Map: L05-00-map

Notation. The slides use A (treatment) and $Y(a)$; MHE uses d_i and y_{1i}, y_{0i}. Both appear below; they name the same objects.

- $A_i \in \{0,1\}$ $(= d_i)$: treatment received; Y_i $(= y_i)$: observed outcome; $i = 1, \dots, N$.
- $Y_i(a)$ $(= y_{ai})$: potential outcome of unit i under treatment a; $Y_i(1) - Y_i(0)$: individual causal effect.
- L (MHE: X_i): observed pre-treatment covariates (confounders); V: covariates defining a subpopulation for the CATE.
- $F_a(y) = P(Y(a) \le y)$: cdf of the potential outcome; $F_a^{-1}(p)$ its p-quantile, $0 < p < 1$.
- $e(l) = P(A = 1 \mid L = l)$: propensity score, $0 < e(l) < 1$ under positivity.

---

1 · U — the question the slides start from (p. 14)

Daniels, Linero and Roy (2024, §1.1, reproduced on slide p. 14) compare two classes of antihypertensives: $A = 1$ if a patient is prescribed an ACE inhibitor, $A = 0$ if an angiotensin-receptor blocker; the outcome is systolic blood pressure (SBP). Prescription is not random: doctors choose by the patient's condition. The slides' motto (p. 15): **"Causal inference requires assumptions that cannot be checked from the data."**

---

2 · C — potential outcomes and the estimands (slide p. 15)

Observed outcome (MHE eq. 2.1.1, p. 11):

$$Y_i = Y_i(0) + (Y_i(1) - Y_i(0))A_i = \begin{cases} Y_i(1) & A_i = 1 \\ Y_i(0) & A_i = 0. \end{cases} \tag{1}$$

Estimands (slide p. 15, the slides' definitions and readings):

Estimand	Definition	Reading on the slide
ATE	$E\{Y(1)\} - E\{Y(0)\}$ (2)	"how much higher would the average SBP be in the population if everyone had been treated with ACEIs compared to if everyone had been treated with ARBs?"
ATT	$E\{Y(1) \mid A = 1\} - E\{Y(0) \mid A = 1\}$ (3)	contrast within the treated; $Y(0)$ for them is "the counterfactual outcome"
QTE	$F_1^{-1}(p) - F_0^{-1}(p)$ (4)	"particularly useful when the outcome of interest is skewed, or when the treatment only has a large effect for a small subset"
Causal relative risk	$E\{Y(1)\}/E\{Y(0)\}$ (5), binary Y	"how many times more likely would high BP[S] be if everyone was treated with ACEIs rather than ARBs?"
CATE	$E\{Y(1) - Y(0) \mid V = v\}$ (6)	the ATE in the subpopulation $V = v$

Note that the QTE (4) is a difference of **quantiles**, not the quantile of the individual differences $Y(1) - Y(0)$: the joint distribution of $(Y(0), Y(1))$ is never observed, but each margin can be.

> **The fundamental problem.** $Y_i(1)$ and $Y_i(0)$ are never both observed, so no estimand above is a function of the data until assumptions link it to (Y_i, A_i, L_i) (slide p. 16: "To link observed data (no potential outcomes!) to potential outcomes, we need causal assumptions").

---

3 · C — the naive comparison: ATT plus selection bias (MHE p. 11)

Start from the observed difference and use (1): within $A = 1$, $Y = Y(1)$; within $A = 0$, $Y = Y(0)$:

$$E[Y \mid A = 1] - E[Y \mid A = 0] = E[Y(1) \mid A = 1] - E[Y(0) \mid A = 0].$$

Add and subtract $E[Y(0) \mid A = 1]$, the treated's unobserved untreated mean:

$$E[Y \mid A = 1] - E[Y \mid A = 0] = \underbrace{E[Y(1) \mid A = 1] - E[Y(0) \mid A = 1]}_{\text{ATT (3)}} + \underbrace{E[Y(0) \mid A = 1] - E[Y(0) \mid A = 0]}_{\text{selection bias}}. \tag{7}$$

MHE's example (p. 10, 2005 NHIS; health 1 = excellent … 5 = poor): hospitalised mean 2.79 ($n = 7774$), not hospitalised 2.07 ($n = 90049$), "the difference in the means is 0.71" with $t = 58.9$ — hospitals seem to make people sicker. In (7) the selection term is large: the sick go to hospital, so $E[y_{0i} \mid d_i = 1]$ is worse than $E[y_{0i} \mid d_i = 0]$, and it "may be so large … that it completely masks a positive treatment effect" (p. 12).

Interactive companions (existing). The decomposition (7) is what the three Class 5+6 pages already do: Selection Bias Decomposer (`../../Aula/class-05-06/companion-selection-bias.html`) (choose who is treated and how effects vary; watch the gap split into effect + bias), Bigger Sample, Same Bias (`../../Aula/class-05-06/companion-sample-size.html`) (500 studies: intervals shrink onto the truth only under random assignment), and The Eight-Person Table (`../../Aula/class-05-06/companion-eight-person-table.html`) (both potential outcomes visible, every random assignment enumerated). The worked notebook is potential-outcomes.ipynb (MHE Ch. 2, STAR, regression on a randomised dummy).

---

4 · C — the identifying assumptions (slide p. 16)

SUTVA. For assignments $\mathbf{a} = (a_1, \ldots, a_N)$ and $\mathbf{a}' = (a'_1, \ldots, a'_N)$ with $a_i = a'_i$:

$$Y_i(\mathbf{a}) = Y_i(\mathbf{a}'), \tag{8}$$

"the potential outcome of any subject i does not depend on the treatment received by the other subjects" (no interference; one version of each treatment). (The slide's §8 heading writes "STUVA".)

Consistency. $Y_i = Y_i(a)$ if $A_i = a$ — the observed outcome is the potential outcome of the treatment actually received. This is what licenses (1).

Randomisation. If A is randomised, $\{Y(0), Y(1)\} \perp A$, hence for $a = 0, 1$:

$$E\{Y_i(a)\} = E\{Y_i(a) \mid A_i = a\} = E(Y_i \mid A_i = a). \tag{9}$$

The first equality is independence, the second consistency. In (7) the selection term vanishes because $E[Y(0) \mid A = 1] = E[Y(0)] = E[Y(0) \mid A = 0]$, and the ATT equals the ATE (MHE p. 12).

Ignorability (no unmeasured confounders, unconfoundedness, exchangeability; MHE's CIA, eq. 3.2.2, p. 40):

$$\{Y(0), Y(1)\} \perp A \mid L. \tag{10}$$

"L is a set of observed covariates that influence both Y and A (remember when we defined a fork?)". And the slides' warning: "The validity of ignorability cannot be assessed from the observed data, regardless of how large the sample size is" (p. 16).

Positivity.

$$P(A = a \mid L = l) > 0 \quad \text{for all } a \text{ and } l. \tag{11}$$

Every covariate cell must contain both treated and untreated units; otherwise one of the conditional means below is undefined.

---

5 · C — identification: the adjustment formula

Iterate expectations over L:

$$E\{Y(a)\} = E_L\big[E\{Y(a) \mid L\}\big].$$

Ignorability (10) lets us condition on $A = a$ inside, since $Y(a) \perp A \mid L$; positivity (11) makes the conditioning event non-empty:

$$E\{Y(a)\} = E_L\big[E\{Y(a) \mid A = a, L\}\big].$$

Consistency replaces $Y(a)$ by Y on the event $A = a$:

$$E\{Y(a)\} = E_L\big[E(Y \mid A = a, L)\big] = \sum_l P(L = l)\, E(Y \mid A = a, L = l). \tag{12}$$

ATE is (12) at $a = 1$ minus (12) at $a = 0$. This is MHE's "averaging all of the X-specific effects" (eq. 3.2.3, p. 42), and it is exactly the standardisation of L05-01-simpson-dags §2.

5.1 · Worked example: the slides' drug table, gender as L

Gender is the fork of slide p. 6 ($D \leftarrow G \rightarrow R$), so take $L = G$. $P(G = 0) = 357/700 = 0.51$, $P(G = 1) = 343/700 = 0.49$.

L	$E(Y \mid A = 1, L)$	$E(Y \mid A = 0, L)$	difference	$P(L)$
men	$81/87 = 0.93103$	$234/270 = 0.86667$	0.06437	0.51
women	$192/263 = 0.73004$	$55/80 = 0.68750$	0.04254	0.49

ATE by (12):

$$\text{ATE} = (0.51)(0.06437) + (0.49)(0.04254) = 0.03283 + 0.02084 = 0.05367.$$

ATT: the same differences weighted by the **treated's** gender mix, $P(G = 0 \mid A = 1) = 87/350 = 0.24857$, $P(G = 1 \mid A = 1) = 263/350 = 0.75143$:

$$\text{ATT} = (0.24857)(0.06437) + (0.75143)(0.04254) = 0.01600 + 0.03196 = 0.04796.$$

Naive: $273/350 - 289/350 = 0.78000 - 0.82571 = -0.04571$. By (7), the selection bias against the ATT is $-0.04571 - 0.04796 = -0.09367$: drug takers are mostly women, whose untreated recovery is low.

> **Same data, three numbers.** Naive −4.57 points, ATE +5.37, ATT +4.80. Only the first is free of assumptions; the other two are correct **if** gender is the only confounder (ignorability) and both genders contain takers and non-takers (positivity: e below is 0.24 and 0.77, safely inside $(0, 1)$).

---

6 · C — the propensity score and inverse weighting (slides pp. 16–17)

Definition. $e(l) = P(A = 1 \mid L = l)$.

Proposition (slide p. 16). Under SUTVA, consistency, ignorability and positivity with binary A,

$$\text{ATE} = E\left[\frac{AY}{e(L)} - \frac{(1-A)Y}{1-e(L)}\right] = E\left[Y\,\frac{A-e(L)}{e(L)\{1-e(L)\}}\right]. \tag{13}$$

Proof of the first term. By consistency $AY = AY(1)$ (when $A = 1$, $Y = Y(1)$; when $A = 0$ both sides are 0):

$$E\left[\frac{AY}{e(L)}\right] = E\left[\frac{AY(1)}{e(L)}\right].$$

Iterate on L:

$$E\left[\frac{AY(1)}{e(L)}\right] = E_L\left[\frac{E\{AY(1) \mid L\}}{e(L)}\right].$$

By ignorability (10), $A \perp Y(1) \mid L$, so the inner expectation factors:

$$E\{AY(1) \mid L\} = E(A \mid L)\,E\{Y(1) \mid L\} = e(L)\,E\{Y(1) \mid L\}.$$

Substitute; positivity (11) makes $e(L) > 0$ so it cancels:

$$E\left[\frac{AY}{e(L)}\right] = E_L\left[\frac{e(L)\,E\{Y(1) \mid L\}}{e(L)}\right] = E_L\big[E\{Y(1) \mid L\}\big] = E\{Y(1)\}.$$

Second term, identically with $1 - A$ and $1 - e(L)$: $E[(1-A)Y/(1-e(L))] = E\{Y(0)\}$. Subtracting gives the first form of (13).

Second form. Put the two fractions over $e(1-e)$:

$$\frac{A}{e} - \frac{1-A}{1-e} = \frac{A(1-e) - (1-A)e}{e(1-e)} = \frac{A - Ae - e + Ae}{e(1-e)} = \frac{A-e}{e(1-e)}. \qquad \blacksquare$$

Plug-in estimator (slide p. 17): $\widehat{\text{ATE}} = \frac{1}{n}\sum_{i=1}^{n} Y_i \frac{A_i - \hat{e}(L_i)}{\hat{e}(L_i)\{1-\hat{e}(L_i)\}}$, consistent "if we can estimate the propensity score sufficiently well".

6.1 · Worked example: IPW on the drug table

With $L = G$ the propensity is a cell frequency: $\hat{e}(\text{men}) = 87/357 = 0.24370$, $\hat{e}(\text{women}) = 263/343 = 0.76676$. Each treated man counts $1/0.24370 = 4.103$ times, each untreated man $1/(1-0.24370) = 1.322$ times, and so on. Evaluating (13) on the 700 rebuilt observations gives **0.05367**, identical to the adjustment formula (L05-10-python §2). This is not a coincidence: with a saturated (cell-frequency) propensity, IPW **is** standardisation — weighting each cell by $1/\hat{e}$ re-creates the population mix $P(L = l)$ in each arm.

The propensity score is a balancing score (slide p. 17): "a useful one-dimensional summary of L". With many covariates, (12) needs a mean for every cell of L; (13) needs only one number per unit.

---

7 · C — regression as an adjustment device (MHE §3.2.1, pp. 38–44)

MHE writes the CIA (10) for schooling, $\{y_{0i}, y_{1i}\} \perp c_i \mid X_i$ (eq. 3.2.2), and the constant-effect linear causal model $f_i(s) = \alpha + \rho s + \eta_i$ (3.2.7). If $\eta_i = X_i'\gamma + v_i$ with $E[\eta_i \mid X_i] = X_i'\gamma$, then

$$y_i = \alpha + \rho s_i + X_i'\gamma + v_i \qquad \text{(MHE 3.2.9, p. 44)}, \tag{14}$$

and "the key assumption here is that the observable characteristics, X_i, are the only reason why η_i and s_i … are correlated" — selection on observables. Note 03 asks what happens to ρ when part of X_i is missing from (14).

---

8 · G — beyond the slides

Matching, doubly robust estimators and the efficiency theory of (13) are not on the slides. The Heckman–Lopes–Piatek (2014) Bayesian treatment-effects paper is listed on slide p. 20 (named only), and Greenberg §11.1 (pp. 168–173) treats binary treatment with an instrument in a Bayesian latent-index model: **Classes 3+4 tools**, named only.

Links

- Neighbours: L05-01-simpson-dags (forks = confounders; the same 5.37 by standardisation) · L05-03-omitted-variables-bad-control (next: what (14) loses when a confounder is omitted).
- Unit: L05-05-instrumental-variables (when (10) fails for every observable L) · L05-06-difference-in-differences (DiD's parallel trends are an ignorability statement about changes) · L05-07-regression-discontinuity (positivity fails by design: no overlap at the cutoff) · L05-08-formulas · L05-09-pitfalls · L05-10-python §2.
- Existing material for this note (draft-plan potential-outcomes set, kept and linked): notebook · quizzes quiz 1, quiz 2, quiz 3 · practice plan pratica-class-01 · narration class-01-narrated.txt · video `Videos/class-01-potential-outcomes/` · audio prompts part 1, part 2.
- Rules pack: rules/02.
- **Practice** (worked examples in the book, not exercises): MHE Table 2.2.1 (p. 14) and Table 2.2.2 (p. 15), Tennessee STAR balance and experimental estimates; G 11.1–11.2 (p. 180), 11.3–11.4 (p. 181) for the Bayesian treatment model of G §11.1.

omitted-variable.html · MHE §3.2.2–3.2.3 · Short equals long plus omitted times included, and the controls you must not add

> **Claim.** Leaving a regressor z out of a regression moves the coefficient on x by exactly **(effect of z on y) × (slope of z on x)**: "short equals long plus the effect of omitted times the regression of omitted on included" (MHE p. 45). The site's simulation ($\beta = 0.75$, $\gamma = 1.2$) is that formula, $E\hat{\beta}_{\text{short}} = 0.75 + 1.2\rho$, plus a variance story: the long regression is centred but its spread blows up as $\rho \to 1$. Adding controls is not always better: a variable **caused by the treatment** (occupation after college; a test score after schooling) is a **bad control** that reintroduces selection bias — the chain of L05-01-simpson-dags, now in regression form. Prev: L05-02-potential-outcomes-identification · Next: L05-04-errors-in-variables · Map: L05-00-map

Notation.

- y, x, z: outcome, included regressor, omitted (or added) regressor; in the site's model all have mean 0 and variance 1.
- σ_{xy}, σ_{yz}: covariances (= correlations) of y with x and z; $\rho = \text{corr}(x, z) \in (-1, 1)$.
- Long regression $y = \beta x + \gamma z + \varepsilon$; short regression $y = \beta^s x + u$; $\delta_{zx} = \text{Cov}(z, x)/\text{Var}(x)$: slope of z on x.
- MHE: y_i log wage, s_i schooling, A_i ability vector, ρ the schooling coefficient (MHE's letter; **not** the correlation above — context disambiguates), γ ability effects, δ_{As} slopes of A_i on s_i.
- $n = 30$ observations per replication, $N = 1000$ replications (the site's design).

---

1 · C — the site's model: a trivariate Gaussian

$$\begin{pmatrix} y_i \\ x_i \\ z_i \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \sigma_{xy} & \sigma_{yz} \\ \sigma_{xy} & 1 & \rho \\ \sigma_{yz} & \rho & 1 \end{pmatrix} \right). \tag{1}$$

Conditional mean. For a Gaussian vector, $E(y \mid x, z) = \Sigma_{y,(x,z)}\Sigma^{-1}_{(x,z)}(x, z)'$. The regressor block and its inverse:

$$\Sigma_{(x,z)} = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \qquad \Sigma^{-1}_{(x,z)} = \frac{1}{1-\rho^2}\begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}. \tag{2}$$

Multiply $(\sigma_{xy}, \sigma_{yz})$ by (2):

$$(\beta, \gamma) = \frac{1}{1-\rho^2}(\sigma_{xy} - \rho\sigma_{yz},\ \sigma_{yz} - \rho\sigma_{xy}). \tag{3}$$

Conditional variance $\tau^2 = 1 - (\sigma_{xy}, \sigma_{yz})\Sigma^{-1}_{(x,z)}(\sigma_{xy}, \sigma_{yz})'$. Substituting (2):

$$\tau^2 = 1 - \frac{\sigma^2_{xy} - 2\rho\sigma_{xy}\sigma_{yz} + \sigma^2_{yz}}{1-\rho^2} = \frac{1 - \rho^2 - \sigma^2_{xy} - \sigma^2_{yz} + 2\sigma_{xy}\sigma_{yz}\rho}{1-\rho^2}. \tag{4}$$

So $y_i \mid x_i, z_i = \beta x_i + \gamma z_i + \varepsilon_i$, $\varepsilon_i \sim N(0, \tau^2)$, as on the page. (The page prints the numerator as $1 - \rho - ...$; it is $1 - \rho^2 - ...$ — the two agree only at $\rho = 0$, the page's next case. Checked against the Schur complement in L05-10-python §3.)

At $\rho = 0$ (the page's §1.2), (3) gives $\beta = \sigma_{xy}$, $\gamma = \sigma_{yz}$ and (4) gives $\tau^2 = 1 - (\sigma_{xy}^2 + \sigma_{yz}^2)$.

---

2 · C — the omitted-variables bias formula

Short slope in the Gaussian model. Regressing y on x alone gives $\beta^s = \mathrm{Cov}(y,x)/\mathrm{Var}(x) = \sigma_{xy}$. Write σ_{xy} through the long model $y = \beta x + \gamma z + \varepsilon$:

$$\sigma_{xy} = \mathrm{Cov}(\beta x + \gamma z + \varepsilon,\ x) = \beta(1) + \gamma(\rho) + 0.$$

Hence

$$\beta^s = \beta + \gamma\rho. \tag{5}$$

(Check with (3): $\beta + \gamma\rho = [\sigma_{xy} - \rho\sigma_{yz} + \rho\sigma_{yz} - \rho^2\sigma_{xy}]/(1-\rho^2) = \sigma_{xy}$.)

MHE's general version (eq. 3.2.11, p. 45). Long regression $y_i = \alpha + \rho s_i + A_i'\gamma + \varepsilon_i$ (3.2.10) with ε_i uncorrelated with s_i and A_i by construction. Plug it into the short-regression formula:

$$\frac{\mathrm{Cov}(y_i, s_i)}{V(s_i)} = \frac{\mathrm{Cov}(\alpha + \rho s_i + A_i'\gamma + \varepsilon_i,\ s_i)}{V(s_i)} = \rho + \gamma' \frac{\mathrm{Cov}(A_i, s_i)}{V(s_i)} = \rho + \gamma'\delta_{As}. \tag{6}$$

> **"Short equals long plus the effect of omitted times the regression of omitted on included"** (MHE p. 45). The formula is mechanical — it links two regressions "whether or not the longer regression is causal" (p. 44). The sign of the bias is sign(γ) × sign(δ): ability raises wages and (probably) schooling, so the short return is too big; if some omitted factor is negatively related to schooling it can be too small (p. 45). Short = long exactly when the omitted and included variables are uncorrelated.

Multivariate form (MHE footnote 10, p. 45): $\beta_1^s = \beta_1^l + E[X_{1i}X_{1i}']^{-1}E[X_{1i}X_{2i}']\beta_2^l$.

---

3 · C — the site's simulation, made exact

Design (omitted-variable.html): $(x_i, z_i) \sim N(0, \left[\begin{smallmatrix}1 & \rho\\ \rho & 1\end{smallmatrix}\right])$, $y_i = 0.75x_i + 1.2z_i + e_i$, $e_i \sim N(0,1)$, $n = 30$, 1000 replications per $\rho \in \{0, 0.05, \ldots, 0.95\}$; both regressions **without intercept** (`lm(y~x-1)`, `lm(y~x+z-1)`).

Short estimator. Write $z_i = \rho x_i + \sqrt{1-\rho^2}\, v_i$ with $v_i \sim N(0,1)$ independent of x_i. Substitute into y_i:

$$y_i = 0.75x_i + 1.2(\rho x_i + \sqrt{1-\rho^2}\, v_i) + e_i = (0.75 + 1.2\rho)x_i + 1.2\sqrt{1-\rho^2}\, v_i + e_i.$$

The no-intercept OLS slope is $\sum x_i y_i / \sum x_i^2$; substitute:

$$\hat{\beta}^s = (0.75 + 1.2\rho) + \frac{\sum x_i(1.2\sqrt{1-\rho^2}v_i + e_i)}{\sum x_i^2}. \tag{7}$$

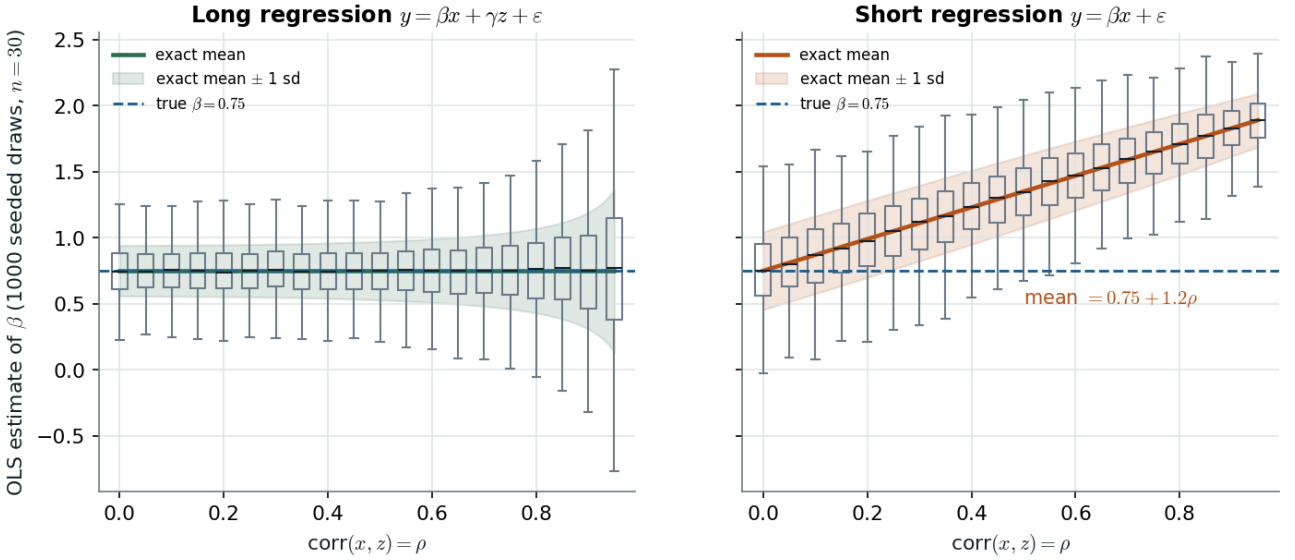
Given x, the last term has mean 0 and variance $(1.44(1-\rho^2) + 1)/\sum x_i^2$. With $\sum x_i^2 \sim \chi^2_{30}$ and $E[1/\chi^2_n] = 1/(n-2)$:

$$E\hat{\beta}^s = 0.75 + 1.2\rho, \qquad \mathrm{sd}(\hat{\beta}^s) = \sqrt{\frac{1 + 1.44(1-\rho^2)}{28}}. \tag{8}$$

Long estimator. Unbiased; given X, Var $= [(X'X)^{-1}]_{11}$ (error variance 1). (G) The inverse-Wishart mean $E[(X'X)^{-1}] = \Sigma^{-1}/(n-p-1)$ with $p = 2$, and $[\Sigma^{-1}]_{11} = 1/(1-\rho^2)$ by (2):

$$E\hat{\beta} = 0.75, \qquad \mathrm{sd}(\hat{\beta}) = \sqrt{\frac{1}{27(1-\rho^2)}}. \tag{9}$$

ρ	$E\hat{\beta}^s$ by (8)	sd short	sd long by (9)	MSE short	MSE long
0	$0.75+1.2(0) = 0.75$	$\sqrt{2.44/28} = 0.2952$	$\sqrt{1/27} = 0.1925$	0.0871	0.0370
0.5	$0.75 + 1.2(0.5) = 1.35$	$\sqrt{2.08/28} = 0.2726$	$\sqrt{1/20.25} = 0.2222$	$0.36+0.0743 = 0.4343$	0.0494
0.95	$0.75 + 1.2(0.95) = 1.89$	$\sqrt{1.1404/28} = 0.2018$	$\sqrt{1/2.6325} = 0.6163$	$1.2996 + 0.0407 = 1.3403$	0.3799



Boxes: a seeded re-run of the site's simulation in Python (not the site's draws). Lines and bands: exact (8)–(9). omitted-variable.html; MHE eq. 3.2.11.

Reading: the right panel is (5) drawn — every box sits on $0.75 + 1.2\rho$, and no amount of data moves it back to 0.75. The left panel is the price of including z when it is nearly collinear with x: centred, but at $\rho = 0.95$ its sd is three times the short regression's. Even so the long regression wins on MSE at every ρ in the table, because the bias 1.2ρ grows faster than the variance it saves.

---

4 · C — reading MHE Table 3.2.1 through the formula (p. 46)

NLSY men, log wage on schooling, $n = 2434$ (MHE p. 46):

Controls	none	age dummies	+ parents' schooling, race, region	+ AFQT	+ occupation
schooling coefficient	0.132	0.131	0.114	0.087	0.066
(s.e.)	(0.007)	(0.007)	(0.007)	(0.009)	(0.010)

By (6), each drop is $\gamma'\delta$ of the variables just added: family background takes $0.132 - 0.114 = 0.018$ off and AFQT (an ability proxy) $0.114 - 0.087 = 0.027$, "a result of the fact that the additional controls are positively correlated with both wages and schooling" (p. 46). The last drop, $0.087 - 0.066 = 0.021$, is **not** an OVB correction: occupation is a bad control (§5).

---

5 · C — bad control (MHE §3.2.3, pp. 47–49)

"Bad controls are variables that are themselves outcome variables in the notional experiment at hand" (p. 47). College c_i randomly assigned; white-collar status w_i and earnings y_i both respond to it:

$$y_i = c_i y_{1i} + (1 - c_i) y_{0i}, \qquad w_i = c_i w_{1i} + (1 - c_i) w_{0i}. \tag{10}$$

Comparison within white-collar workers (MHE eq. 3.2.12, p. 48). On $c_i = 1$ the observed (y, w) are (y_1, w_1), on $c_i = 0$ they are (y_0, w_0):

$$E[y_i \mid w_i = 1, c_i = 1] - E[y_i \mid w_i = 1, c_i = 0] = E[y_{1i} \mid w_{1i} = 1, c_i = 1] - E[y_{0i} \mid w_{0i} = 1, c_i = 0].$$

Random assignment of c_i (independent of all four potential outcomes) drops the c_i conditions:

$$= E[y_{1i} \mid w_{1i} = 1] - E[y_{0i} \mid w_{0i} = 1].$$

Add and subtract $E[y_{0i} \mid w_{1i} = 1]$ (MHE p. 49):

$$= \underbrace{E[y_{1i} - y_{0i} \mid w_{1i} = 1]}_{\text{causal effect on those white-collar with college}} + \underbrace{E[y_{0i} \mid w_{1i} = 1] - E[y_{0i} \mid w_{0i} = 1]}_{\text{selection bias}}. \tag{11}$$

> **Even with college randomly assigned, conditioning on occupation is apples to oranges.** College changes who is in the white-collar pool; the selection term need not be zero even if $y_{1i} = y_{0i}$ for everyone (p. 49). In DAG language $c \to w \to y$ is a chain and w is also a collider on $c \to w \leftarrow$ (ability) $\to y$: conditioning on it both blocks part of the effect and opens a back door.

5.1 · C — proxy control (MHE pp. 49–50)

Wanted: $y_i = \alpha + \rho s_i + \gamma a_i + \varepsilon_i$ (3.2.13), with a_i ability measured before schooling. Available only: late ability $al_i = \pi_0 + \pi_1 s_i + \pi_2 a_i$ (3.2.14), raised by schooling ($\pi_1 > 0$).

Solve (3.2.14) for a_i: $a_i = (al_i - \pi_0 - \pi_1 s_i)/\pi_2$. **Substitute into (3.2.13):**

$$y_i = \alpha + \rho s_i + \gamma \left(\frac{al_i - \pi_0 - \pi_1 s_i}{\pi_2} \right) + \varepsilon_i.$$

Collect terms (MHE eq. 3.2.15):

$$y_i = \left(\alpha - \gamma \frac{\pi_0}{\pi_2}\right) + \left(\rho - \gamma \frac{\pi_1}{\pi_2}\right) s_i + \frac{\gamma}{\pi_2}\, al_i + \varepsilon_i. \tag{12}$$

With $\rho, \gamma, \pi_1, \pi_2 > 0$ the schooling coefficient $\rho - \gamma\pi_1/\pi_2$ is **too small**; e.g. $\rho = 0.10$, $\gamma = 0.05$, $\pi_1 = 0.2$, $\pi_2 = 1$ gives $0.10 - 0.05(0.2)/1 = 0.09$. With no control at all, (6) gives $\rho + \gamma\delta_{as}$, too big when $\delta_{as} > 0$; so "the causal effect of interest lies between these two" (p. 50).

> **Timing decides.** "Variables measured before the variable of interest was determined are generally good controls" (p. 50). In DAG terms: condition on forks (confounders); never on chains out of the treatment (mediators, outcomes) or on colliders.

Links

- Neighbours: L05-02-potential-outcomes-identification · L05-04-errors-in-variables (next: a regressor that is *there* but measured with noise).
- Unit: L05-01-simpson-dags (§4: the Z_3 collider is a bad control of the third kind) · L05-05-instrumental-variables (IV as the cure when the omitted variable cannot be measured; slide §9.2 shows that simply adding the instrument does not work) · L05-08-formulas · L05-09-pitfalls · L05-10-python §3.
- Class 1: L01 §6 (OLS as a projection; partialling out). Class 2: L02 §2 (the Mroz return to education that moves with controls).
- Rules pack: rules/02 (omitted variable bias formula; bad controls) · rules/01 (endogeneity sources).
- **Practice:** MHE Table 3.2.1 (p. 46), read as in §4.

error-in-variables.pdf · W §9.4 · A noisy regressor shrinks the slope towards zero, by exactly the reliability ratio

> **Claim.** If the regressor is observed with classical noise, $w = x + u$, OLS of y on w estimates $\beta\,\sigma_x^2/(\sigma_x^2+\sigma_u^2)$: the slope is **attenuated** towards zero by the share of the observed variance that is signal. Noise in y costs only precision; noise in x costs consistency. Controls that explain part of the true regressor make attenuation **worse**, since they remove signal and leave the noise. A second, independent measurement used as an **instrument** restores β — the first appearance of IV in the unit, and historically one of its origins (MHE p. 84).
> Prev: L05-03-omitted-variables-bad-control · Next: L05-05-instrumental-variables · Map: L05-00-map

Notation (the slide's; Wooldridge's letters in brackets).

- x_i [x_1^*]: true regressor, variance σ_x^2; w_i [x_1]: observed regressor; u_i [e_1]: measurement error, mean 0, variance σ_u^2.
- $y_i = \beta x_i + \varepsilon_i$ [$y = \beta_0 + \beta_1 x_1^* + u$]: true model, ε_i independent of x_i.
- $\lambda = \sigma_x^2/(\sigma_x^2+\sigma_u^2) \in (0,1]$: reliability (attenuation factor); $\kappa = \sigma_u^2/\sigma_x^2$: noise-to-signal ratio.
- r_1^*: error of the regression of x_1^* on the other regressors (W 9.36), variance σ_r^2.

---

1 · C — the slide's derivation (error-in-variables.pdf)

Model. True: $y_i = \beta x_i + \varepsilon_i$. Observed: $w_i = x_i + u_i$, with u_i independent of x_i and ε_i.

Naive slope. Regressing y on w estimates

$$\hat{\beta}_{\text{naive}} \;\to\; \frac{\text{Cov}(y,w)}{\text{Var}(w)}. \tag{1}$$

Numerator. Substitute both definitions and use the independence of x, u, ε:

$$\text{Cov}(y,w) = \text{Cov}(\beta x+\varepsilon,\; x+u) = \beta\,\text{Var}(x) + \beta\,\text{Cov}(x,u) + \text{Cov}(\varepsilon,x) + \text{Cov}(\varepsilon,u) = \beta\sigma_x^2. \tag{2}$$

Denominator.

$$\text{Var}(w) = \text{Var}(x) + \text{Var}(u) = \sigma_x^2 + \sigma_u^2. \tag{3}$$

Substitute (2) and (3) into (1):

$$\hat{\beta}_{\text{naive}} \;\to\; \beta\,\frac{\sigma_x^2}{\sigma_x^2+\sigma_u^2} = \beta\lambda = \frac{\beta}{1+\kappa}. \tag{4}$$

"The slope is biased toward zero (attenuation bias)" (slide). Since $\lambda < 1$ whenever $\sigma_u^2 > 0$, $|\beta\lambda| < |\beta|$ with the same sign: the estimate never flips sign, it only shrinks.

Worked numbers. $\sigma_x = \sigma_u = 1$: $\lambda = 1/(1+1) = 0.5$ — half the slope. $\beta = 1.5$, $\sigma_x = 1$, $\sigma_u = 0.75$: $\lambda = 1/(1+0.5625) = 0.64$, so OLS converges to $1.5(0.64) = 0.96$ (L05-10-python §4 gets 0.959 on 400,000 draws).

---

2 · C — Wooldridge's two polar assumptions (W §9.4b, pp. 289–291)

W writes $e_1 = x_1 - x_1^*$ (9.28) and substitutes $x_1^* = x_1 - e_1$ into $y = \beta_0 + \beta_1 x_1^* + u$ (9.27):

$$y = \beta_0 + \beta_1 x_1 + (u - \beta_1 e_1). \tag{5}$$

Whether OLS on (5) works depends on the correlation of x_1 with the composite error $u - \beta_1 e_1$:

Assumption	Statement	Consequence
(9.29)	$\mathrm{Cov}(x_1, e_1) = 0$: error uncorrelated with the **observed** measure	$u - \beta_1 e_1$ uncorrelated with x_1: OLS **consistent**, only the error variance rises to $\sigma_u^2 + \beta_1^2 \sigma_{e_1}^2$ (W p. 290)
(9.31) **CEV**	$\mathrm{Cov}(x_1^*, e_1) = 0$: error uncorrelated with the **true** value (the slide's case)	$\mathrm{Cov}(x_1, e_1) = \sigma_{e_1}^2$ (9.32), so OLS is inconsistent

CEV algebra (W eq. 9.33, p. 291). The composite error's covariance with x_1, by (9.32):

$$\mathrm{Cov}(x_1, u - \beta_1 e_1) = -\beta_1 \,\mathrm{Cov}(x_1, e_1) = -\beta_1 \sigma_{e_1}^2 .$$

Then

$$\operatorname{plim} \hat{\beta}_1 = \beta_1 + \frac{\mathrm{Cov}(x_1, u - \beta_1 e_1)}{\mathrm{Var}(x_1)} = \beta_1 - \frac{\beta_1 \sigma_{e_1}^2}{\sigma_{x_1^*}^2 + \sigma_{e_1}^2} = \beta_1 \left(\frac{\sigma_{x_1^*}^2}{\sigma_{x_1^*}^2 + \sigma_{e_1}^2} \right), \tag{6}$$

the slide's (4) in W's letters. "If the variance of x_1^* is large relative to the variance in the measurement error, then the inconsistency in OLS will be small" (p. 291).

2.1 · Error in the dependent variable is harmless for consistency (W §9.4a, pp. 287–289)

If $y = y^* + e_0$ with e_0 uncorrelated with the regressors, $y = \beta_0 + \beta_1 x + (u + e_0)$: OLS stays unbiased and consistent; only $\mathrm{Var}(u + e_0) = \sigma_u^2 + \sigma_0^2 > \sigma_u^2$ inflates the standard errors (W p. 288). W's Example 9.6 (p. 289) shows the exception: if the error in y is correlated with a regressor (grant recipients under-reporting scrap), the coefficient is biased.

---

3 · C — other regressors make it worse (W eq. 9.36, p. 291)

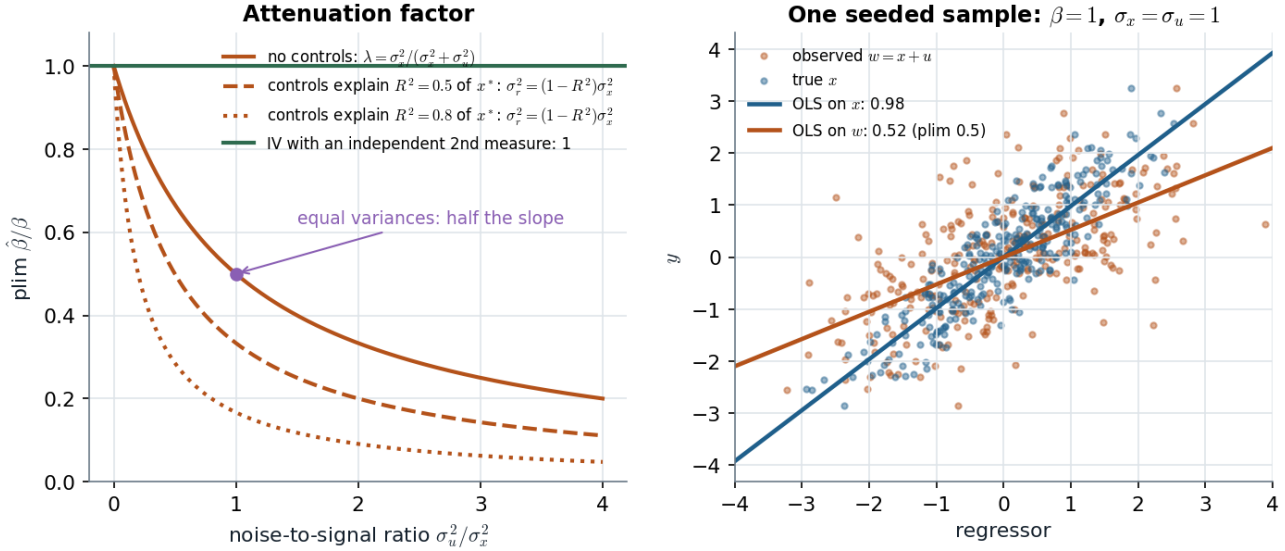
With $y = \beta_0 + \beta_1 x_1^* + \beta_2 x_2 + \beta_3 x_3 + u$ and only x_1^* mismeasured (CEV),

$$\operatorname{plim} \hat{\beta}_1 = \beta_1 \left(\frac{\sigma_{r_1^*}^2}{\sigma_{r_1^*}^2 + \sigma_{e_1}^2} \right), \tag{7}$$

where r_1^* is the error in $x_1^* = \alpha_0 + \alpha_1 x_2 + \alpha_2 x_3 + r_1^*$. If the other regressors explain a share R^2 of $\mathrm{Var}(x_1^*)$, then $\sigma_r^2 = (1 - R^2)\sigma_x^2$ while the noise $\sigma_{e_1}^2$ is untouched, so (7) is (4) with the signal cut:

$$\lambda_r = \frac{(1-R^2)\sigma_x^2}{(1-R^2)\sigma_x^2 + \sigma_u^2} = \frac{1-R^2}{1-R^2+\kappa} \le \lambda. \tag{8}$$

Worked numbers (L05-10-python §4): $x^* = 0.8x_2 + r$, $\mathrm{Var}(r) = 0.36$, $\sigma_u = 0.75$: $\lambda_r = 0.36/(0.36 + 0.5625) = 0.390$, against $\lambda = 0.64$ without the control. "Generally, measurement error in a single variable causes inconsistency in all estimators" (W p. 291): the other coefficients are biased too, in directions not easily derived.



Left: (4) and (8) as exact curves; right: a seeded illustration (not the slide's data, which has none). error-in-variables.pdf; W eqs. 9.33 and 9.36.

On the left, attenuation depends only on the noise-to-signal ratio: at $\kappa = 1$ half the slope is lost, at $\kappa = 4$ four fifths. Each control that soaks up signal pushes the curve down. On the right the noisy cloud is **wider horizontally** than the true one (variance $1 + 1 = 2$ instead of 1) with the same vertical relation to the signal, so the fitted line through it is flatter: 0.52 against 0.98 in this draw, plim 0.5.

---

4 · C — the cure: an instrument that is a second measurement (W §15.4, pp. 479–481; MHE p. 84)

Suppose a second report $w_2 = x + u_2$ with u_2 independent of u_1, x and ε. Use w_2 as an instrument for $w_1 = x + u_1$ (IV formula, W eq. 15.9; note 05):

$$\frac{\mathrm{Cov}(w_2, y)}{\mathrm{Cov}(w_2, w_1)} = \frac{\mathrm{Cov}(x+u_2,\ \beta x+\varepsilon)}{\mathrm{Cov}(x+u_2,\ x+u_1)} = \frac{\beta\sigma_x^2}{\sigma_x^2} = \beta. \tag{9}$$

The noise cancels because the two errors are independent: the instrument is correlated with the true signal (relevance) and uncorrelated with the composite error $\varepsilon - \beta u_1$ (exogeneity). W's examples: parents' education for a mismeasured `educ` (p. 480); two test scores as indicators of ability (Example 15.6, p. 481). MHE p. 84: "Today, we are more likely to find IV used to address measurement error problems than to estimate the parameters of an SEM"; Wald (1940) proposed the grouping estimator for exactly this problem (MHE footnote 8, p. 95). MHE p. 169 gives the panel version: Ashenfelter and Krueger (1994) instrument each twin's schooling with the brother's report.

Measurement error is endogeneity in disguise. In (5) the regressor is correlated with its own error term, $\text{Cov}(x_1, u - \beta_1 e_1) = -\beta_1 \sigma^2_{e_1}$ — exactly the condition note 05 handles with instruments.

Links

- Neighbours: L05-03-omitted-variables-bad-control · L05-05-instrumental-variables (next: the IV estimator in general; (9) is its special case).
- Unit: L05-06-difference-in-differences (differencing removes fixed effects but amplifies measurement error: MHE p. 169) · L05-08-formulas · L05-09-pitfalls · L05-10-python §4.
- Class 1: L01 §6 (OLS slope as Cov / Var).
- Rules pack: rules/01 (measurement error and attenuation bias).
- **Practice** (statements in the book): W 15.C3 (p. 492) and 15.C5 (p. 493) use the CARD data of note 05; W Example 15.6 (p. 481) is the worked two-indicator case.

Slides §9 · Instrumental variables: a variable that moves the treatment and nothing else identifies the effect as a ratio

> **Claim.** When the treatment x shares unobserved causes with the outcome y, OLS converges to β_1 plus a confounding term. An **instrument** z that shifts x (**relevance**, $\delta_1 \neq 0$) and touches y only through x (**exclusion**, $\mathrm{Cov}(z,u)=0$) identifies $\beta_1 = \mathrm{Cov}(z,y)/\mathrm{Cov}(z,x)$ — the **reduced form over the first stage**. With a binary instrument this is the **Wald estimator**; with covariates it is **2SLS**. Simply adding z to the outcome regression does **not** work. The price is variance (the IV SE is $1/|\mathrm{corr}(x,z)|$ times larger) and fragility: a weak first stage magnifies any exclusion violation. **Card:** OLS 0.0747, IV 0.1315 (SE 0.0550), first-stage $F = 13.26$. Prev: L05-04-errors-in-variables · Next: L05-06-difference-in-differences · Map: L05-00-map

Notation (the slides'; MHE and W letters where they differ).

- x_i: treatment (schooling; MHE s_i, W y_2); y_i: outcome (log wage); z_i: instrument; $i = 1, \ldots, n$.
- Outcome equation $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i^y$; treatment (first-stage) equation $x_i = \delta_0 + \delta_1 z_i + \varepsilon_i^x$.
- $(\varepsilon_i^x, \varepsilon_i^y)$ bivariate normal, mean 0, variances σ_x^2, σ_y^2, covariance $\sigma_{xy} = \rho\sigma_x\sigma_y$; z_i independent of both errors; $\sigma_z^2 = \mathrm{Var}(z)$.
- Reduced form $y_i = \theta_0 + \theta_1 z_i + u_i$, $u_i \sim N(0, \tau^2)$.
- φ: direct effect of z on y (an exclusion violation, used in §6 and the companion); $p = P(z=1)$ for a binary instrument.
- MHE: ρ is the causal schooling coefficient in (4.1.2); in this note ρ is always the error correlation, MHE's coefficient is written β_1.

---

1 · U — the IV graph and the examples (slide p. 17)

$$\text{IV} \;\rightarrow\; T \;\rightarrow\; O, \qquad T \leftarrow C \rightarrow O.$$

"IV: Instrumental variable; T: Treatment/Program/Policy/Risk factors; C: Unmeasured confounders …; O: Outcome." The instrument opens a causal path into T and has **no** arrow into O or C. The slide's table:

Confounders (C)	Treatment (T)	Outcome (O)	IV
maternal characteristics	smoking during pregnancy	low birth weight	cigarette taxes
smoking, caffeine, alcohol	body mass index	Parkinson's disease	FTO gene variant
prognostic factors	catheter use	mortality	patients with catheter at facility
ability	education	earnings	mother's education
proximity	tutoring programme	GPA	library hours

The path IV → T → O is a chain and IV ← … → O does not exist: IV and O are dependent **only** through T, which is the whole argument.

---

2 · C — the two-equation system and the IV ratio (slides pp. 17–18)

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i^y, \tag{1}$$

$$x_i = \delta_0 + \delta_1 z_i + \varepsilon_i^x. \tag{2}$$

The slide's two switches: $\rho = 0 \Rightarrow$ no endogeneity; $\delta_1 = 0 \Rightarrow$ no instrument.

Why OLS fails. The OLS slope of y on x converges to $\mathrm{Cov}(x, y)/\mathrm{Var}(x)$. From (2), $\mathrm{Var}(x) = \delta_1^2\sigma_z^2 + \sigma_x^2$ and $\mathrm{Cov}(x, \varepsilon^y) = \mathrm{Cov}(\varepsilon^x, \varepsilon^y) = \rho\sigma_x\sigma_y$. Substitute (1) into the covariance:

$$\operatorname{plim}\hat{\beta}_1^{OLS} = \frac{\mathrm{Cov}(x,\ \beta_0 + \beta_1 x + \varepsilon^y)}{\mathrm{Var}(x)} = \beta_1 + \frac{\rho\sigma_x\sigma_y}{\delta_1^2\sigma_z^2 + \sigma_x^2}. \tag{3}$$

Reduced form. Substitute (2) into (1) (slide p. 18):

$$y_i = \beta_0 + \beta_1\left[\delta_0 + \delta_1 z_i + \varepsilon_i^x\right] + \varepsilon_i^y = (\beta_0 + \beta_1\delta_0) + (\beta_1\delta_1)z_i + (\beta_1\varepsilon_i^x + \varepsilon_i^y),$$

so

$$\theta_0 = \beta_0 + \beta_1\delta_0, \qquad \theta_1 = \beta_1\delta_1, \qquad \tau^2 = \beta_1^2\sigma_x^2 + \sigma_y^2 + 2\beta_1\rho\sigma_x\sigma_y. \tag{4}$$

$u_i = \beta_1\varepsilon_i^x + \varepsilon_i^y$ is independent of z_i, so OLS of y on z is consistent for θ_1, and OLS of x on z for δ_1.

The two-stage estimator (slide p. 18): (1) obtain $\hat{\delta}_1$ from the treatment equation; (2) obtain $\hat{\theta}_1$ from the outcome-on-instrument regression; (3) since $\theta_1 = \beta_1\delta_1$ by (4), with $\delta_1 \neq 0$:

$$\hat{\beta}_1^{IV} = \frac{\hat{\theta}_1}{\hat{\delta}_1} = \frac{\widehat{\mathrm{Cov}}(z,y)/\widehat{\mathrm{Var}}(z)}{\widehat{\mathrm{Cov}}(z,x)/\widehat{\mathrm{Var}}(z)} = \frac{\widehat{\mathrm{Cov}}(z,y)}{\widehat{\mathrm{Cov}}(z,x)} \quad \Big(= \frac{\text{effect of } z \text{ on } y}{\text{effect of } z \text{ on } x}\Big). \tag{5}$$

The covariance route (W eq. 15.9, p. 465; MHE eq. 4.1.3, p. 86). Take the covariance of (1) with z:

$$\mathrm{Cov}(z, y) = \beta_1\,\mathrm{Cov}(z, x) + \mathrm{Cov}(z, \varepsilon^y).$$

Exogeneity sets the last term to 0 and relevance makes $\mathrm{Cov}(z, x) = \delta_1\sigma_z^2 \neq 0$, so

$$\beta_1 = \frac{\mathrm{Cov}(z, y)}{\mathrm{Cov}(z, x)}. \tag{6}$$

The sample analogue (W eq. 15.10) is $\hat{\beta}_1 = \sum(z_i - \bar{z})(y_i - \bar{y})/\sum(z_i - \bar{z})(x_i - \bar{x})$; with $z = x$ it is OLS (W p. 465).

> **Two assumptions, one checkable.** Relevance ($\delta_1 \neq 0$) is a regression of x on z — "something you can check in the data" (MHE p. 86). Exclusion ($\mathrm{Cov}(z, \varepsilon^y) = 0$) involves the unobserved error and is argued, not tested. MHE adds that exclusion "really has two parts": z as good as randomly assigned, and no effect on y except through x (p. 86).

(The slide's p. 18 sentence "$z_i \to x_i \to y_i$ is a fork" describes a **chain**; the conclusion, $y \perp z \mid x$ when $\rho = 0$, is Rule 1 of L05-01-simpson-dags.)

---

3 · C — why adding z to the outcome regression does not work (slides pp. 18–19; G p. 173)

Condition one error on the other. For the bivariate normal $(\varepsilon^x, \varepsilon^y)$ (slide p. 18):

$$\varepsilon_i^y \mid \varepsilon_i^x \sim N\Big(\frac{\sigma_{xy}}{\sigma_x^2}\varepsilon_i^x,\ \sigma_y^2(1-\rho^2)\Big) \quad \Longleftrightarrow \quad \varepsilon_i^y = \frac{\sigma_{xy}}{\sigma_x^2}\varepsilon_i^x + \omega_i,\ \omega_i \sim N(0, \sigma_y^2(1-\rho^2)). \tag{7}$$

Replace ε_i^x by its value from (2), $\varepsilon_i^x = x_i - \delta_0 - \delta_1 z_i$:

$$\varepsilon_i^y = \frac{\sigma_{xy}}{\sigma_x^2}[x_i - \delta_0 - \delta_1 z_i] + \omega_i.$$

Substitute into (1) and collect (slide p. 19):

$$y_i = \Big(\beta_0 - \frac{\sigma_{xy}}{\sigma_x^2}\delta_0\Big) + \Big(\beta_1 + \frac{\sigma_{xy}}{\sigma_x^2}\Big)x_i + \Big(-\frac{\sigma_{xy}}{\sigma_x^2}\delta_1\Big)z_i + \omega_i. \tag{8}$$

ω_i is independent of (x_i, z_i), so (8) **is** the population regression of y on (x, z): the coefficient on x is $\beta_1 + \sigma_{xy}/\sigma_x^2$, biased whenever $\rho \neq 0$. "In words, by simply including the instrumental variable z_i in the outcome regression will not suffice to estimate β_1" (p. 19). Greenberg (§11.2, p. 173) reaches the same expression and concludes that from $y \mid x$ alone "β_s is not identifiable. We have no way of separating β_s from $\beta_s + \sigma_{12}/\sigma_{22}$." The instrument must be used **through the first stage**, not beside it.

---

4 · C — the Wald estimator (MHE §4.1.2, pp. 94–96)

Binary z with $P(z=1) = p$, model $y_i = \alpha + \beta_1 s_i + \eta_i$ (4.1.11).

Covariance with a dummy. $E[yz] = pE[y \mid z=1]$ and $E[y] = pE[y \mid z=1] + (1-p)E[y \mid z=0]$, so

$$\mathrm{Cov}(y,z) = E[yz] - E[y]E[z] = pE[y \mid z=1] - p\{pE[y \mid z=1] + (1-p)E[y \mid z=0]\} = \{E[y \mid z=1] - E[y \mid z=0]\}p(1-$$

The same holds for $\mathrm{Cov}(s,z)$. **Substitute both into (6)**; $p(1-p)$ cancels (MHE eq. 4.1.12):

$$\beta_1 = \frac{E[y_i \mid z_i = 1] - E[y_i \mid z_i = 0]}{E[s_i \mid z_i = 1] - E[s_i \mid z_i = 0]}. \tag{9}$$

Direct route (4.1.13): $E[y_i \mid z_i] = \alpha + \beta_1 E[s_i \mid z_i]$ because $E[\eta_i \mid z_i] = 0$; difference at $z=1$ and $z=0$ and divide.

Worked example, MHE Table 4.1.2 (p. 96; Angrist–Krueger quarter of birth, 1980 census, $n =$ 329,509):

	born Q1–Q2	born Q3–Q4	difference
ln weekly wage	5.8916	5.9051	−0.01349 (0.00337)
years of education	12.6881	12.8394	−0.1514 (0.0162)

$$\hat{\beta}_1^{Wald} = \frac{-0.01349}{-0.1514} = 0.0891 \ (0.0210), \qquad \text{vs OLS } 0.0703 \ (0.0005).$$

On the Card data without controls: $\bar{y}$ is 6.311401 near a college and 6.155494 not near; $\bar{x}$ is 13.527034 and 12.698015, so

$$\hat{\beta}^{Wald} = \frac{6.311401 - 6.155494}{13.527034 - 12.698015} = \frac{0.155907}{0.829019} = 0.1881,$$

against a no-controls OLS of 0.0521 (L05-10-python §5). The controls of §7 matter: region and race shift both college proximity and wages.

---

5 · C — two-stage least squares (MHE §4.1.1, pp. 89–91; W §15.2–15.3, pp. 471–477)

With exogenous covariates X_i (W's $z_1, \ldots, z_{k-1}$) and instrument(s) z_i:

$$y_i = X_i'\alpha + \beta_1 s_i + \eta_i \quad \text{(structural, MHE 4.1.6)}, \qquad s_i = X_i'\pi_{10} + \pi_{11} z_i + \xi_{1i} \quad \text{(first stage)}. \tag{10}$$

Substitute the first stage into the structural equation (MHE 4.1.7):

$$y_i = X_i'\alpha + \beta_1[X_i'\pi_{10} + \pi_{11} z_i + \xi_{1i}] + \eta_i = X_i'(\alpha + \beta_1\pi_{10}) + (\beta_1\pi_{11}) z_i + (\beta_1\xi_{1i} + \eta_i),$$

the **reduced form** with z-coefficient $\pi_{21} = \beta_1\pi_{11}$, so $\beta_1 = \pi_{21}/\pi_{11}$ (indirect least squares, MHE 4.1.5). **2SLS:** regress s_i on (X_i, z_i), form $\hat{s}_i$, then regress y_i on $(X_i, \hat{s}_i)$ (MHE 4.1.9). With one instrument, 2SLS = IV = ILS exactly (L05-10-python §5 checks all three on card.csv). The second-stage OLS standard errors are wrong (they use $y - \hat{\beta}\hat{s}$ instead of $y - \hat{\beta}s$); software computes the right ones (MHE p. 91).

Requirements in W's statement (15.29)–(15.31): $\mathrm{Cov}(z_j, u_1) = 0$ for all exogenous variables, and $\pi_k \neq 0$ — the **partial** correlation of the excluded instrument with y_2 given the included exogenous variables.

---

6 · C — the price and the fragility of IV (W pp. 466–470, 478)

Asymptotic variance under $E(u^2 \mid z) = \sigma^2$ (W eq. 15.12):

$$\mathrm{Avar}(\hat{\beta}_1^{IV}) = \frac{\sigma^2}{n\sigma_x^2\rho_{x,z}^2}, \qquad \widehat{\mathrm{se}} = \sqrt{\frac{\hat{\sigma}^2}{\mathrm{SST}_x\, R_{x,z}^2}} \ \ (15.13). \tag{11}$$

OLS has σ^2/SST_x; IV divides further by $R_{x,z}^2 < 1$: "the IV variance is always larger than the OLS variance (when OLS is valid)" (W p. 466). The SE ratio is $1/|\,\mathrm{corr}(x, z)|$.

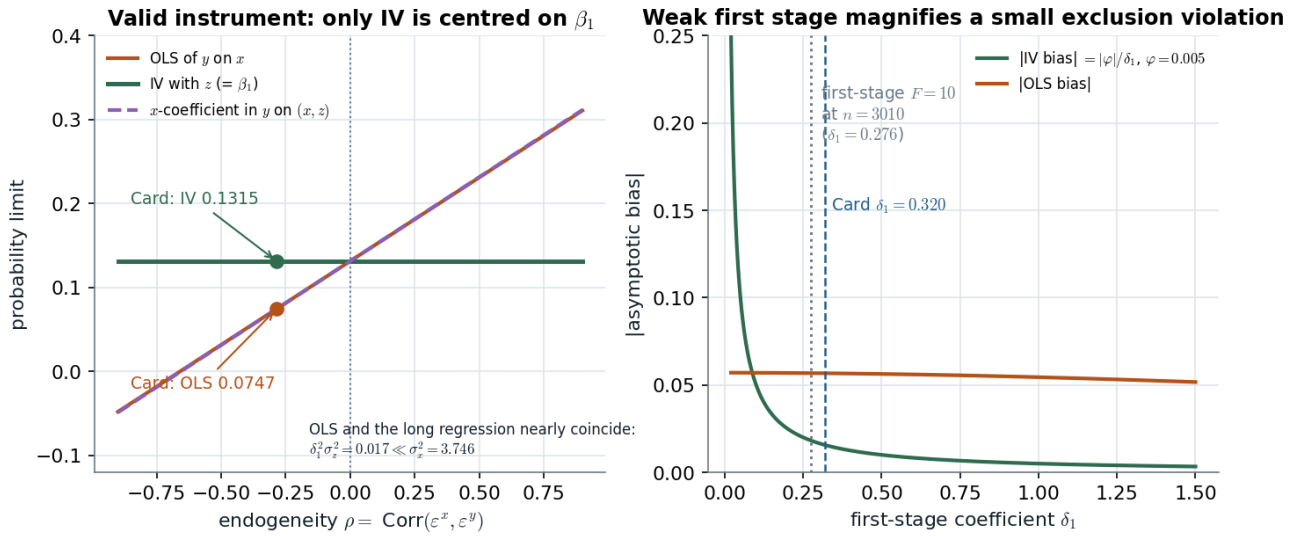
Bias with an imperfect instrument (W eqs. 15.19–15.20, p. 469):

$$\mathrm{plim}\,\hat{\beta}_{1,IV} = \beta_1 + \frac{\mathrm{Corr}(z, u)}{\mathrm{Corr}(z, x)} \cdot \frac{\sigma_u}{\sigma_x}, \qquad \mathrm{plim}\,\hat{\beta}_{1,OLS} = \beta_1 + \mathrm{Corr}(x, u) \cdot \frac{\sigma_u}{\sigma_x}. \tag{12}$$

A small $\text{Corr}(z,u)$ divided by a small $\text{Corr}(z,x)$ can exceed the OLS bias: with $\text{Corr}(z,x)=0.2$, IV is less biased only if $\text{Corr}(z,u)$ is under one fifth of $\text{Corr}(x,u)$ (W p. 470; the book's sentence repeats "$\text{Corr}(z,u)$" where it means $\text{Corr}(x,u)$). In the companion's parametrisation, a direct effect φ of z on y gives $\text{plim}\,\hat{\beta}^{IV}=\beta_1+\varphi/\delta_1$.

When relevance fails. W Example 15.3 (p. 470): mother's packs per day on cigarette price, coefficient 0.0003 (s.e. 0.0008), $R^2=0.0000$; using price anyway gives a packs coefficient of +2.99 (s.e. 8.70) — "meaningless because cigprice fails the one requirement of an IV that we can always test" (also the slide's example C, p. 19).

Weak instruments. Staiger–Stock / Stock–Yogo rule (W p. 478): proceed with standard IV inference if the first-stage $|t|>\sqrt{10}\approx 3.2$ (one instrument) or $F>10$ (several), "not the overall F statistic", only the excluded instruments.



Exact plims (3), (5)/(12) and the x-coefficient of (8), calibrated to card.csv moments after partialling out the site's 14 controls (so the dots reproduce 0.0747 and 0.1315). Slides pp. 18–19; W eqs. 15.19–15.20.

Reading the left panel: IV is flat because it does not depend on ρ; OLS crosses it only at $\rho=0$. The long regression is no better than OLS — here they almost coincide because the instrument explains only $\delta_1^2\sigma_z^2=0.017$ of a schooling variance of 3.746. The Card point sits at $\rho=-0.285$: the data imply OLS **below** IV. On the right, with an exclusion violation of only 0.005 log points, IV is less biased than OLS to the right of $\delta_1\approx 0.09$ and much worse to the left; Card's first stage (0.320) is barely past the $F=10$ line (0.276 at $n=3010$).

> **Interactive companion.** The Instrument Dial (`companions/companion-iv-plims.html`) — move β_1, the endogeneity ρ, the first stage δ_1, an exclusion violation φ and n; watch the OLS, IV and long-regression plims with their 95% bands, the IV bias φ/δ_1, and the first-stage F against 10. Defaults reproduce Card.

---

7 · C — the worked example: Card's college proximity (slide p. 19; example1-iv-card.html; W Example 15.4)

Outcome `lwage`, treatment `educ`, instrument `nearc4` (grew up near a four-year college), controls `exper`, `expersq`, `black`, `smsa`, `south`, `smsa66`, `reg662`–`reg669`; $n=3010$ (the site's printout, reproduced in L05-10-python §5).

Step	Estimate (s.e.)	Source
OLS, `educ`	0.0746933 (0.0034983)	site; W Table 15.1: .075 (.003)
First stage, `nearc4` on `educ`	0.320 (0.088), $t = 3.64$, $R^2 = 0.477$	W p. 473
First-stage $F = t^2$	$3.64^2 \approx 13.256$	site "Weak instruments 13.256"
Reduced form, `nearc4` on `lwage`	0.04207	computed
ILS = IV	$0.04207/0.31990 = 0.1315$	(5)
IV (2SLS), `educ`	0.1315038 (0.0549637)	site; W Table 15.1: .132 (.055)
95% CI	$0.1315 \pm 1.96(0.0550) =$ $(0.024,\ 0.239)$	W p. 474
Wu–Hausman	1.168 ($p = 0.280$)	site
BIC-selected controls: IV	0.1365294 (0.0495914), $F = 16.621$	site §4

Readings.

- **IV is almost twice OLS** (W p. 474) and its SE is $0.0549637/0.0034983 = 15.7$ times larger. The slide rounds to 0.075 and 0.132, SEs 0.003 and 0.055 ("twenty times as big!"), CIs (0.069, 0.081) and (0.022, 0.242) — the latter is $0.132 \pm 2(0.055)$. "Larger CIs: price paid for consistent estimator of the return to (endogenous) education" (p. 19).
- **First stage passes the rule of thumb** ($t = 3.64 > 3.2$, $F = 13.26 > 10$), but not by much — see the right panel above.
- **Wu–Hausman** (W §15.5a, p. 481) adds the first-stage residual $\hat{v}_2$ to the OLS equation and tests it: $t^2 = 1.168$, $p = 0.28$. OLS and IV are not statistically distinguishable here, despite the factor of two, because the IV interval is so wide.
- **Why might IV exceed OLS?** Omitted ability (MHE p. 45) predicts OLS **above** the truth; the data say the opposite (implied $\rho = -0.285$). Measurement error in `educ` (note 04) biases OLS down, and IV with an instrument independent of that error removes it (W p. 480). The heterogeneous-effects reading is in §8.

---

8 · G — heterogeneous effects: LATE (MHE §4.4.1, Theorem 4.4.1, p. 114)

With individual effects and potential treatments d_{1i}, d_{0i} (treatment status if $z_i = 1$ or 0), assume (A1) independence of z_i from all potential outcomes and treatments, (A2) exclusion $y_i(d, 0) = y_i(d, 1)$, (A3) a first stage $E[d_{1i} - d_{0i}] \neq 0$, (A4) monotonicity $d_{1i} \geq d_{0i}$ for all i (or the reverse). Then

$$\frac{E[y_i \mid z_i = 1] - E[y_i \mid z_i = 0]}{E[d_i \mid z_i = 1] - E[d_i \mid z_i = 0]} = E[y_{1i} - y_{0i} \mid d_{1i} > d_{0i}],$$

the average effect on **compliers** — those whose treatment the instrument changes. Proved in MHE pp. 114–115; not on the slides, so not examinable here beyond the statement. For Card, the compliers are men whose schooling depended on having a college nearby, plausibly those with high marginal returns.

Bayesian IV (Greenberg §11.2, pp. 173–174: the system (11.9)–(11.10) with a Wishart prior and a Gibbs sampler; slide p. 20: Hahn, He and Lopes 2018, Lopes and Polson 2014): Classes 3+4 tools, named only.

Links

- Neighbours: L05-04-errors-in-variables (IV as the cure for attenuation, (9) there) · L05-06-difference-in-differences (next).
- Unit: L05-01-simpson-dags (chains, forks; the IV graph) · L05-03-omitted-variables-bad-control (the bias IV removes is (6) there) · L05-07-regression-discontinuity (fuzzy RD is IV with the cutoff as instrument) · L05-08-formulas · L05-09-pitfalls · L05-10-python §5.
- Class 2: L02 §2 (log-level reading: $0.1315 \approx 13.2\%$ per year, exactly $e^{0.1315} - 1 = 14.1\%$).
- Rules pack: rules/01 (2SLS and the Wald estimator; LATE; weak instruments).
- **Practice** (statements in the books): W 15.C3 (p. 492) and W 15.C5 (p. 493), both on CARD; MHE Table 4.1.1 (p. 92), Table 4.1.2 (p. 96), Table 4.1.3 (p. 97) as worked examples; G 11.1–11.4 (pp. 180–181) for the Bayesian versions.

Slides §10 · Difference-in-differences: subtract the control group's change, and assume it is the treated group's counterfactual change

> **Claim.** A before–after comparison of the treated mixes the policy with everything else that changed over time; a treated–control comparison mixes it with permanent group differences. The **difference of the two differences** removes both, under one assumption: **without the policy, both groups would have moved in parallel** (MHE eq. 5.2.1). The four cell means and the saturated regression with an interaction (MHE 5.2.3) give the same number. On the site's Card–Krueger file New Jersey's employment rose by 0.467 FTE while Pennsylvania's fell by 2.283, so **DiD = 2.750** (SE 1.731, $t = 1.59$): no sign that the higher minimum wage cut employment. Prev: L05-05-instrumental-variables · Next: L05-07-regression-discontinuity · Map: L05-00-map

Notation.

- y_{ist}: full-time-equivalent (FTE) employment of store i in state s and period t.
- T_i (`treatment`): 1 for New Jersey (treated), 0 for Pennsylvania (control); P_t (`time`): 1 for November 1992 (after), 0 for February 1992 (before).
- $\bar{y}_{T,t}$: mean of group T in period t (the site's `pretreatment.treated` etc.).
- MHE: γ_s state effect, λ_t period effect, d_{st} the treatment dummy ($= T_i P_t$); MHE calls the causal effect β in (5.2.2), written δ here to keep β for the regression (6).
- W: dT, $d2$, $\delta_0, \beta_1, \delta_1$ in (13.10).
- Δ: a differential trend (treated minus control) that would have occurred without the policy.

---

1 · U — two back doors, one design (slides pp. 21–22)

The slides follow Huntington-Klein: an office switches from cubicles to an open plan on 1 January 2017; productivity Y rises, but "the economy also improved from 2016 to 2017" (p. 21). With a DAG $X \to Y$, $X \leftarrow \text{Time} \to Y$, the before–after difference picks up "the way that Y may have changed over Time for reasons unrelated to Treatment". Adding a never-treated **control group** closes the Time back door but opens another: treatment T now also depends on being **in the treated group** (ITG), and the groups may differ: $T \leftarrow \text{Time} \to Y$ and $T \leftarrow \text{ITG} \to Y$ (p. 22). "We can close both back-door paths through Time and In Treated Group using Difference-in-Differences" — difference within group over time, then across groups.

---

2 · C — the identifying assumption (MHE §5.2, p. 170)

Potential outcomes y_{1ist} (high minimum wage) and y_{0ist} (low). **Common trends**:

$$E(y_{0ist} \mid s, t) = \gamma_s + \lambda_t. \tag{1}$$

"In the absence of a minimum wage change, employment is determined by the sum of a time-invariant state effect and a year effect that is common across states." With a constant effect $E(y_{1ist} - y_{0ist} \mid s, t) = \delta$:

$$y_{ist} = \gamma_s + \lambda_t + \delta d_{st} + \varepsilon_{ist}, \qquad E(\varepsilon_{ist} \mid s, t) = 0. \tag{2}$$

Control state's change. Pennsylvania is never treated ($d = 0$ both periods); from (2):

$$E(y \mid PA, Nov) - E(y \mid PA, Feb) = (\gamma_{PA} + \lambda_{Nov}) - (\gamma_{PA} + \lambda_{Feb}) = \lambda_{Nov} - \lambda_{Feb}. \tag{3}$$

Treated state's change. New Jersey has $d = 1$ in November only:

$$E(y \mid NJ, Nov) - E(y \mid NJ, Feb) = (\gamma_{NJ} + \lambda_{Nov} + \delta) - (\gamma_{NJ} + \lambda_{Feb}) = \lambda_{Nov} - \lambda_{Feb} + \delta. \tag{4}$$

Difference (4) minus (3): the state effects cancelled inside each change and the common trend cancels now:

$$[E(y \mid NJ, Nov) - E(y \mid NJ, Feb)] - [E(y \mid PA, Nov) - E(y \mid PA, Feb)] = \delta. \tag{5}$$

(MHE prints the population DD on p. 170 with the brackets in the opposite order — the PA change minus the NJ change — which as written equals $-\beta$, not β; the order in (5) is the one MHE uses for the regression parameter on p. 175.)

> **"The key identifying assumption here is that employment trends would be the same in both states in the absence of treatment"** (MHE p. 171). Groups may differ in level (γ_s); they may not differ in trend. Like ignorability, this is about an unobserved counterfactual — New Jersey's November employment without the increase.

---

3 · C — the slides' two strategies and the regression (slide p. 22; W Table 13.3)

Strategy 1: change of the treated over time minus change of the control over time. **Strategy 2:** treated–control gap after minus treated–control gap before. They are the same four numbers rearranged (W p. 410: "the estimate does not depend on how we do the differencing").

Regression (slide p. 22; W eq. 13.10; MHE eq. 5.2.3):

$$y_{ti} = \beta_0 + \beta_1 P_t + \beta_2 T_i + \beta_3 P_t T_i + x'_{ti}\gamma + \varepsilon_{ti}. \tag{6}$$

Cell means without covariates. Set $P_t, T_i \in \{0, 1\}$ in (6):

	Before ($P = 0$)	After ($P = 1$)	After − Before
Control ($T = 0$)	β_0	$\beta_0 + \beta_1$	β_1
Treated ($T = 1$)	$\beta_0 + \beta_2$	$\beta_0 + \beta_1 + \beta_2 + \beta_3$	$\beta_1 + \beta_3$
Treated − Control	β_2	$\beta_2 + \beta_3$	β_3

(W Table 13.3, p. 411, with $\delta_0 = \beta_1$, $\beta_1^W = \beta_2$, $\delta_1 = \beta_3$.) **Four cells, four parameters: the model is saturated**, so OLS reproduces the four sample means exactly and

$$\hat{\beta}_3 = (\bar{y}_{1,1} - \bar{y}_{1,0}) - (\bar{y}_{0,1} - \bar{y}_{0,0}) \qquad \text{(W eq. 13.11)}. \tag{7}$$

The regression adds standard errors, more groups and periods, and covariates (slide p. 23).

---

4 · C — the worked example: the site's Card–Krueger file

"On April 1, 1992, New Jersey raised the state minimum wage from \$4.25 to \$5.05 while the minimum wage in Pennsylvania stays the same at \$4.25" (slide p. 24); 384 restaurants observed in February and November 1992, 768 rows; 309 NJ stores, 75 PA. (`treatment = 1` is New Jersey: its pre-period mean matches MHE's NJ column. The slide's phrase "NJ (0) and PA (1)" is a slip.)

Cell means (site output; L05-10-python §6):

	February 1992	November 1992	change
PA (control)	23.380	21.097	$B = 21.097 - 23.380 = -2.283$
NJ (treated)	20.431	20.897	$A = 20.897 - 20.431 = +0.467$
NJ − PA	$20.431 - 23.380 = -2.949$	$20.897 - 21.097 = -0.199$	

Strategy 1: $A - B = 0.4666667 - (-2.2833333) = 2.7500000$ (the site prints `0.4666667 -2.2833333 2.7500000`). **Strategy 2:** $-0.199 - (-2.949) = 2.750$.

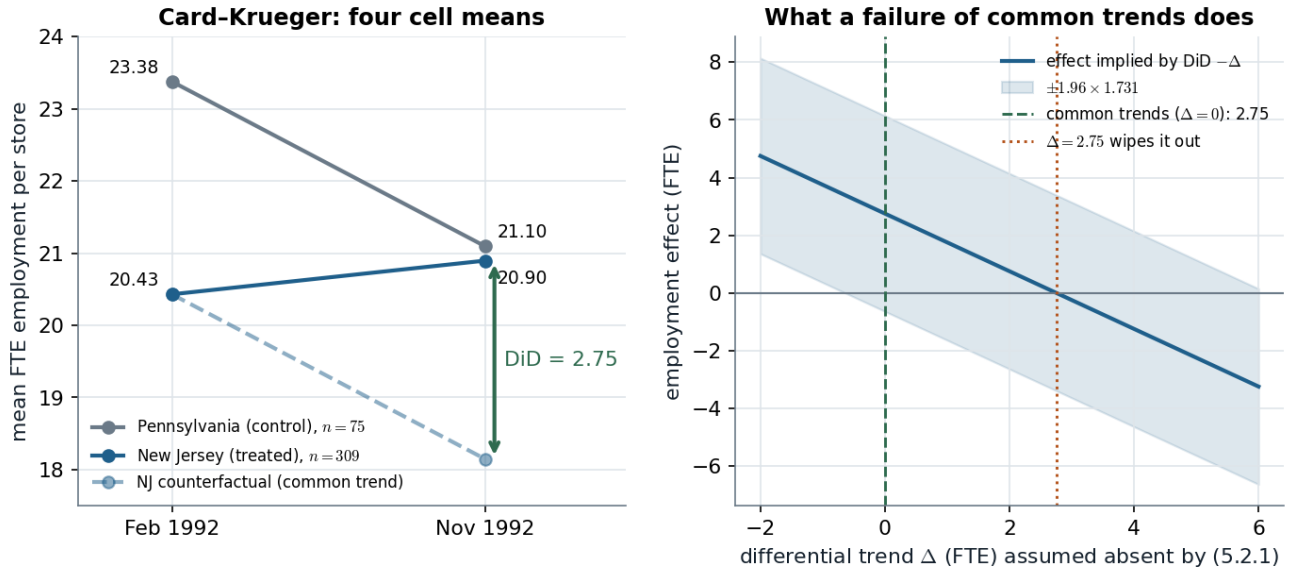
Regression (site `lm(outcome ~ time + treatment + interaction)`):

	Estimate	SE	t	p
(Intercept) $= \bar{y}_{0,0}$	23.380	1.098	21.29	<2e-16
time $= B$	−2.283	1.553	−1.47	0.142
treatment = NJ−PA before	−2.949	1.224	−2.41	0.016
interaction = DiD	**2.750**	**1.731**	1.59	0.113

Residual SE 9.511 on 764 df, $R^2 = 0.0076$. Every coefficient is a cell-mean contrast from the table in §3.

Compare MHE Table 5.2.1 (p. 171, Card–Krueger's own Table 3, all available observations): PA 23.33 → 21.17 (−2.16), NJ 20.44 → 21.03 (+0.59); DiD printed **2.76** (s.e. 1.36); from the rounded entries $0.59 - (-2.16) = 2.75$. The site's balanced 384-store file gives almost the same point estimate with a larger SE (1.731: the pooled-OLS formula ignores that each store appears twice).

> **The sign is the finding.** A competitive labour market predicts employment **falls** in New Jersey relative to Pennsylvania; the DiD is **positive**, "the opposite of what we might expect if a higher minimum wage pushes businesses up the labor demand curve" (MHE p. 171). Statistically, the 95% interval $2.75 \pm 1.96(1.731) = (-0.64, 6.14)$ includes 0 but excludes any sizeable job loss.



Left: card-krueger.txt cell means and the common-trend counterfactual $20.431 + (-2.283) = 18.147$ *(MHE Fig. 5.2.1, p. 172, drawn with the data). Right: if (1) fails by a differential trend* Δ*, the estimate of (5) is* $\delta + \Delta$*, so the effect is* $2.75 - \Delta$*.*

The dashed line is the whole assumption made visible: New Jersey would have lost 2.28 FTE per store, as Pennsylvania did. On the right, the band already contains zero at $\Delta = 0$ (the interval $(-0.64, 6.14)$); a differential trend of $\Delta = 2.75$ — the size of the estimate itself — centres it on zero, and only $\Delta > 6.14$ would turn the data into significant evidence of job **loss**. Card and Krueger's (2000) payroll data (MHE p. 172, Fig. 5.2.2) show substantial differential swings between the two states in other years, so "Pennsylvania may not provide a very good measure of counterfactual employment rates in New Jersey".

---

5 · C — checking and extending (MHE pp. 172–178)

- **Pre-trends.** With several pre-periods, (1) predicts parallel lines before the policy; MHE's Bavaria example (Fig. 5.2.3, p. 174) is the encouraging case.
- **More groups, continuous intensity.** $y_{ist} = \gamma_s + \lambda_t + \beta(fa_s \cdot d_t) + \varepsilon_{ist}$ (MHE 5.2.4, p. 175): Card (1992) uses the fraction of teenagers affected by the federal minimum as treatment intensity.
- **Logs vs levels** (MHE footnote 7, p. 171): "if there is a common trend in logs, there will not be one in levels and vice versa" — the assumption is scale-dependent.

---

6 · G — the slides' Bayesian version (slide p. 25)

The R code draws $\beta \sim N(\hat{\beta},\ s^2(X'X)^{-1})$ with $s^2 = \text{SSR}/(n-4)$ and plots the four cell posteriors and the posterior of β_4 (the interaction): a flat-prior normal approximation, so the posterior is centred at 2.750 with sd 1.731, and $P(\beta_4 > 0 \mid y) \approx \Phi(2.750/1.731) = \Phi(1.589) = 0.944$ (L05-10-python §6). (The snippet uses a `beta1` computed earlier in the session.) Posterior simulation and priors are **Classes 3+4**.

Links

- Neighbours: L05-05-instrumental-variables · L05-07-regression-discontinuity (next).

- Unit: L05-01-simpson-dags (back-door paths) · L05-02-potential-outcomes-identification (DiD's estimand is an ATT: the effect on New Jersey stores) · L05-04-errors-in-variables (differencing and measurement error) · L05-08-formulas · L05-09-pitfalls · L05-10-python §6.
- Class 7: panel fixed effects and hierarchical models generalise $\gamma_s + \lambda_t$ (`Map/class-plan.md`, Class 7 note on FE/RE).
- Rules pack: rules/03 (difference-in-differences and parallel trends).
- **Practice** (statements in the books): W 13.2 (p. 425), W 13.6 (p. 426), W 13.C3 (p. 427, KIELMC incinerator, the data of W Example 13.3); MHE Table 5.2.1 (p. 171) as a worked example.

Slides §11 · Regression discontinuity: when a rule switches treatment on at a cutoff, the jump in the outcome is the effect at the cutoff

> **Claim.** In a **sharp** design treatment is a deterministic, discontinuous function of a running variable, $D_a = 1(a \geq 21)$, so treated and untreated never share a value of a and ignorability holds trivially while **positivity fails by construction**. Identification comes instead from **continuity**: if $E[y_0 \mid a]$ is smooth at the cutoff, the jump in $E[y \mid a]$ there is $E[y_1 - y_0 \mid a = a_0]$ (MHE eq. 6.1.7). Estimating it means modelling the trend on each side — linear, quadratic with interactions, or locally linear in a window — and the trend model is where RD goes wrong: curvature can masquerade as a jump. On the site's MLDA data, motor-vehicle deaths jump by **4.53 per 100,000** at 21 (SE 0.77; 3.41 residual sds), deaths from internal causes do not (0.39, SE 0.60). Prev: L05-06-difference-in-differences · Next: L05-08-formulas · Map: L05-00-map

Notation (the slides'; MHE's in brackets).

- a [x_i]: running variable, age in years; a_0 [x_0]: cutoff, 21; D_a [d_i]: treatment, legal access to alcohol.
- y_a [y_i]: death rate per 100,000 in the 30-day cell at age a; y_0, y_1: potential outcomes without/with treatment.
- ρ: the jump (the causal effect at the cutoff); γ [MHE β]: slope of the trend; $\gamma_1, \gamma_2, \delta_1, \delta_2$: quadratic and interaction terms of slide p. 31.
- b [MHE Δ]: bandwidth, half-width of the window $[a_0 - b, a_0 + b]$; $\hat{\sigma}$: residual standard deviation.

---

1 · U — birthdays and funerals (slides pp. 26–28)

"Human behavior is constrained by rules" (p. 26): class sizes capped at 32, pensions from age 62, the US **minimum legal drinking age** of 21. Daily deaths of Americans aged 20–22 (1997–2003) spike on and right after the 21st birthday: "This spike adds about 100 deaths to a baseline level of about 150 per day" (p. 27). By month of age, "few rates to the left of the age-21 cutoff are above 95 … few of those to the right … are below 95" (p. 28).

---

2 · C — the sharp design (slide p. 28; MHE §6.1, p. 189)

$$D_a = \begin{cases} 1 & a \geq 21 \\ 0 & a < 21 \end{cases} \qquad \text{(MHE 6.1.1: } d_i = 1(x_i \geq x_0)\text{).} \tag{1}$$

The slide's three properties: treatment is a **deterministic** function of a ("once we know a, we know D_a"), a **discontinuous** one ("no matter how close a gets to the cutoff, D_a remains unchanged until the cutoff is reached"), and a is the **running variable**. Sharp RD: "treatment switches cleanly off or on as the running variable passes a cutoff". (The site's code sets `over21[age>21]=1`; no cell sits exactly at 21, so $>$ and $\geq$ agree.)

Why regression still matters (slide p. 29). "Mortality clearly changes with the running variable, a, for reasons unrelated to the MLDA": internal-cause deaths rise through the late teens, external-cause deaths fall. "RD gets its name from the practice of using regression models to implement this control."

---

3 · C — the linear model (slide p. 29; MHE eq. 6.1.2, p. 190)

Assume a linear untreated trend and a constant effect (MHE p. 190):

$$E[y_0 \mid a] = \alpha + \gamma a, \qquad y_1 = y_0 + \rho. \tag{2}$$

Observed mean. By (1) the observed outcome is $y = y_0 + (y_1 - y_0)D_a = y_0 + \rho D_a$; take expectations given a and substitute (2):

$$E[y_a \mid a] = \alpha + \gamma a + \rho D_a \quad \Longrightarrow \quad y_a = \alpha + \rho D_a + \gamma a + \varepsilon_a. \tag{3}$$

"The negative slope, captured by γ, reflects smoothly declining death rates among young people as they mature. The parameter ρ captures the jump in deaths at age 21. The regression generates an estimate of ρ equal to 7.7" (p. 29) — for **all** deaths in Mastering 'Metrics. The site's file has the two components of Fig. 4.5, motor-vehicle accidents (MVA) and internal causes, worked in §6.

> **What makes (3) identify anything.** D_a is perfectly determined by a, yet not collinear with it: RD "distinguish[es] the nonlinear and discontinuous function, $1(x_i \geq x_0)$, from the smooth and (in this case) linear function, x_i" (MHE p. 190). Everything rests on the trend being correctly modelled on both sides.

---

4 · C — identification without functional form (MHE eq. 6.1.7, p. 193)

If $E[y_0 \mid a]$ and $E[y_1 \mid a]$ are continuous at a_0, then just below the cutoff everyone is untreated and just above everyone is treated:

$$E[y \mid a_0 - \Delta < a < a_0] \simeq E[y_0 \mid a = a_0], \qquad E[y \mid a_0 < a < a_0 + \Delta] \simeq E[y_1 \mid a = a_0].$$

Subtract and let the window shrink:

$$\lim_{\Delta \to 0} \Big\{ E[y \mid a_0 < a < a_0 + \Delta] - E[y \mid a_0 - \Delta < a < a_0] \Big\} = E[y_1 - y_0 \mid a = a_0]. \tag{4}$$

The estimand is an **average effect at the cutoff**, valid without the constant-effect assumption in (2) (MHE p. 193) — and only there: RD says nothing about 25-year-olds. Since no age has both treated and untreated cells, RD "turns on our willingness to extrapolate across covariate values, at least in a neighborhood of the discontinuity" (MHE p. 190).

---

5 · C — polynomials and interactions (slides pp. 30–31; MHE eqs. 6.1.4–6.1.6, p. 192)

Different trends on each side. Model both conditional means with polynomials in $\tilde{a} = a - a_0$:

$$E[y_0 \mid a] = \alpha + \gamma_1 \tilde{a} + \gamma_2 \tilde{a}^2, \qquad E[y_1 \mid a] = \alpha + \rho + \gamma_1^{(1)} \tilde{a} + \gamma_2^{(1)} \tilde{a}^2.$$

Combine with $E[y \mid a] = E[y_0 \mid a] + E[y_1 - y_0 \mid a]D_a$ (MHE p. 192):

$$E[y \mid a] = \alpha + \gamma_1 \tilde{a} + \gamma_2 \tilde{a}^2 + D_a[\rho + (\gamma_1^{(1)} - \gamma_1)\tilde{a} + (\gamma_2^{(1)} - \gamma_2)\tilde{a}^2].$$

Name the differences $\delta_k = \gamma_k^{(1)} - \gamma_k$ — the slide's quadratic regression with interactions (p. 31):

$$y_a = \alpha + \rho D_a + \gamma_1(a - a_0) + \gamma_2(a - a_0)^2 + \delta_1[(a - a_0)D_a] + \delta_2[(a - a_0)^2 D_a] + \varepsilon_a. \tag{5}$$

Centring at a_0 makes ρ the jump **at the cutoff** (at $\tilde{a} = c > 0$ the effect is $\rho + \delta_1 c + \delta_2 c^2$, MHE p. 192). The site's code does not centre: it regresses on `age`, `age2`, `over21*age`, `over21*age2` and recovers the jump as `right - left` $= \widehat{\beta}_{over21} + 21\widehat{\beta}_{over21 \cdot age} + 441\widehat{\beta}_{over21 \cdot age^2}$, the same number.

The danger (slide p. 30; MHE Fig. 6.1.1 panel C, p. 191): "nonlinearity mistaken for a discontinuity" — a sharp turn in a smooth $E[y_0 \mid a]$, fitted with straight lines, produces a spurious jump. Gelman and Imbens (2019, slide p. 38) warn the other way: high-order global polynomials give erratic jumps. Hence windows (§7).

---

6 · C — the worked example: the site's MLDA data

48 monthly cells, ages 19.07 to 22.93, deaths per 100,000. Fits on all cells (L05-10-python §7 reproduces every number):

Linear (3), MVA: $\widehat{mva} = 95.4814 - 3.1488\,a + 4.534\,D_a$. The two fitted limits at 21:

$$\text{left} = 95.4814 - 3.1488(21) = 95.4814 - 66.1248 = 29.3566, \qquad \text{right} = 29.3566 + 4.534 = 33.8906.$$

Outcome, model	$\widehat{\rho}$ (SE)	$\widehat{\sigma}$	$\widehat{\rho}/\widehat{\sigma}$ (site: "stdevs")	t
MVA, linear (3)	**4.534** (0.768)	1.3293	4.534/1.3293 = **3.411**	5.90
MVA, quadratic with interactions (5)	4.663 (1.155)	1.3292	**3.508**	4.04
Internal, linear	0.392 (0.596)	1.0318	**0.380**	0.66
Internal, quadratic with interactions	1.073	1.0458	**1.026**	—

Readings.

- **MVA jumps, robustly.** Linear and quadratic agree (4.53 vs 4.66); the curvature terms cost precision (SE 0.77 → 1.15) without moving the estimate. A 4.5-point rise on a left limit of 29.4 is about 15%.
- **Internal causes do not, on the full sample.** Deaths from disease should not respond to legal drinking; a jump of 0.39 (SE 0.60) there is a built-in **placebo outcome** that supports the design. (The quadratic gives 1.07, and some narrow windows give more — §7.)

- **The slide's "stdevs" legend** divides the jump by the residual sd, a scale-free size (not a t statistic). In the slide's p. 33 code the linear DIC legend uses `delta/sigma1` — the MVA jump over the DIC sd, 4.394; the site page corrects it to `delta1/sigma1` = 0.380.

---

7 · C — local linear regression and the bandwidth (slide p. 36; MHE p. 193)

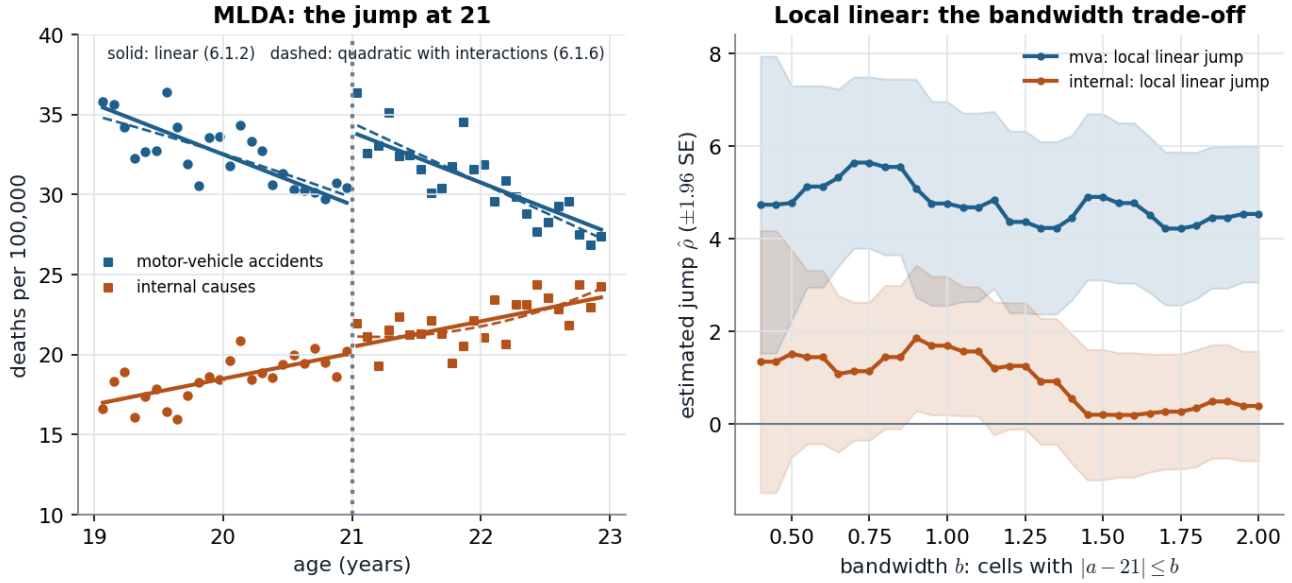
"For observations in $[a_0 - b, a_0 + b]$, for small b, nonlinear trends need not concern us at all. Strategy: comparing averages in a narrow window just to the left and just to the right of the cutoff. High variance: very narrow window → few observations → too imprecise estimates. Trade-off: bias reduction near the boundary against the increased variance" (slide p. 36). The estimator is (3) — or (5) with linear terms only — on the cells with $|a - a_0| \leq b$. MHE (p. 193) describes it as weighted least squares with more weight near the cutoff; the uniform window is the simplest kernel.

MVA, linear with separate slopes, by bandwidth (L05-10-python §7):

b (years)	$\hat{\rho}$	SE
0.5	4.772	1.291
1.0	4.759	1.123
2.0 (all cells)	4.534	0.751

Placebo cutoffs (same estimator, $b = 1$, at ages where no rule changes): $a_0 = 20$: 0.828 (SE 1.096); $a_0 = 22$: −0.103 (SE 1.106). Neither is distinguishable from zero.

> **The jump survives every specification.** Narrowing the window from all 48 cells to one year either side moves $\hat{\rho}$ by a quarter of a death and widens the SE by half; fake cutoffs show nothing. That stability, not any one number, is the RD evidence (MHE p. 193: "Hopefully, as you zero in on x_0 with fewer and fewer controls, the estimated effect of d_i remains stable").



Left: (3) and (5) fitted to age-mva-internal.txt (the site's two figures in one). Right: local linear with separate slopes on $|a - 21| \leq b$, $b = 0.40, 0.45, \ldots, 2.00$. Slides pp. 29–36; MHE eqs. 6.1.2, 6.1.6.

On the left the discontinuity is visible by eye for MVA and absent for internal causes, and the two specifications nearly coincide. On the right the bias–variance trade-off of slide p. 36 is literal: narrow windows give wide bands. The MVA band stays well above zero at every bandwidth (its lower edge never falls below 1.5). The internal-cause band covers zero at most bandwidths but **not** for b between 0.90 and 1.10 (22–26 cells), where $\hat{\rho} \approx 1.6$–1.9 with a lower edge of 0.17–0.28: scanning 33 bandwidths of a noisy placebo outcome turns up a "significant" window. Report the whole curve, not the bandwidth that looks best.

> **Interactive companion.** The Discontinuity Window (`companions/companion-rdd-window.html`) — choose the outcome, the bandwidth b, the polynomial order, separate slopes on/off and the cutoff a_0 (move it off 21 for a placebo test); watch the fitted pieces, $\hat{\rho} \pm 1.96$SE, the jump in residual sds, and $\hat{\rho}$ against every bandwidth. Buttons reset to the site's linear and quadratic fits.

---

8 · G — beyond the sharp parametric design

- **The slides' Bayesian version** (pp. 34–35): draws $\beta \sim N(\hat{\beta}, \hat{\sigma}^2(X'X)^{-1})$ (with $\hat{\sigma}^2$ the ML variance) and plots the posteriors of the left and right limits at 21 for each model. A flat-prior normal approximation; Classes 3+4.
- **Fuzzy RD is IV** (MHE §6.2, pp. 196–198): when crossing the cutoff changes only the **probability** of treatment, the jump in y divided by the jump in $P(d = 1 \mid x)$ (MHE eq. 6.2.5) is a Wald estimator (L05-05-instrumental-variables §4) with $1(x \geq x_0)$ as the instrument. Named only; not on the slides.
- **Bayesian nonparametric RD** (slides pp. 36–37): Branson et al. (2019), Gaussian-process regression on 1,238 NBA draftees — "the treatment effect of being a second-round pick significantly reduces the number of games played". Data-driven bandwidths (Imbens–Kalyanaraman 2012; Calonico–Cattaneo–Titiunik 2014) are in the reference list (p. 38). Named only.

Links

- Neighbours: L05-06-difference-in-differences · L05-08-formulas (next).
- Unit: L05-02-potential-outcomes-identification (positivity fails by design; (4) is a CATE at $a = a_0$) · L05-05-instrumental-variables (fuzzy RD) · L05-09-pitfalls · L05-10-python §7.
- Class 2: L02 §3 (polynomial order and over-fitting: the same trade-off as the bandwidth here).
- Rules pack: rules/02; RDD is a site topic beyond the syllabus table and in scope (Map/class-plan, end notes).
- **Practice** (worked examples in the book): MHE Figure 6.1.2 (p. 195, Lee 2008 elections) and Table 6.2.1 (p. 202, Angrist–Lavy class size, fuzzy RD).

Classes 5+6 · Formula sheet: every definition, result and number of the unit

What this sheet is. §0 maps the class's stated goals to the notes that derive them; §1 lists every definition and result with its source numbering and the note where it is proved; §2 lists every number worked in the unit. No derivations here — follow the links. Prev: L05-07-regression-discontinuity · Next: L05-09-pitfalls · Map: L05-00-map

---

0 · The exam specification, item by item

Goals of the Classes 5+6 block (Map/class-plan), and the final-exam line "derivations (… bias formulas) and interpretation of output from the site's examples" (no coding):

Goal	Where	What to be able to do
Simpson's paradox	L05-01-simpson-dags §1–2	write the pooled rate as a weighted average; decompose the gap; say which table answers the causal question from the graph
DAGs: paths, chains, forks, colliders, d-separation	L05-01-simpson-dags §3–6	classify junctions; list paths; decide $X \perp Y \mid S$ by the blocking rule; pick an adjustment set
Potential outcomes, SUTVA, ATE, CATE, QTE	L05-02-potential-outcomes-identification §2, §4	define each estimand; state each assumption and what it buys
Selection bias	L05-02-potential-outcomes-identification §3; existing notebook and companions	derive naive = ATT + selection bias; show randomisation removes it
Identification, propensity score	L05-02-potential-outcomes-identification §5–6	derive the adjustment formula; prove the IPW identity
Omitted-variable bias	L05-03-omitted-variables-bad-control §1–4	derive short = long + $\gamma\delta$; sign the bias; read Table 3.2.1
Bad control	L05-03-omitted-variables-bad-control §5	derive (3.2.12)'s selection term; the proxy-control coefficient
Errors-in-variables (attenuation)	L05-04-errors-in-variables	derive $\beta\sigma_x^2/(\sigma_x^2+\sigma_u^2)$; CEV vs (9.29); effect of controls; IV fix
IV (Card)	L05-05-instrumental-variables	reduced form, $\beta_1 = \theta_1/\delta_1$; why y on (x, z) fails; Wald; 2SLS; IV variance; weak instruments; read the Card printout
DiD (Card–Krueger)	L05-06-difference-in-differences	common trends ⇒ DD = δ; regression ↔ cell means; read the printout; what a trend violation does
Sharp RDD (MLDA)	L05-07-regression-discontinuity	sharp design; continuity ⇒ jump = effect at cutoff; linear/quadratic/local linear; read the fits; placebo checks

1 · Definitions and results

Simpson and DAGs (L05-01-simpson-dags)

#	Statement	Source
(1)	$\bar{r}_d = (1-\pi_d)r_{d0} + \pi_d r_{d1}$	law of total probability; slide p. 6 data
(2)	$\bar{r}_1 - \bar{r}_0 = [(1-\pi_0)(r_{10}-r_{00}) + \pi_0(r_{11}-r_{01})] + (\pi_1-\pi_0)(r_{11}-r_{10})$	add and subtract
Rule 1	chain $X \to Y \to Z$: $X \perp Z \mid Y$	slide p. 8
Rule 2	fork $Y \leftarrow X \to Z$: $Y \perp Z \mid X$	slide p. 8
Rule 3	collider $X \to Z \leftarrow Y$: $X \perp Y$, but dependent given Z or any descendant of Z	slide p. 9
(4)	$\mathrm{Corr}(X, Z \mid W) = -ab/\sqrt{(a^2+s^2)(b^2+s^2)}$ for $W = aX + bZ + e$	Gaussian conditioning
d-sep.	path blocked by Z iff a chain/fork middle is in Z, or a collider with neither it nor a descendant in Z; all paths blocked ⇒ independence given Z	slide p. 11

Potential outcomes (L05-02-potential-outcomes-identification)

#	Statement	Source
(1)	$Y_i = Y_i(0) + (Y_i(1) - Y_i(0))A_i$	MHE 2.1.1, p. 11
(2)–(6)	ATE $E\{Y(1)\} - E\{Y(0)\}$; ATT $E\{Y(1) - Y(0) \mid A = 1\}$; QTE $F_1^{-1}(p) - F_0^{-1}(p)$; RR $E\{Y(1)\}/E\{Y(0)\}$; CATE $E\{Y(1) - Y(0) \mid V = v\}$	slide p. 15
(7)	$E[Y \mid A = 1] - E[Y \mid A = 0] =$ ATT $+ \{E[Y(0) \mid A = 1] - E[Y(0) \mid A = 0]\}$	MHE p. 11
(8)	SUTVA: $Y_i(\mathbf{a}) = Y_i(\mathbf{a}')$ if $a_i = a_i'$	slide p. 16
(9)	randomisation: $E\{Y(a)\} = E(Y \mid A = a)$	slide p. 16; MHE p. 12
(10)	ignorability $\{Y(0), Y(1)\} \perp A \mid L$ (MHE CIA 3.2.2)	slide p. 16; MHE p. 40
(11)	positivity $P(A = a \mid L = l) > 0$	slide p. 16
(12)	$E\{Y(a)\} = \sum_l P(L = l)E(Y \mid A = a, L = l)$	derived

#	Statement	Source
(13)	ATE $= E[AY/e(L) - (1 - A)Y/(1 - e(L))] = E[Y(A - e(L))/\{e(L)(1 - e(L))\}]$	slide pp. 16–17 (proved)
(14)	$y_i = \alpha + \rho s_i + X_i'\gamma + v_i$	MHE 3.2.9, p. 44

Omitted variables, bad control (L05-03-omitted-variables-bad-control)

#	Statement	Source
(3)	$\beta = (\sigma_{xy} - \rho\sigma_{yz})/(1 - \rho^2)$, $\gamma = (\sigma_{yz} - \rho\sigma_{xy})/(1 - \rho^2)$	omitted-variable.html
(4)	$\tau^2 = (1 - \rho^2 - \sigma_{xy}^2 - \sigma_{yz}^2 + 2\sigma_{xy}\sigma_{yz}\rho)/(1 - \rho^2)$	idem (page's "$1 - \rho$" corrected)
(5)/(6)	short $= \beta + \gamma\rho$; generally $\mathrm{Cov}(y, s)/V(s) = \rho + \gamma'\delta_{As}$	MHE 3.2.11, p. 45
(8)/(9)	site simulation: $E\hat{\beta}^s = 0.75 + 1.2\rho$, sd $\sqrt{(1 + 1.44(1 - \rho^2))/28}$; long sd $\sqrt{1/(27(1 - \rho^2))}$	derived
(11)	bad control: conditional comparison $= E[y_1 - y_0 \mid w_1 = 1] + \{E[y_0 \mid w_1 = 1] - E[y_0 \mid w_0 = 1]\}$	MHE 3.2.12, p. 49
(12)	proxy control: coefficient $\rho - \gamma\pi_1/\pi_2$	MHE 3.2.15, p. 50

Errors in variables (L05-04-errors-in-variables)

#	Statement	Source
(4)	$\operatorname{plim}\hat{\beta}_{\text{naive}} = \beta\sigma_x^2/(\sigma_x^2 + \sigma_u^2)$	error-in-variables.pdf; W 9.33, p. 291
(5)	$y = \beta_0 + \beta_1 x_1 + (u - \beta_1 e_1)$; CEV $\Rightarrow \mathrm{Cov}(x_1, e_1) = \sigma_{e_1}^2$	W 9.30–9.32
(7)/(8)	with controls: $\beta_1\sigma_r^2/(\sigma_r^2 + \sigma_{e_1}^2)$, $\sigma_r^2 = (1 - R^2)\sigma_x^2$	W 9.36, p. 291
(9)	second measurement as IV: $\mathrm{Cov}(w_2, y)/\mathrm{Cov}(w_2, w_1) = \beta$	W §15.4

Instrumental variables (L05-05-instrumental-variables)

#	Statement	Source
(3)	$\operatorname{plim}\hat{\beta}^{OLS} = \beta_1 + \rho\sigma_x\sigma_y/(\delta_1^2\sigma_z^2 + \sigma_x^2)$	derived from slide p. 18 system

#	Statement	Source
(4)	$\theta_0 = \beta_0 + \beta_1\delta_0$, $\theta_1 = \beta_1\delta_1$, $\tau^2 = \beta_1^2\sigma_x^2 + \sigma_y^2 + 2\beta_1\rho\sigma_x\sigma_y$	slide p. 18
(5)/(6)	$\beta_1 = \theta_1/\delta_1 = \mathrm{Cov}(z,y)/\mathrm{Cov}(z,x)$	slide p. 18; W 15.9; MHE 4.1.3
(8)	y on (x,z): coefficient on x $= \beta_1 + \sigma_{xy}/\sigma_x^2$, on z $= -\sigma_{xy}\delta_1/\sigma_x^2$	slide p. 19; G p. 173
(9)	Wald: $\beta_1 = \dfrac{E[y \mid z=1] - E[y \mid z=0]}{E[s \mid z=1] - E[s \mid z=0]}$	MHE 4.1.12, p. 94
(10)	2SLS: reduced-form z coefficient $= \beta_1\pi_{11}$	MHE 4.1.7
(11)	$\mathrm{Avar}(\hat{\beta}^{IV}) = \sigma^2/(n\sigma_x^2\rho_{xz}^2)$	W 15.12, p. 466
(12)	$\operatorname{plim}\hat{\beta}_{IV} = \beta_1 + \frac{\mathrm{Corr}(z,u)}{\mathrm{Corr}(z,x)}\frac{\sigma_u}{\sigma_x}$; OLS: $\beta_1 + \mathrm{Corr}(x,u)\frac{\sigma_u}{\sigma_x}$	W 15.19–15.20, p. 469
rule	first-stage $\lvert t\rvert > 3.2$ / $F > 10$	W p. 478
G	LATE theorem (A1–A4 ⇒ Wald = effect on compliers)	MHE Thm 4.4.1, p. 114

DiD (L05-06-difference-in-differences)

#	Statement	Source
(1)	$E(y_{0ist} \mid s,t) = \gamma_s + \lambda_t$	MHE 5.2.1, p. 170
(2)	$y_{ist} = \gamma_s + \lambda_t + \delta d_{st} + \varepsilon_{ist}$	MHE 5.2.2
(5)	(NJ change) − (PA change) $= \delta$	MHE p. 170 (order corrected)
(6)	$y_{ti} = \beta_0 + \beta_1 P_t + \beta_2 T_i + \beta_3 P_t T_i + x'\gamma + \varepsilon$	slide p. 22; W 13.10; MHE 5.2.3
(7)	$\hat{\beta}_3 = (\bar{y}_{1,1} - \bar{y}_{1,0}) - (\bar{y}_{0,1} - \bar{y}_{0,0})$	W 13.11, Table 13.3

RDD (L05-07-regression-discontinuity)

#	Statement	Source
(1)	$D_a = 1(a \geq a_0)$	slide p. 28; MHE 6.1.1
(3)	$y_a = \alpha + \rho D_a + \gamma a + \varepsilon_a$	slide p. 29; MHE 6.1.2
(4)	$\lim_{\Delta\to 0}\{E[y \mid a_0 < a < a_0 + \Delta] - E[y \mid a_0 - \Delta < a < a_0]\} = E[y_1 - y_0 \mid a = a_0]$	MHE 6.1.7, p. 193
(5)	quadratic with interactions, centred at a_0	slide p. 31; MHE 6.1.6
G	fuzzy RD: jump in y / jump in $P(d=1 \mid x)$	MHE 6.2.5, p. 198

Case	Numbers	Note
Drug table	93/87% (men), 73/69% (women), 78/83% pooled; same-mix effect +5.94, composition −10.51, gap −4.57 points	01 §2
Standardised / adjusted	0.833 vs 0.779 ⇒ ATE **+0.0537**; ATT **+0.0480**; selection bias −0.0937; $e = 0.2437$ (men), 0.7668 (women)	01 §2, 02 §5–6
Collider, $a = b = s = 1$	$\text{Corr}(X, Z \mid W) = -0.5$; given descendant with $t = 1$: $-1/3$	01 §3.1
Harrison DAG	5 paths; 15 junctions = 7 chains, 3 forks, 5 colliders; $X \to Y$ effect 6: adjust $\{Z_1, Z_3\}$ or $\{Z_2, Z_3\}$ 6.0000, $\{Z_3\}$ 6.0051, none 5.4381	01 §4
NHIS (MHE p. 10)	2.79 vs 2.07, difference 0.71, $t = 58.9$	02 §3
OVB simulation	$E\hat{\beta}^s = 0.75, 1.35, 1.89$ at $\rho = 0, 0.5, 0.95$; long sd 0.1925, 0.2222, 0.6163	03 §3
MHE Table 3.2.1	0.132, 0.131, 0.114, 0.087, 0.066	03 §4
Proxy control example	$0.10 - 0.05(0.2)/1 = 0.09$	03 §5.1
Attenuation	$\sigma_x = \sigma_u$: $\lambda = 0.5$; $\beta = 1.5, \sigma_u = 0.75$: $\lambda = 0.64 \to 0.96$; with control $\lambda_r = 0.390$	04
Wald, MHE Table 4.1.2	$-0.01349/-0.1514 = 0.0891$ (0.0210) vs OLS 0.0703	05 §4
Card, no controls	Wald 0.155907/0.829019 = 0.1881; OLS 0.0521	05 §4
Card, site controls	OLS 0.0746933 (0.0034983); first stage 0.320 (0.088), t 3.64, F 13.256, R^2 0.477; RF 0.04207; IV 0.1315038 (0.0549637); CI (0.024, 0.239); SE ratio 15.7; Wu–Hausman 1.168 (p 0.28); selected IV 0.1365294 (0.0495914), F 16.621	05 §7
Card calibration (companion)	$\sigma_z = 0.4026$, $\sigma_x = 1.9354$, $\sigma_y = 0.3873$, $\delta_1 = 0.3199$, $\rho = -0.2851$; $F = 10$ at $\delta_1 = 0.276$ ($n = 3010$)	05 §6
W Example 15.3	packs on cigprice 0.0003, $R^2 = 0$; IV 2.99 (8.70)	05 §6

Case	Numbers	Note
Card–Krueger (site)	PA 23.380 → 21.097 (−2.283); NJ 20.431 → 20.897 (+0.467); DiD **2.750** (SE 1.731, t 1.59, p 0.113); treatment −2.949 (1.224); residual SE 9.511; R^2 0.0076; 95% CI (−0.64, 6.14); $P(\beta_4 > 0)$ 0.944	06 §4, §6
MHE Table 5.2.1	PA 23.33 → 21.17, NJ 20.44 → 21.03; DD 2.76 (1.36)	06 §4
MLDA (slide)	+100 deaths/day on 150; $\hat{\rho} = 7.7$ (all deaths)	07 §1, §3
MLDA (site data)	MVA linear 4.534 (0.768), $\hat{\sigma}$ 1.3293, 3.411 sd, left/right limits 29.357/33.891; MVA quadratic 4.663 (1.155), 3.508 sd; internal 0.392 (0.596), 0.380 sd; internal quadratic 1.073, 1.026 sd	07 §6
Local linear MVA	$b = 0.5$: 4.772 (1.291); $b = 1$: 4.759 (1.123); $b = 2$: 4.534 (0.751); placebo $a_0 = 20$: 0.828 (1.096), $a_0 = 22$: −0.103 (1.106)	07 §7

Links

- Unit: L05-00-map · L05-09-pitfalls · L05-10-python (every number above is asserted there).
- Rules packs: rules/01 · rules/02 · rules/03.

Classes 5+6 · Pitfalls and a True/False self-test

Claim. Most mistakes in causal inference are one of four: reading the wrong table (conditioning on a mediator or collider, or failing to condition on a confounder), confusing a mechanical identity with a causal one (OVB, regression on a randomised dummy), mistaking precision for identification (a tight interval around a biased number), and forgetting the assumption that does all the work (exclusion, common trends, continuity at the cutoff). Part I gives each confusion with the smallest counterexample; Part II is an original 25-item True/False test. Prev: L05-08-formulas · Next: L05-10-python · Map: L05-00-map

---

Part I — the confusions

1 · "Control for everything" vs control for confounders only

More controls are not always better. **Counterexample:** in the drug table with post-treatment BP as the stratifier (slide p. 7), stratifying removes the part of the effect that works through BP; in MHE Table 3.2.1 the occupation dummies move the return from 0.087 to 0.066 by bad control, not by removing bias (L05-03-omitted-variables-bad-control §5). Condition on **forks**; never on **chains out of the treatment** or **colliders**.

2 · A collider blocks; conditioning on it opens

Two independent causes X, Z of W: $\mathrm{Corr}(X, Z) = 0$, but $\mathrm{Corr}(X, Z \mid W) = -0.5$ when $W = X + Z + e$, $\mathrm{Var}(e) = 1$ (L05-01-simpson-dags (4)). Conditioning on a **descendant** of W also opens the path ($-1/3$ with unit noise).

3 · d-separation is not "no arrow between them"

Z and Y in slide p. 12's second graph share no arrow, yet are dependent through the fork at T. Conversely Z and Y in the first graph are linked by a path and are still independent, because the path has a collider.

4 · Simpson: the arithmetic never decides

The same four rates give the right answer in the stratified table for gender and in the pooled table for post-treatment BP (slide p. 7). A pooled rate is a mix; which mix to use is a causal question about the arrows.

5 · Naive difference = ATE

It equals ATT + selection bias (MHE p. 11). In the drug data: naive -4.57 points, ATT $+4.80$, so selection bias -9.37 (L05-02-potential-outcomes-identification §5.1). The ATE ($+5.37$) is a third number.

6 · QTE = quantile of individual effects

$F_1^{-1}(p) - F_0^{-1}(p)$ compares two **margins**. Illustrative counterexample: $Y(0) \in \{0, 1\}$ equally likely and $Y(1) = 1 - Y(0)$: both margins are identical, so the QTE is 0 at every p, yet every individual effect is ± 1.

7 · Ignorability can be tested with enough data

"The validity of ignorability cannot be assessed from the observed data, regardless of how large the sample size is" (slide p. 16). Balance checks test observed covariates only.

8 · OVB means the long regression is causal

The OVB formula is "a mechanical link between coefficient vectors that applies to short and long regressions whether or not the longer regression is causal" (MHE p. 44).

9 · Measurement error biases towards zero, always

Only under **classical** error in a **regressor**. Error in y uncorrelated with x leaves OLS consistent (W §9.4a); error correlated with x (W Example 9.6) can bias either way; and with other regressors, the other coefficients are biased in directions "not easily derived" (W p. 291).

10 · Add the instrument to the regression

y on (x, z) gives $\beta_1 + \sigma_{xy}/\sigma_x^2$ on x — the same confounding (slide p. 19; G p. 173). The instrument works only through the first stage.

11 · A significant first stage is a strong first stage

$t = 2$ rejects $\delta_1 = 0$ at 5% but fails the Stock–Yogo $|t| > 3.2$ rule (W p. 478). With an exclusion violation φ, IV is off by φ/δ_1: weak δ_1 magnifies it (L05-05-instrumental-variables §6 figure).

12 · IV is "less biased" than OLS

Only if exclusion holds exactly. With $\text{Corr}(z, x) = 0.2$, IV beats OLS on bias only if $\text{Corr}(z, u) < \text{Corr}(x, u)/5$ (W p. 470).

13 · DiD needs equal levels

It needs equal **trends** without treatment. NJ and PA differ by −2.949 FTE before (MHE: −2.89); the state effect absorbs that. A differential trend Δ moves the estimate one-for-one (L05-06-difference-in-differences figure).

14 · The DiD regression's β_2 is the effect

β_2 is the treated–control gap **before** treatment; the effect is the interaction β_3 (W Table 13.3).

15 · RD identifies the average effect

It identifies $E[y_1 - y_0 \mid a = a_0]$, the effect **at the cutoff** (MHE 6.1.7). And because no age has both treated and untreated units, it rests on extrapolating the trend into the cutoff.

16 · A jump in a polynomial fit is a discontinuity

A curved $E[y_0 \mid a]$ fitted with lines produces a spurious jump (slide p. 30); high-order polynomials produce erratic ones (Gelman & Imbens 2019, slide p. 38). Check windows, orders, placebo cutoffs and placebo outcomes — and do not pick the bandwidth that looks best: the internal-cause placebo is "significant" for $b \in [0.90, 1.10]$ (L05-07-regression-discontinuity §7).

Part II — True/False self-test (original items)

[**E**] easy · [**M**] medium · [**P**] hard. Scoring as in rules/00: $\max\{0,\ \text{right} - \text{wrong}/2\}$ per block. Numbers marked "illustrative" are invented for the item; all others come from the notes.

Block A — Simpson and DAGs

1. [**E**] If a treatment raises the recovery rate within every stratum of a variable, it must raise the pooled recovery rate. **False.** The pooled rate is a weighted average with group-specific weights (01 (1)); the drug wins within gender (93 vs 87, 73 vs 69) and loses pooled (78 vs 83) because 75% of drug takers but 23% of non-takers are women. The composition term −10.5 points beats the within effect +5.9 (01 (2)).

2. [**M**] In a chain $X \to M \to Y$, conditioning on M makes X and Y independent, so a regression of Y on X and M estimates the total effect of X on Y. **False.** The first half is Rule 1 (true); the conclusion is the reciprocal trap. Conditioning on the mediator blocks the causal path, so the X coefficient is the direct effect only (zero in a pure chain). That is why the BP table must be read pooled.

3. [**M**] Two variables that are marginally independent can become dependent after conditioning on a third variable. **True.** Rule 3: for a collider $X \to W \leftarrow Z$ with $X \perp Z$, conditioning on W (or a descendant) creates dependence; with unit coefficients and noise, $\text{Corr}(X, Z \mid W) = -0.5$ (01 (4)).

4. [**M**] In slide p. 12's second graph ($T \to Z$, $T \to Y$, $Z \to W \leftarrow X \to Y$, $W \to U$), conditioning on $\{T, U\}$ makes Z and Y independent. **False.** T blocks the fork path, but U is a descendant of the collider W and opens $Z \to W \leftarrow X \to Y$; X is not conditioned on, so the path is open. The slide lists $\{U, T\}$ among the d-connecting sets.

5. [**P**] In Harrison's DAG, adjusting for Z_3 alone is a valid back-door adjustment for the effect of X on Y, because Z_3 lies on every back-door path. **False.** Z_3 lies on every back-door path, but on $X \leftarrow Z_1 \to Z_3 \leftarrow Z_2 \to Y$ it is a **collider**: conditioning on it opens that path. $\{Z_1, Z_3\}$ or $\{Z_2, Z_3\}$ are valid; in the slide's SEM $\{Z_3\}$ gives 6.0051 against the true 6 (01 §4).

Block B — potential outcomes and identification

6. [**E**] Under random assignment the average treatment effect on the treated equals the average treatment effect. **True.** Independence of A from $(Y(0), Y(1))$ gives $E[Y(1) - Y(0) \mid A = 1] = E[Y(1) - Y(0)]$ (MHE p. 12; 02 (9)).

7. [**M**] If hospitalised people report worse health than the non-hospitalised (2.79 vs 2.07 on a 1–5 scale where 5 is poor), the selection-bias term in MHE's decomposition is negative in health terms. **True.** People who go to hospital have worse y_{0i} — they would be sicker without hospital too — so $E[y_{0i} \mid d_i = 1]$ is worse than $E[y_{0i} \mid d_i = 0]$: MHE calls the selection bias "negative in this example" (p. 12), large enough to mask a positive effect.

8. [**M**] Ignorability $\{Y(0), Y(1)\} \perp A \mid L$ implies $Y \perp A \mid L$. **False.** The observed $Y = Y(A)$ depends on A through the treatment effect; ignorability concerns the **potential** outcomes. If A has any effect, Y and A are dependent given L — which is how the effect is estimated.

9. [**M**] Positivity can fail even when ignorability holds. **True.** They are separate conditions. A sharp RD is the extreme case: D_a is a function of a, so ignorability given a holds trivially while $P(D = 1 \mid a) \in \{0, 1\}$ violates positivity everywhere (07 §4).

10. [**P**] With a propensity score estimated as the treated share in each covariate cell, the IPW estimate of the ATE equals the adjustment-formula estimate. **True.** Weighting by $1/\hat{e}(l)$ re-creates the population

cell weights in each arm; on the drug table both give 0.05367 (02 §6.1). With a parametric propensity model the two differ in finite samples.

11. **[M]** The QTE at the median, $F_1^{-1}(0.5) - F_0^{-1}(0.5)$, is the median of the individual effects $Y(1) - Y(0)$. **False.** It is a difference of marginal quantiles; the median of individual effects depends on the unobserved joint distribution. The swap counterexample of Part I §6 has QTE 0 and individual effects ± 1.

Block C — omitted variables, bad control, measurement error

12. **[E]** If the omitted variable is uncorrelated with the included regressor, the short and long coefficients on the included regressor coincide. **True.** $\delta = 0$ in "short = long + $\gamma\delta$" (MHE 3.2.11; 03 (5)–(6)), whatever γ.

13. **[M]** In the site's simulation ($\beta = 0.75$, $\gamma = 1.2$), the long regression has a smaller sampling variance than the short one at every $\text{corr}(x, z)$. **False.** At $\rho = 0$ the long sd is 0.1925 vs 0.2952 short, but at $\rho = 0.95$ it is 0.6163 vs 0.2018 (03 (8)–(9)): collinearity inflates the long variance. The long regression still wins on MSE (0.380 vs 1.340) because the short bias 1.2ρ is larger.

14. **[M]** Controlling for a test score taken after schooling, as a proxy for ability, gives a schooling coefficient below the causal effect when schooling raises the score. **True.** By MHE (3.2.15) the coefficient is $\rho - \gamma\pi_1/\pi_2 < \rho$ when $\gamma, \pi_1, \pi_2 > 0$; e.g. 0.09 instead of 0.10 (illustrative, 03 §5.1).

15. **[E]** Classical measurement error in the dependent variable biases the OLS slope towards zero. **False.** Error in y uncorrelated with the regressors goes into the error term: OLS stays unbiased, only less precise (W §9.4a). Attenuation is a property of error in a **regressor**.

16. **[M]** If the noise variance in a regressor equals the variance of its true value, OLS converges to half the true slope. **True.** $\lambda = \sigma_x^2/(\sigma_x^2 + \sigma_u^2) = 1/2$ (04 (4)).

17. **[P]** Adding a correctly measured control that is correlated with a mismeasured regressor reduces the attenuation of the latter's coefficient. **False.** It does the opposite: the control removes signal from the regressor but none of the noise, so $\lambda_r = \sigma_r^2/(\sigma_r^2 + \sigma_u^2) \leq \lambda$ (W 9.36; 04 (8)): 0.390 vs 0.64 in the note's example.

Block D — instrumental variables

18. **[E]** With one instrument and one endogenous regressor, the IV estimate equals the reduced-form coefficient divided by the first-stage coefficient. **True.** $\theta_1 = \beta_1\delta_1$ (05 (4)–(5)); on Card $0.04207/0.31990 = 0.1315$, identical to 2SLS (05 §7).

19. **[M]** If $\text{Cov}(z, x) \neq 0$ and $\text{Cov}(z, u) = 0$, the IV estimator is unbiased in finite samples. **False.** It is consistent, but "essentially never unbiased" when IV is needed (W p. 465); small samples can carry substantial bias.

20. **[M]** In Card's data the IV standard error of the return to schooling is about 16 times the OLS one, so the IV estimate is useless. **False.** The ratio is $0.0549637/0.0034983 = 15.7$, the price of consistency (W eq. 15.12: IV divides by $R^2_{x,z}$). The interval (0.024, 0.239) still excludes zero; it is wide, not useless.

21. **[P]** A first-stage t statistic of 2.5 on the excluded instrument is enough, by the Stock–Yogo rule of thumb, to proceed with standard IV inference. **False.** The rule asks $|t| > \sqrt{10} \approx 3.2$, i.e. $F > 10$ (W p. 478); 2.5 rejects at 5% but is weak. Card's 3.64 passes, barely.

22. **[P]** With an instrument whose direct effect on the outcome is $\varphi = 0.005$ (illustrative), the IV asymptotic bias falls as the first stage gets weaker. **False.** The bias is φ/δ_1, which **grows** as $\delta_1 \to 0$; at the Card calibration IV is worse than OLS for δ_1 below about 0.09 (05 §6).

Block E — DiD and RDD

23. **[E]** In the regression $y = \beta_0 + \beta_1 P + \beta_2 T + \beta_3 PT + \varepsilon$, the coefficient β_2 measures the effect of the policy. **False.** β_2 is the treated–control difference before the policy (-2.949 for NJ vs PA); the DiD effect is $\beta_3 = 2.750$ (06 §3–4).

24. **[M]** If New Jersey's employment would have grown 1 FTE per store faster than Pennsylvania's without the minimum-wage increase, the DiD estimate of 2.750 overstates the policy effect by 1 FTE. **True.** A differential trend Δ enters additively: the DiD estimates $\delta + \Delta$, so the effect is $2.750 - 1 = 1.750$ (06 §4 figure).

25. **[M]** A sharp RD design estimates the average effect of legal drinking on mortality for all young adults aged 19 to 23. **False.** It estimates $E[y_1 - y_0 \mid a = 21]$, the effect at the cutoff (MHE 6.1.7). Extending it to other ages needs the constant-effect assumption of (6.1.2), which the design does not test.

Links

- Unit: L05-00-map · L05-08-formulas · L05-10-python.
- Existing self-tests on the potential-outcomes part: quiz 1 · quiz 2 · quiz 3; the notebook's §12–13 (notebook).
- Rules: rules/00 (V/F scoring) · rules/01 · rules/02 · rules/03.

Classes 5+6 · Python — the reference implementation, asserted against the sources

> Every routine the unit needs, each followed by the `asserts` that tie it to a printed number: d-separation, the adjustment formula and IPW, OVB, attenuation, IV/2SLS/Wald, DiD and RDD. numpy, scipy, pandas, statsmodels and linearmodels — **no scikit-learn**. The lecturer's R is the reference, never the code. **The file is the source of truth:** `Notas/L05/L05_python_checks.py`; the blocks below are copied from it verbatim. Prev: L05-09-pitfalls · Map: L05-00-map · Formulas: L05-08-formulas

Run: `python Notas/L05/L05_python_checks.py` — it prints one line per section and ends with `ALL L05 CHECKS PASSED` (§8 shows the run). The first disagreement aborts with an `AssertionError` naming the quantity and the printed value.

Reused, not rewritten.

Existing code	What it supplies	Used in
`figures/make_figures.py` (imported as `mf`)	`DATA` path, `ovb_exact`, `ovb_mc`, `iv_plims` and the calibration constants `IV_*`, `did_cells`, `rdd_fit`	§3, §5, §6, §7
`companions/check_companions.py`	the two pages' JavaScript under `node` against numpy at three control vectors	cited, not imported: it checks the **pages**; this file checks the **notes**
`Aula/class-05-06/potential-outcomes.ipynb`	selection-bias simulations, STAR, regression on a randomised dummy	cited for note 02 §3; not re-implemented

What is new here, and why: no existing project code computes d-separation, IPW, IV or RDD (the notebook stops at randomised experiments), so these routines are written once here and asserted.

---

0 · Set-up: the comparison helper

`check(x, printed, dec)` passes when `x` rounds to the printed value at `dec` decimals.

```
"""Classes 5+6 (L05) reference implementation with self-checks against the sources'
 ↪  printed numbers.

Run: python Notas/L05/L05_python_checks.py   (numpy, scipy, pandas, statsmodels,
 ↪  linearmodels)
Every section asserts the numbers of one section note and prints a one-line summary;
 ↪  the first
mismatch aborts with an AssertionError naming it. Shown section by section in
 ↪  L05-10-python.md.

Reused, not rewritten: figures/make_figures.py (imported as mf) supplies the Simpson
 ↪  weights, the
collider correlation, the exact OVB moments and their Monte Carlo, the IV plims and
 ↪  calibration
```

```
constants, the DiD cell means and the RDD local-polynomial fit. Importing it draws
↪ nothing.
"""
import itertools
import pathlib
import sys

import numpy as np
import pandas as pd
import statsmodels.formula.api as smf
from linearmodels.iv import IV2SLS
from scipy import stats

sys.path.insert(0, str(pathlib.Path(__file__).resolve().parent / "figures"))
import make_figures as mf  # noqa: E402

DATA = mf.DATA


def near(x, printed, dec):
    """True when x rounds to the printed value at `dec` decimals (half a unit in the
    ↪ last place)."""
    return abs(x - printed) <= 0.5 * 10.0 ** -dec + 1e-9


def check(x, printed, dec, what):
    assert near(x, printed, dec), f"{what}: computed {x!r}, source prints {printed}"
```

---

1 · DAGs and d-separation (L05-01-simpson-dags §3–6)

`d_separated` uses Lauritzen's criterion (moral graph of the ancestral set, conditioning nodes deleted), which is equivalent to the path-blocking definition of slide p. 11 and needs no path enumeration. `paths` and `junctions` reproduce the slide's lists on Harrison's DAG. The back-door check deletes X's outgoing edge and asks whether $X \perp Y \mid S$. The last block builds the **exact population covariance** of the slide's SEM (equations as written, noise propagating, noise variance equal to the systematic variance as in the R code's `sig.*` rule) and computes population regression coefficients — no Monte Carlo error, so the 6.0051 of the collider-only adjustment is a real bias, not noise.

```
def ancestors(dag, nodes):
    """All ancestors of `nodes` (inclusive) in `dag`, a dict child -> set of
    ↪ parents."""
    out, stack = set(nodes), list(nodes)
    while stack:
        for p in dag.get(stack.pop(), set()):
            if p not in out:
                out.add(p)
                stack.append(p)
    return out
```

```
def d_separated(dag, x, y, z):
    """True when x and y are d-separated given the set z (moral graph of the
     ↪  ancestral set, z deleted).

    dag: dict child -> set of parents (every node a key). Lauritzen's criterion,
 ↪  equivalent to the
    path-blocking definition on slide p. 11.
    """
    keep = ancestors(dag, {x, y} | set(z))
    adj = {v: set() for v in keep}
    for c in keep:
        ps = dag[c] & keep
        for p in ps:
            adj[c].add(p)
            adj[p].add(c)
        for p, q in itertools.combinations(ps, 2):      # marry the parents
            adj[p].add(q)
            adj[q].add(p)
    seen, stack = {x}, [x]
    while stack:
        for v in adj[stack.pop()] - set(z):
            if v == y:
                return False
            if v not in seen:
                seen.add(v)
                stack.append(v)
    return True


def paths(dag, x, y):
    """All simple paths between x and y in the skeleton, each as a list of (node,
     ↪  arrow-to-next) pairs."""
    nb = {v: set() for v in dag}
    for c, ps in dag.items():
        for p in ps:
            nb[c].add(p)
            nb[p].add(c)
    out = []

    def walk(path):
        if path[-1] == y:
            out.append(list(path))
            return
        for v in nb[path[-1]]:
            if v not in path:
                walk(path + [v])
    walk([x])
    return out
```

```
def junctions(dag):
    """Every (a, b, c) with a - b - c adjacent and a, c distinct, classified chain /
     ↪  fork / collider."""
    nb = {v: set() for v in dag}
    for c, ps in dag.items():
        for p in ps:
            nb[c] |= {p}
            nb[p] |= {c}
    kinds = {"chain": 0, "fork": 0, "collider": 0}
    for b in dag:
        for a, c in itertools.combinations(sorted(nb[b]), 2):
            into = (a in dag[b]) + (c in dag[b])               # arrows pointing into
 ↪  b
            kinds["collider" if into == 2 else "fork" if into == 0 else "chain"] += 1
    return kinds


# slide p. 11 (example 1) and p. 12 (example 2); slide p. 9 (Harrison's DAG)
EX1 = {"Z": set(), "X": set(), "W": {"Z", "X"}, "U": {"W"}, "Y": {"X"}}
EX2 = {"T": set(), "Z": {"T"}, "X": set(), "W": {"Z", "X"}, "U": {"W"}, "Y": {"X",
 ↪  "T"}}
HARRISON = {"Z1": set(), "Z2": set(), "Z3": {"Z1", "Z2"}, "X": {"Z1", "Z3"}, "W":
 ↪  {"X"}, "Y": {"W", "Z2", "Z3"}}


def section1():
    # example 1
    assert d_separated(EX1, "Z", "Y", [])
    assert not d_separated(EX1, "Z", "Y", ["W"]) and not d_separated(EX1, "Z", "Y",
     ↪  ["U"])
    assert d_separated(EX1, "Z", "Y", ["W", "X"])
    # example 2: the slide's two lists, every set checked
    connected = [["W"], ["U"], ["W", "U"], ["W", "T"], ["U", "T"], ["W", "U", "T"],
 ↪  ["W", "X"], ["U", "X"], ["W", "U", "X"]]
    separated = [["T"], ["X", "T"], ["W", "X", "T"], ["U", "X", "T"], ["W", "U", "X",
 ↪  "T"]]
    assert not d_separated(EX2, "Z", "Y", [])
    assert all(not d_separated(EX2, "Z", "Y", s) for s in connected)
    assert all(d_separated(EX2, "Z", "Y", s) for s in separated)
    # exhaustive: these are ALL the separating subsets of {T, W, U, X}
    allsep = [set(s) for k in range(5) for s in itertools.combinations("TWUX", k) if
 ↪  d_separated(EX2, "Z", "Y", s)]
    assert sorted(map(sorted, allsep)) == sorted(map(sorted, map(set, separated)))
    # Harrison: 5 paths X ... Y and 15 junctions = 7 chains, 3 forks, 5 colliders
     ↪  (slides pp. 9-10)
    assert len(paths(HARRISON, "X", "Y")) == 5
    assert junctions(HARRISON) == {"chain": 7, "fork": 3, "collider": 5}
    # back-door sets for X -> Y: delete X's outgoing edge and test X indep Y | S
    bd = {k: (v - {"X"} if k == "W" else set(v)) for k, v in HARRISON.items()}
```

```
    bd["W"] = set()
    valid = {s: d_separated(bd, "X", "Y", list(s)) for s in [(), ("Z3",), ("Z1",
↪   "Z3"), ("Z2", "Z3"), ("Z1",), ("Z2",)]}
    assert valid == {(): False, ("Z3",): False, ("Z1", "Z3"): True, ("Z2", "Z3"):
     ↪  True, ("Z1",): False, ("Z2",): False}

    # the SEM of slide p. 10 with noise propagating (the equations as written), noise
     ↪  variance = variance of
    # the systematic part (the R code's sig.* rule): exact population covariance,
     ↪  then population regressions
    names = ["z1", "z2", "z3", "x", "w", "y"]
    B = np.zeros((6, 6))
    B[2, 0], B[2, 1] = 3, -1.5
    B[3, 0], B[3, 2] = 2, 2.5
    B[4, 3] = 3
    B[5, 4], B[5, 1], B[5, 2] = 2, 2, -3
    S = np.zeros((6, 6))
    S[0, 0], S[1, 1] = 1.72**2, 1.89**2
    for k in range(2, 6):
        b = B[k, :k]
        S[k, :k] = S[:k, :k] @ b
        S[:k, k] = S[k, :k]
        S[k, k] = 2 * (b @ S[:k, :k] @ b)                              # systematic +
↪   equal noise variance

    def coef_x(adj):
        idx = [3] + [names.index(c) for c in adj]
        return np.linalg.solve(S[np.ix_(idx, idx)], S[idx, 5])[0]
    pop = {tuple(a): coef_x(a) for a in ([], ["z3"], ["z1", "z3"], ["z2", "z3"],
↪   ["z1"], ["z2"])}
    assert abs(pop[("z1", "z3")] - 6) < 1e-10 and abs(pop[("z2", "z3")] - 6) < 1e-10
     ↪  # total effect 3 x 2 = 6
    check(pop[("z3",)], 6.0051, 4, "M-bias from conditioning on the collider Z3
↪   alone")
    check(pop[()], 5.4381, 4, "no adjustment")
    print(f"1 DAGs: slide d-separation lists confirmed (exhaustive); Harrison 7/3/5
     ↪  junctions; X->Y effect 6: "
          f"adjust {{Z1,Z3}} or {{Z2,Z3}} 6.0000, {{Z3}} alone {pop[('z3',)]:.4f},
           ↪  {{Z1}} {pop[('z1',)]:.4f}, "
          f"{{Z2}} {pop[('z2',)]:.4f}, none {pop[()]:.4f}")
    return pop
```

---

2 · Potential outcomes, adjustment and IPW (L05-02-potential-outcomes-identification §5–6)

The 700 patients are rebuilt from the slide p. 6 counts; the adjustment formula (12), both forms of the IPW identity (13) and the ATT by standardisation to the treated mix are computed on them.

```
def drug_data():
```

```
    """700 rows (A = drug, G = 1 woman, Y = recovered) rebuilt from the slide p. 6
     ↪ counts."""
    cells = [  # (A, G, recovered, n)
        (1, 0, 81, 87), (0, 0, 234, 270), (1, 1, 192, 263), (0, 1, 55, 80)]
    rows = []
    for a, g, r, n in cells:
        rows += [(a, g, 1)] * r + [(a, g, 0)] * (n - r)
    return pd.DataFrame(rows, columns=["A", "G", "Y"])


def section2():
    d = drug_data()
    assert len(d) == 700 and (d.G == 1).sum() == 343 and (d.A == 1).sum() == 350
    naive = d[d.A == 1].Y.mean() - d[d.A == 0].Y.mean()
    check(d[d.A == 1].Y.mean(), 0.78, 2, "drug 78%")
    check(d[d.A == 0].Y.mean(), 0.83, 2, "no drug 83%")
    pg = d.G.value_counts(normalize=True)
    strat = sum(pg[g] * (d[(d.A == 1) & (d.G == g)].Y.mean() - d[(d.A == 0) & (d.G
    ↪ == g)].Y.mean()) for g in (0, 1))
    e = d.groupby("G").A.mean()                                    # propensity
    ↪ e(g) = P(A = 1 | G = g)
    eg = d.G.map(e)
    ipw = np.mean(d.A * d.Y / eg - (1 - d.A) * d.Y / (1 - eg))
    ipw2 = np.mean(d.Y * (d.A - eg) / (eg * (1 - eg)))             # second form
    ↪ of the proposition
    check(naive, -0.0457, 4, "naive gap")
    check(strat, 0.0537, 4, "adjustment formula")
    assert abs(ipw - strat) < 1e-12 and abs(ipw2 - strat) < 1e-12
    check(e[0], 0.2437, 4, "e(men) = 87/357")
    check(e[1], 0.7668, 4, "e(women) = 263/343")
    # ATT by standardising to the treated's gender mix
    pg1 = d[d.A == 1].G.value_counts(normalize=True)
    att = sum(pg1[g] * (d[(d.A == 1) & (d.G == g)].Y.mean() - d[(d.A == 0) & (d.G ==
    ↪ g)].Y.mean()) for g in (0, 1))
    check(att, 0.0480, 4, "ATT")
    print(f"2 potential outcomes: naive {naive:+.4f}; adjusted ATE {strat:+.4f} = IPW
     ↪ {ipw:+.4f}; ATT {att:+.4f}; "
          f"e(men) {e[0]:.4f}, e(women) {e[1]:.4f}")
```

---

3 · Omitted-variable bias (L05-03-omitted-variables-bad-control §1–5)

The conditional-mean coefficients (3) and variance (4) are checked against the Schur complement, including the slip on omitted-variable.html; the site's simulation reuses `mf.ovb_mc` and `mf.ovb_exact` from the figure script.

```
def section3():
    # omitted-variable.html: (y, x, z) ~ N(0, [[1, sxy, syz], [sxy, 1, rho], [syz,
     ↪ rho, 1]])
    sxy, syz, rho = 0.4, 0.5, 0.3
```

```
    S = np.array([[1, sxy, syz], [sxy, 1, rho], [syz, rho, 1]])
    s, Sxx = S[0, 1:], S[1:, 1:]
    coef = np.linalg.solve(Sxx, s)
    var = 1 - s @ coef
    assert np.allclose(coef, [(sxy - syz * rho) / (1 - rho**2), (syz - sxy * rho) /
     ↪  (1 - rho**2)])
    assert abs(var - (1 - rho**2 - sxy**2 - syz**2 + 2 * sxy * syz * rho) / (1 -
     ↪  rho**2)) < 1e-12
    assert abs(var - (1 - rho - sxy**2 - syz**2 + 2 * sxy * syz * rho) / (1 -
     ↪  rho**2)) > 0.05   # the page's "1 - rho" is a slip
    # OVB identity: short slope sxy = beta + gamma * rho
    b, g = coef
    assert abs(sxy - (b + g * rho)) < 1e-12
    # the site's simulation: exact mean beta + gamma rho and the Monte Carlo (shared
     ↪  with the figure)
    rhos = np.array([0.0, 0.5, 0.95])
    short, long_ = mf.ovb_mc(rhos)
    ms, ss, ml, sl = mf.ovb_exact(rhos)
    assert np.all(np.abs(short.mean(1) - ms) < 4 * ss / np.sqrt(mf.OVB_REPS))
    assert np.allclose(ms, [0.75, 1.35, 1.89]) and np.allclose(sl, [0.19245, 0.22222,
     ↪  0.61633], atol=5e-5)
    # MHE Table 3.2.1: the drops attributed to the added controls
    check(0.132 - 0.114, 0.018, 3, "background controls")
    check(0.114 - 0.087, 0.027, 3, "AFQT")
    check(0.087 - 0.066, 0.021, 3, "occupation (bad control)")
    # proxy control (3.2.15): coefficient beta - gamma*pi1/pi2; with beta=.1,
     ↪  gamma=.05, pi1=.2, pi2=1
    beta, gamma, pi1, pi2 = 0.10, 0.05, 0.2, 1.0
    check(beta - gamma * pi1 / pi2, 0.09, 3, "proxy-control coefficient")
    print(f"3 OVB: conditional-mean coefficients and Var(y|x,z) checked (page's '1 -
     ↪  rho' is a slip); "
          f"short means {np.round(short.mean(1), 3)} vs exact {ms}")
```

---

4 · Errors in variables (L05-04-errors-in-variables §1–4)

Seeded simulations with 400,000 draws: naive slope against $\beta\lambda$, IV with an independent second measurement against β, error in y only, and W (9.36) with one control.

```
def section4():
    rng = np.random.default_rng(11)
    n = 400_000
    beta, sx, su = 1.5, 1.0, 0.75
    lam = sx**2 / (sx**2 + su**2)
    check(lam, 0.64, 2, "reliability 1/(1+0.5625)")
    x = rng.normal(0, sx, n)
    y = 2 + beta * x + rng.normal(0, 1, n)
    w1 = x + rng.normal(0, su, n)
    w2 = x + rng.normal(0, su, n)                                    # second,
 ↪  independent measurement
```

```
    b_naive = np.cov(w1, y)[0, 1] / np.var(w1, ddof=1)
    b_iv = np.cov(w2, y)[0, 1] / np.cov(w2, w1)[0, 1]                # w2 instruments
↪   w1 (W section 15-4)
    assert abs(b_naive - beta * lam) < 0.01 and abs(b_iv - beta) < 0.01
    # error in y only: unbiased, noisier
    b_y = np.cov(x, y + rng.normal(0, 2, n))[0, 1] / np.var(x, ddof=1)
    assert abs(b_y - beta) < 0.02
    # with a control (W 9.36): x* = 0.8 x2 + r, Var(r) = 0.36
    x2 = rng.normal(size=n)
    xs = 0.8 * x2 + rng.normal(0, 0.6, n)
    y2 = beta * xs + 0.5 * x2 + rng.normal(size=n)
    x1 = xs + rng.normal(0, su, n)
    b = smf.ols("y ~ x1 + x2", pd.DataFrame(dict(y=y2, x1=x1,
↪   x2=x2))).fit().params.x1
    lam_r = 0.36 / (0.36 + su**2)
    assert abs(b - beta * lam_r) < 0.01
    print(f"4 attenuation: lambda {lam:.3f} -> naive {b_naive:.3f} (beta {beta}); IV
     ↪  with 2nd measure {b_iv:.3f}; "
          f"with a control lambda_r {lam_r:.3f} -> {b:.3f}")
```

---

5 · Instrumental variables on card.csv (L05-05-instrumental-variables §2–7)

`linearmodels.iv.IV2SLS` with `cov_type='unadjusted', debiased=True` reproduces R's `ivreg` standard errors. Also: ILS = IV, manual 2SLS, the control-function Wu–Hausman statistic, the BIC-selected specification, the companion's calibration (partialled moments) and a seeded Gaussian check of slide p. 19's long regression.

```
CONTROLS = "exper + expersq + black + smsa + south + smsa66 + reg662 + reg663 +
↪   reg664 + reg665 + reg666 + reg667 + reg668 + reg669"


def card():
    return pd.read_csv(DATA / "card.csv", na_values=".")


def section5():
    c = card()
    assert len(c) == 3010
    ols = smf.ols("lwage ~ educ + " + CONTROLS, c).fit()
    check(ols.params.educ, 0.0746933, 7, "OLS educ")
    check(ols.bse.educ, 0.0034983, 7, "OLS se")
    fs = smf.ols("educ ~ nearc4 + " + CONTROLS, c).fit()
    check(fs.params.nearc4, 0.320, 3, "first stage (W p. 473)")
    check(fs.tvalues.nearc4, 3.64, 2, "first-stage t")
    check(fs.rsquared, 0.477, 3, "first-stage R2")
    check(fs.tvalues.nearc4**2, 13.256, 3, "weak-instrument F")
    rf = smf.ols("lwage ~ nearc4 + " + CONTROLS, c).fit()
    ils = rf.params.nearc4 / fs.params.nearc4                        # indirect
↪   least squares (MHE 4.1.5)
```

```
iv = IV2SLS.from_formula("lwage ~ 1 + " + CONTROLS + " + [educ ~ nearc4]", c).fit(cov_type="unadjusted", debiased=True)
check(iv.params.educ, 0.1315038, 7, "IV educ")
check(iv.std_errors.educ, 0.0549637, 7, "IV se")
assert abs(ils - iv.params.educ) < 1e-10
# 2SLS by hand: second stage on fitted educ gives the same point estimate (SEs differ, MHE p. 91)
c2 = c.assign(educ_hat=fs.fittedvalues)
check(smf.ols("lwage ~ educ_hat + " + CONTROLS, c2).fit().params.educ_hat, 0.1315038, 7, "manual 2SLS")
# control-function (Wu-Hausman) test: t^2 on the first-stage residual
c3 = c.assign(vhat=fs.resid)
t = smf.ols("lwage ~ educ + vhat + " + CONTROLS, c3).fit().tvalues.vhat
check(t**2, 1.168, 3, "Wu-Hausman")
lo, hi = iv.params.educ - 1.96 * iv.std_errors.educ, iv.params.educ + 1.96 * iv.std_errors.educ
check(lo, 0.024, 3, "CI low (W p. 474)")
check(hi, 0.239, 3, "CI high")
check(iv.std_errors.educ / ols.bse.educ, 15.7, 1, "SE ratio")
# BIC-selected specification of the site
sel = "exper + expersq + black + smsa + south + reg663"
iv_s = IV2SLS.from_formula("lwage ~ 1 + " + sel + " + [educ ~ nearc4]", c).fit(cov_type="unadjusted", debiased=True)
check(iv_s.params.educ, 0.1365294, 7, "IV selected")
check(smf.ols("educ ~ nearc4 + " + sel, c).fit().tvalues.nearc4**2, 16.621, 3, "F selected")
# MHE Table 4.1.2 Wald estimate
check(-0.01349 / -0.1514, 0.0891, 4, "Wald (MHE p. 96)")
# simple (no-controls) Wald with nearc4
m = c.groupby("nearc4")"lwage", "educ".mean()
wald = (m.lwage[1] - m.lwage[0]) / (m.educ[1] - m.educ[0])
check(wald, 0.188, 3, "simple Wald")
# companion calibration: moments of the residuals after partialling out the controls
X = np.column_stack([np.ones(len(c))] + [c[v].values for v in CONTROLS.split(" + ")])
res = lambda v: v - X @ np.linalg.lstsq(X, v, rcond=None)[0]
xr, yr, zr = res(c.educ.values * 1.0), res(c.lwage.values), res(c.nearc4.values * 1.0)
V = lambda a, b: np.mean(a * b)
sz2, d1 = V(zr, zr), V(zr, xr) / V(zr, zr)
sx2 = V(xr, xr) - d1**2 * sz2
b1 = V(zr, yr) / V(zr, xr)
cxy = V(xr, yr) - b1 * V(xr, xr)
sy2 = V(yr, yr) - b1**2 * V(xr, xr) - 2 * b1 * cxy
rho = cxy / np.sqrt(sx2 * sy2)
got = np.array([np.sqrt(sz2), np.sqrt(sx2), np.sqrt(sy2), b1, d1, rho])
want = np.array([mf.IV_SZ, mf.IV_SX, mf.IV_SY, mf.IV_B1, mf.IV_D1, mf.IV_RHO])
assert np.allclose(got, want, atol=5e-7), got
```

```
    o, i, lx = mf.iv_plims(rho, d1=d1, b1=b1, sz=np.sqrt(sz2), sx=np.sqrt(sx2),
↪   sy=np.sqrt(sy2))
    check(o, 0.0746933, 7, "calibrated OLS plim")
    check(i, 0.1315038, 7, "calibrated IV plim")
    # slide p. 19 / Greenberg p. 173: y on (x, z) gives beta1 + sigma_xy / sigma_x^2
     ↪  on x (seeded Gaussian check)
    rng = np.random.default_rng(3)
    n = 400_000
    zz = rng.normal(size=n)
    ex, ey = rng.multivariate_normal([0, 0], [[1, 0.6], [0.6, 1]], n).T
    xx = 0.5 * zz + ex
    yy = 1.0 * xx + ey
    bxz = np.linalg.lstsq(np.column_stack([xx, zz]), yy, rcond=None)[0]
    assert abs(bxz[0] - 1.6) < 0.01 and abs(bxz[1] + 0.3) < 0.01          # 1 + 0.6,
     ↪  -0.6 * 0.5
    assert abs(np.cov(zz, yy)[0, 1] / np.cov(zz, xx)[0, 1] - 1.0) < 0.01
    print(f"5 IV: OLS {ols.params.educ:.7f}, IV {iv.params.educ:.7f} (se
     ↪  {iv.std_errors.educ:.7f}), first stage "
          f"{fs.params.nearc4:.3f} (t {fs.tvalues.nearc4:.2f}, F
           ↪  {fs.tvalues.nearc4**2:.3f}), Wu-Hausman {t**2:.3f}, "
          f"CI ({lo:.3f}, {hi:.3f}); simple Wald {wald:.4f}; calibration rho
           ↪  {rho:.4f}")
```

---

6 · Difference-in-differences (L05-06-difference-in-differences §3–6)

Cell means, both orders of differencing, the saturated regression with every coefficient and SE of the site's printout, the cell-mean ↔ coefficient map, and the flat-prior $P(\beta_4 > 0)$.

```
def section6():
    d, m = mf.did_cells()
    A = m[1, 1] - m[1, 0]
    B = m[0, 1] - m[0, 0]
    check(A, 0.4666667, 7, "A")
    check(B, -2.2833333, 7, "B")
    check(A - B, 2.75, 7, "DiD")
    # the other order of differencing (W p. 410)
    assert abs(((m[1, 1] - m[0, 1]) - (m[1, 0] - m[0, 0])) - (A - B)) < 1e-12
    d = d.assign(inter=d.time * d.treatment)
    r = smf.ols("outcome ~ time + treatment + inter", d).fit()
    for k, v, se in (("Intercept", 23.380, 1.098), ("time", -2.283, 1.553),
     ↪  ("treatment", -2.949, 1.224), ("inter", 2.750, 1.731)):
        check(r.params[k], v, 3, k)
        check(r.bse[k], se, 3, k + " se")
    check(np.sqrt(r.scale), 9.511, 3, "residual se")
    check(r.rsquared, 0.007587, 6, "R2")
    # regression parameters = cell-mean contrasts (MHE p. 175, W Table 13.3)
    assert abs(r.params.Intercept - m[0, 0]) < 1e-9 and abs(r.params.treatment -
     ↪  (m[1, 0] - m[0, 0])) < 1e-9
    assert abs(r.params.time - (m[0, 1] - m[0, 0])) < 1e-9
    # MHE Table 5.2.1 (Card-Krueger's own sample)
```

```
    check((21.03 - 20.44) - (21.17 - 23.33), 2.75, 2, "MHE table arithmetic")
    # flat-prior posterior P(beta4 > 0) under the normal approximation used on slide
     ↪ p. 25
    p = stats.norm.cdf(r.params.inter / r.bse.inter)
    check(p, 0.944, 3, "P(effect > 0)")
    print(f"6 DiD: cells PA {m[0, 0]:.3f}->{m[0, 1]:.3f}, NJ {m[1, 0]:.3f}->{m[1,
     ↪ 1]:.3f}; DiD {A - B:.4f} "
          f"(se {r.bse.inter:.3f}, t {r.tvalues.inter:.3f}); P(beta4>0) {p:.3f}")
```

---

7 · Regression discontinuity (L05-07-regression-discontinuity §5–7)

`mf.rdd_fit` (centred local polynomial, uniform window) reproduces the site's linear and quadratic jumps and "stdevs", the slide's `delta/sigma1` slip, and the bandwidth and placebo tables of the note.

```
def section7():
    d = pd.read_csv(DATA / "age-mva-internal.txt", sep=r"\s+")
    age = d.age.values
    out = {}
    for col in ("mva", "internal"):
        y = d[col].values
        j, se, s, n, b = mf.rdd_fit(age, y)
        jq, seq, sq, _, _ = mf.rdd_fit(age, y, p=2, inter=True)
        out[col] = (j, se, s, jq, seq, sq)
    j, se, s, jq, seq, sq = out["mva"]
    check(j, 4.534, 3, "MVA linear jump")
    check(j / s, 3.411, 3, "MVA stdevs (site legend)")
    check(jq / sq, 3.508, 3, "MVA quadratic stdevs")
    ji, sei, si, jqi, seqi, sqi = out["internal"]
    check(ji / si, 0.380, 3, "DIC stdevs")
    check(jqi / sqi, 1.026, 3, "DIC quadratic stdevs")
    # the slide p. 33 code divides the MVA jump by the DIC sigma for the linear DIC
     ↪ legend
    check(j / si, 4.394, 3, "slide's nsig1 = delta/sigma1 slip")
    # linear fit without interaction: jump = coefficient on over21 directly (D
     ↪ defined as age > 21)
    f = smf.ols("mva ~ age + over21", d.assign(over21=(d.age >
↪ 21).astype(int))).fit()
    assert abs(f.params.over21 - j) < 1e-10
    check(f.params.age, -3.149, 3, "MVA age slope")
    # local linear with separate slopes at a few bandwidths (note 07 table)
    ll = {b: mf.rdd_fit(age, d.mva.values, b=b, inter=True)[:2] for b in (0.5, 1.0,
↪ 2.0)}
    # placebo cutoffs (no rule changes there): linear with interactions on the side's
     ↪ own data
    plac = {a0: mf.rdd_fit(age, d.mva.values, a0=a0, b=1.0, inter=True)[:2] for a0
↪ in (20.0, 22.0)}
    print(f"7 RDD: MVA linear {j:.3f} (se {se:.3f}, {j / s:.3f} sd), quadratic
     ↪ {jq:.3f} (se {seq:.3f}, {jq / sq:.3f} sd); "
```

```
          f"internal {ji:.3f} / {jqi:.3f}; local linear MVA b=0.5
           ↪ {ll[0.5][0]:.3f}({ll[0.5][1]:.3f}), "
          f"b=1 {ll[1.0][0]:.3f}({ll[1.0][1]:.3f}), b=2
           ↪ {ll[2.0][0]:.3f}({ll[2.0][1]:.3f}); "
          f"placebo a0=20 {plac[20.0][0]:.3f}({plac[20.0][1]:.3f}), a0=22
           ↪ {plac[22.0][0]:.3f}({plac[22.0][1]:.3f})")
    return ll, plac
```

---

8 · Running it

```
if __name__ == "__main__":
    section1()
    section2()
    section3()
    section4()
    section5()
    section6()
    section7()
    print("ALL L05 CHECKS PASSED")
```

Output of `python Notas/L05/L05_python_checks.py` (2026-10-25):

```
1 DAGs: slide d-separation lists confirmed (exhaustive); Harrison 7/3/5 junctions;
 ↪ X->Y effect 6: adjust {Z1,Z3} or {Z2,Z3} 6.0000, {Z3} alone 6.0051, {Z1} 5.5666,
 ↪ {Z2} 5.4959, none 5.4381
2 potential outcomes: naive -0.0457; adjusted ATE +0.0537 = IPW +0.0537; ATT +0.0480;
 ↪ e(men) 0.2437, e(women) 0.7668
3 OVB: conditional-mean coefficients and Var(y|x,z) checked (page's '1 - rho' is a
 ↪ slip); short means [0.753 1.347 1.885] vs exact [0.75 1.35 1.89]
4 attenuation: lambda 0.640 -> naive 0.959 (beta 1.5); IV with 2nd measure 1.500;
 ↪ with a control lambda_r 0.390 -> 0.588
5 IV: OLS 0.0746933, IV 0.1315038 (se 0.0549637), first stage 0.320 (t 3.64, F
 ↪ 13.256), Wu-Hausman 1.168, CI (0.024, 0.239); simple Wald 0.1881; calibration rho
 ↪ -0.2851
6 DiD: cells PA 23.380->21.097, NJ 20.431->20.897; DiD 2.7500 (se 1.731, t 1.588);
 ↪ P(beta4>0) 0.944
7 RDD: MVA linear 4.534 (se 0.768, 3.411 sd), quadratic 4.663 (se 1.155, 3.508 sd);
 ↪ internal 0.392 / 1.073; local linear MVA b=0.5 4.772(1.291), b=1 4.759(1.123),
 ↪ b=2 4.534(0.751); placebo a0=20 0.828(1.096), a0=22 -0.103(1.106)
ALL L05 CHECKS PASSED
```

The figure script (`python Notas/L05/figures/make_figures.py`) carries its own asserts, one or more per figure, and the companion check (`python Notas/L05/companions/check_companions.py`) prints the three control vectors per page.

Links

- Unit: L05-00-map · L05-08-formulas · L05-09-pitfalls.
- Class 2 reference implementation (same layout): L02 Python.